

Distorted Waves, Intranuclear Cascade and Nuclear Transparency



Grant **PID2021-127098NA-I00**

funded by MICIU/AEI/10.13039/501100011033, “ERDF a way of making Europe” and FSE+.

Ayudas "Atracción de Investigadores con Alto Potencial". VII Plan Propio de Investigación y Transferencia, 2025, Universidad de Sevilla.

We acknowledge to RES resources provided by BSC to “Quantum Nuclear Structure and Dynamics” [FI-2026-1-0045]

Raúl González Jiménez

University of Seville, Spain

NuWro Workshop, Wroclaw.
16-18 September, 2026

Overview

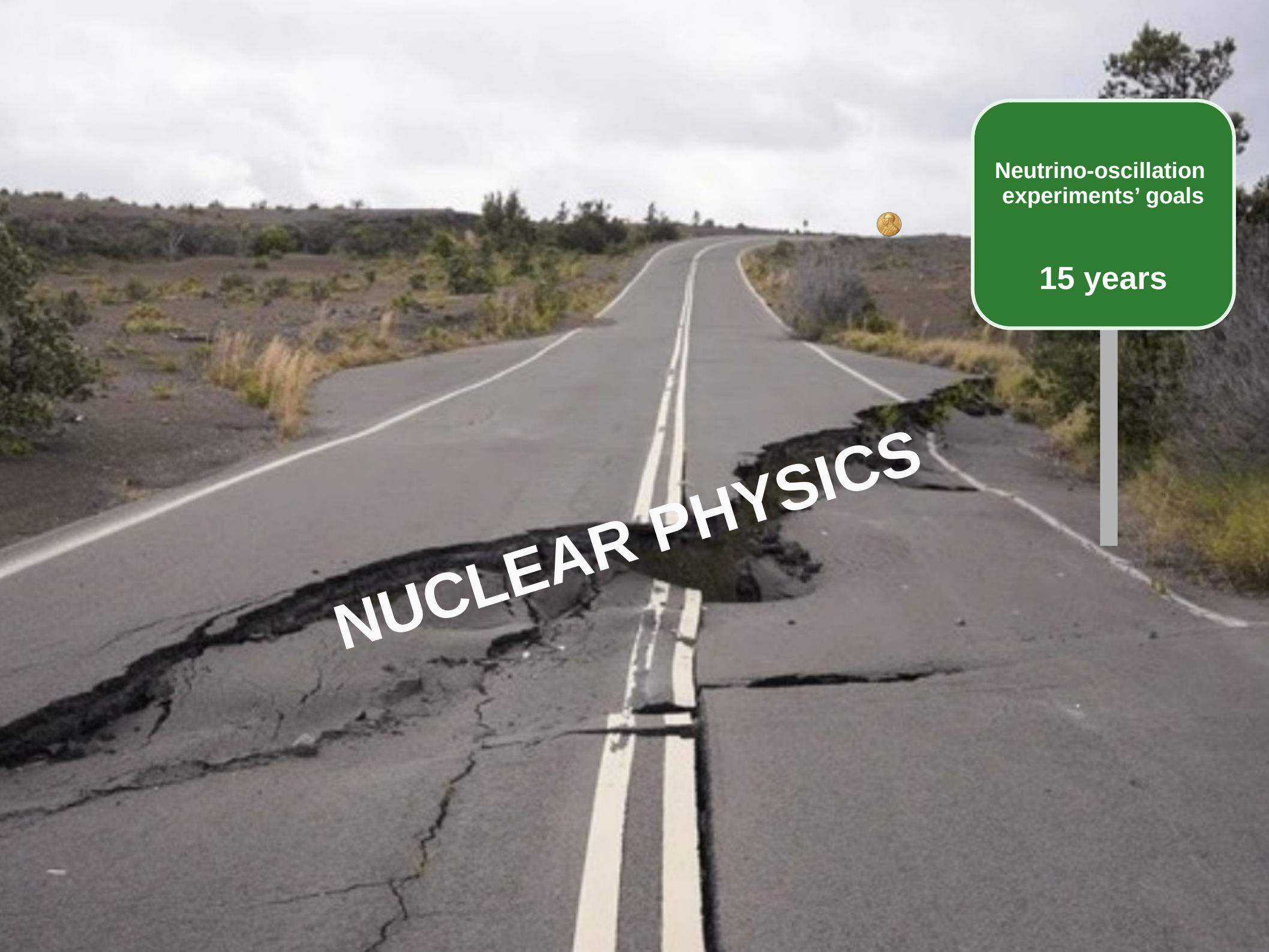
1. Motivation: **neutrino energy reconstruction**
2. The *two-step* approach of **neutrino event generators**
3. **Inclusive** electron-scattering cross section
4. **Exclusive** electron-scattering cross section

Why Nuclear Models in Long-Baseline neutrino-oscillation programs?



Neutrino-oscillation
experiments' goals

15 years

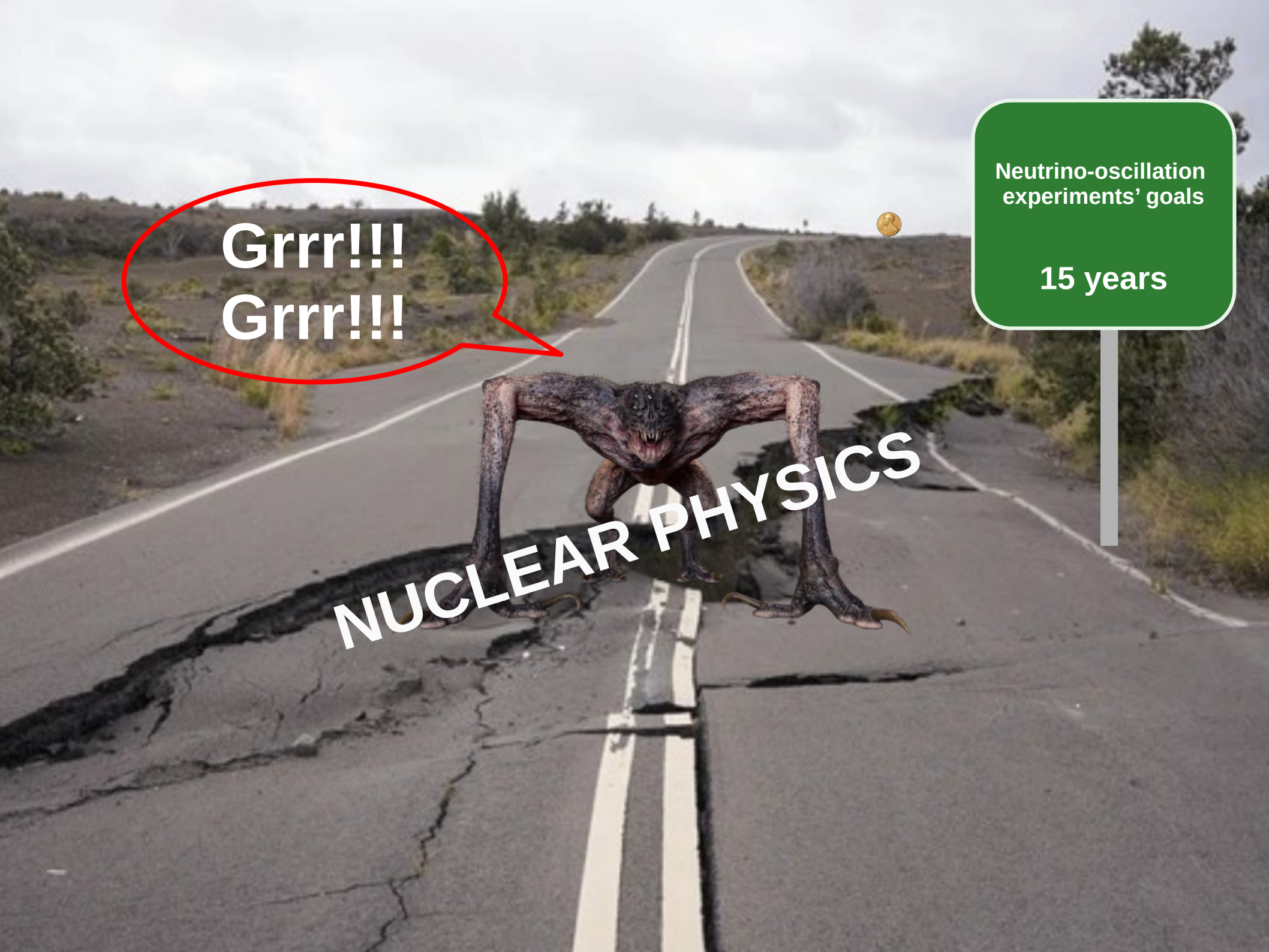


NUCLEAR PHYSICS



Neutrino-oscillation
experiments' goals

15 years

A large, dark, scaly monster with long, thin arms and legs is standing in the middle of a two-lane asphalt road. The road has a white double line down the center and is surrounded by dry, scrubby vegetation under a cloudy sky. The monster is facing the viewer with a menacing expression, showing sharp teeth. A red speech bubble is on the left, and a green sign is on the right.

**Grrr!!!
Grrr!!!**

NUCLEAR PHYSICS

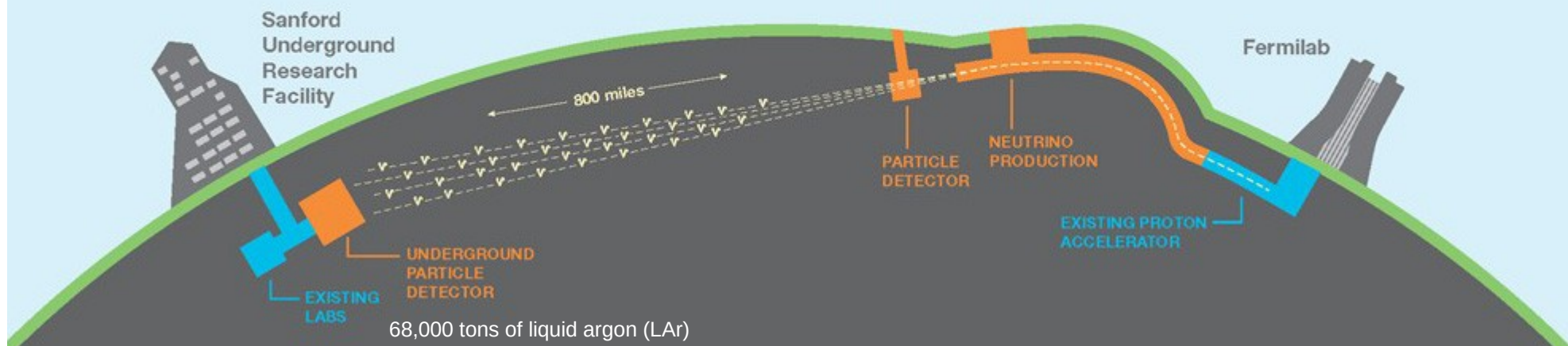
A small, gold-colored Lincoln cent is positioned above the green sign.

**Neutrino-oscillation
experiments' goals**

15 years

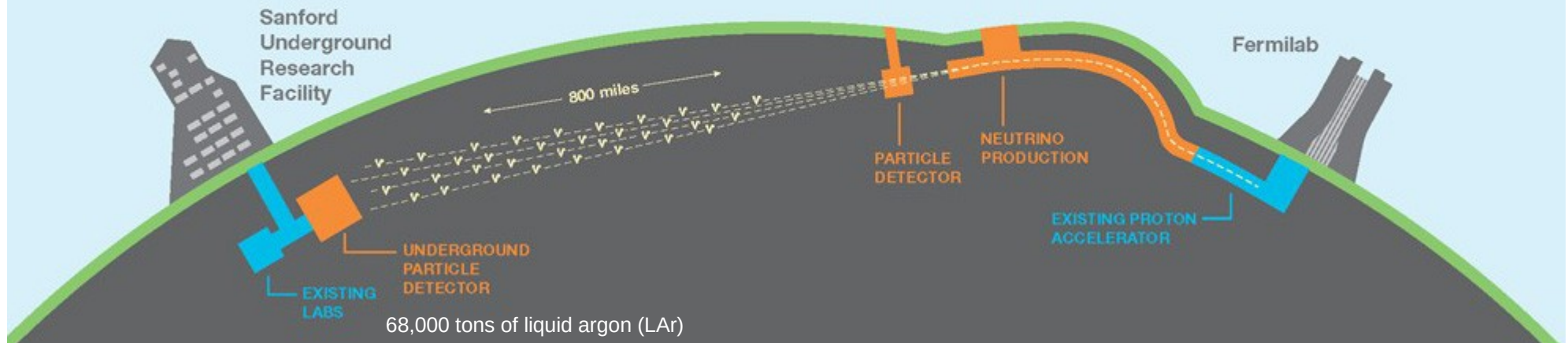
**Why Nuclear Models
in Long-Baseline
neutrino-oscillation programs?
neutrino energy reconstruction**

DUNE: Deep Underground Neutrino Experiment

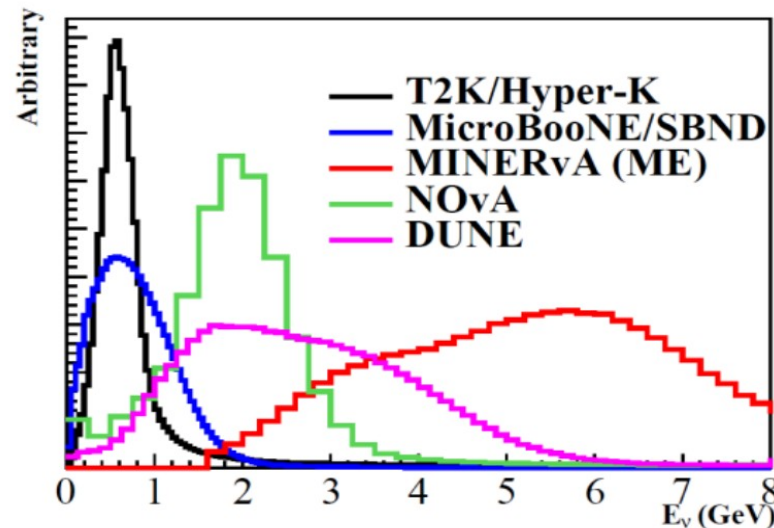


$$P_{a \rightarrow b} = \sin^2 2\theta \sin^2 \left(\frac{1.27 \Delta m^2 (\text{eV}^2) L (\text{km})}{E_\nu (\text{GeV})} \right),$$

DUNE: Deep Underground Neutrino Experiment



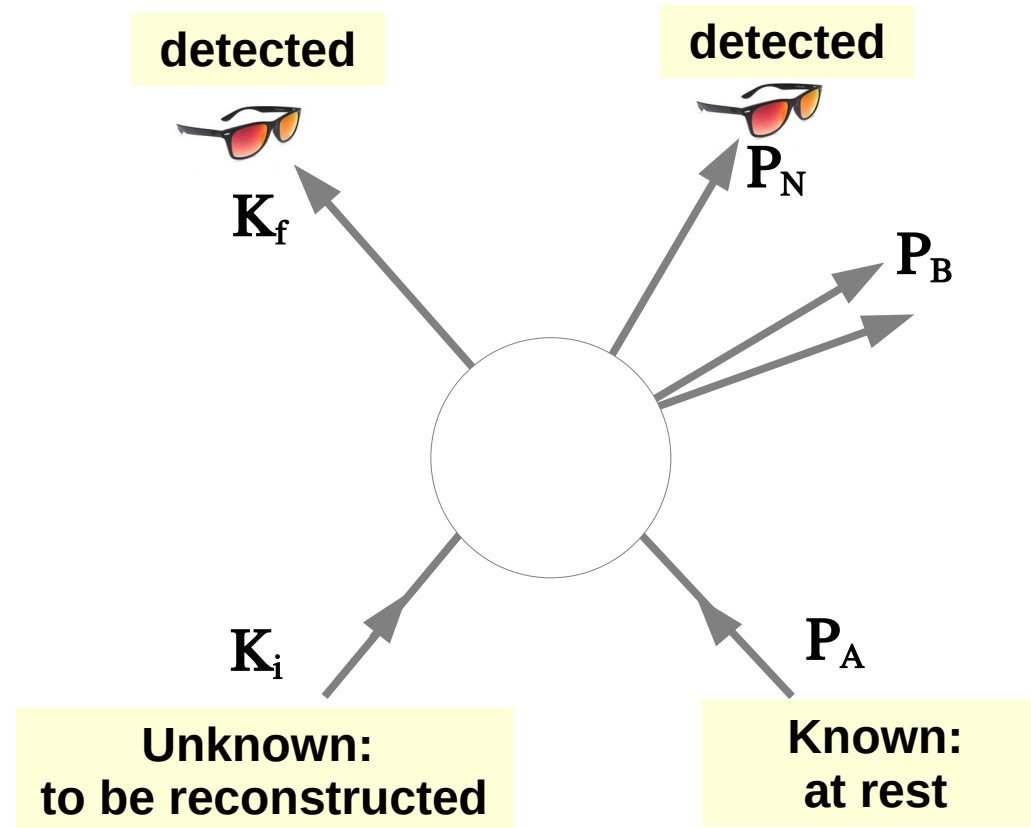
$$P_{a \rightarrow b} = \sin^2 2\theta \sin^2 \left(\frac{1.27 \Delta m^2 (\text{eV}^2) L (\text{km})}{E_\nu (\text{GeV})} \right),$$



neutrino fluxes

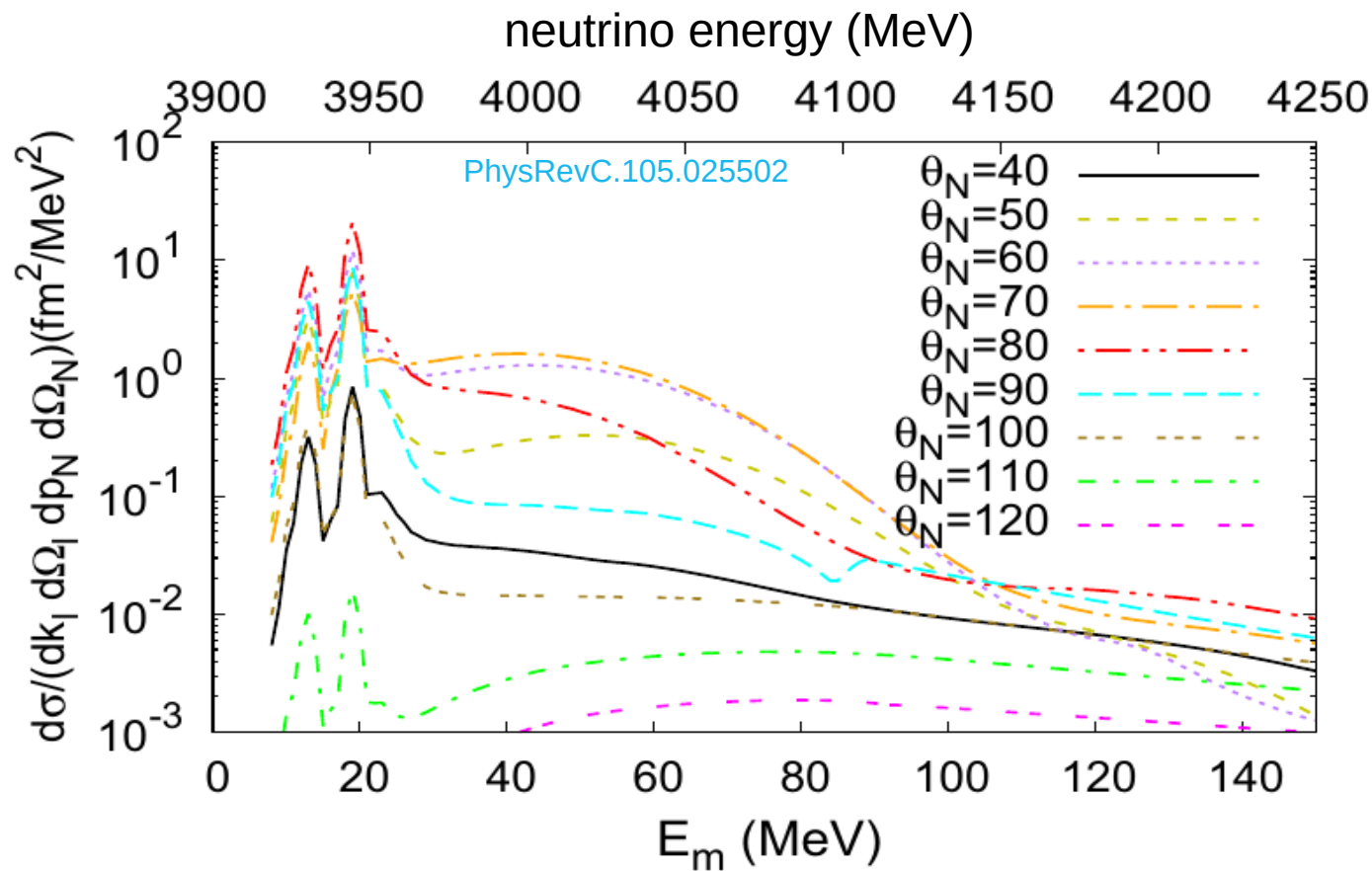
Neutrino energy reconstruction

E.g., consider an event consisting in a muon and a proton



Example of a kinematics:

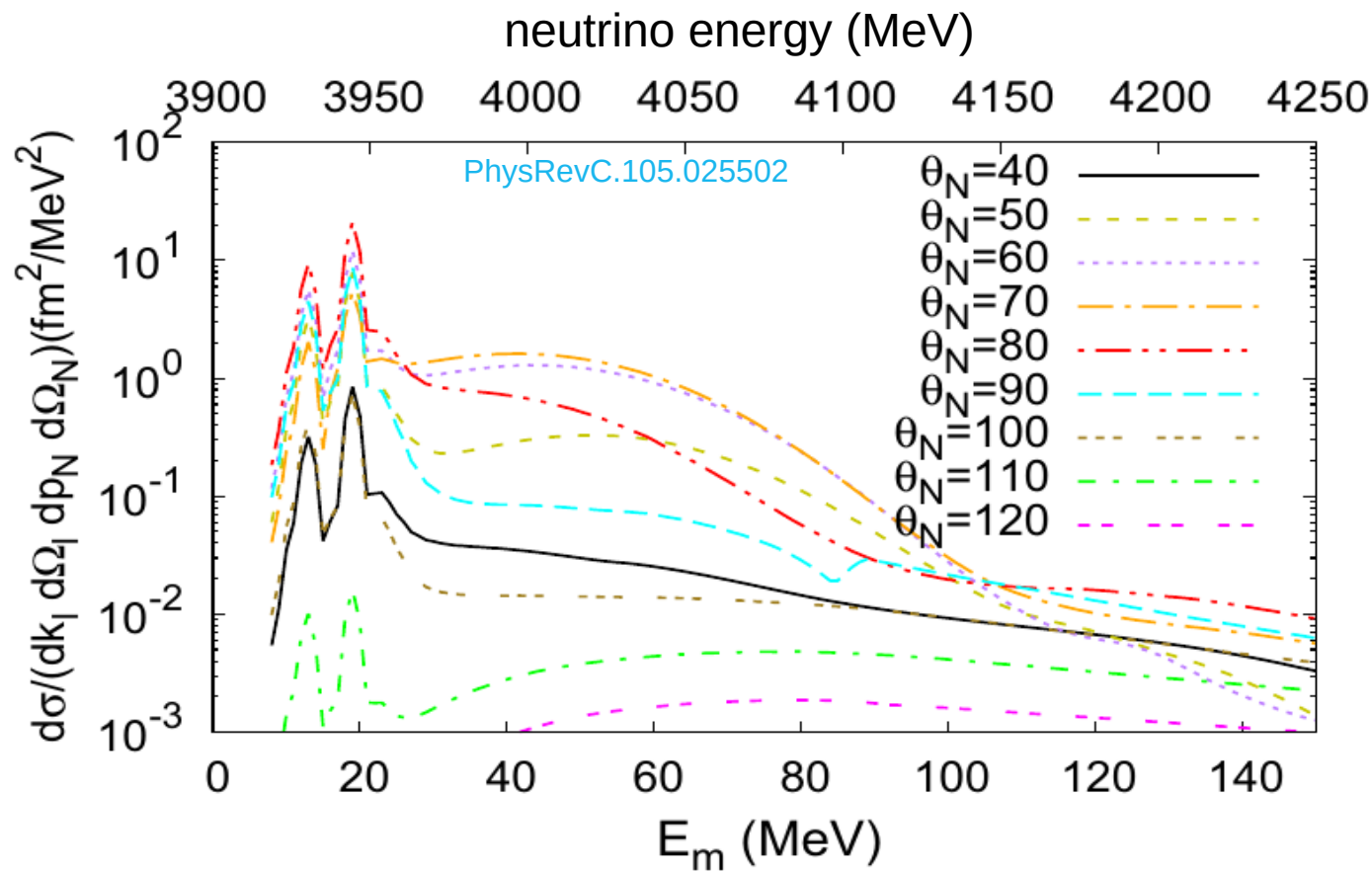
$$E_f = 3800 \text{ MeV}, \theta_f = 7 \text{ deg}, T_N = 140 \text{ MeV}$$



6-differential cross section using the DUNE flux, shown as a function of the neutrino energy (top) and missing energy E_m (bottom axis).

QE kinematics:
 $E_f = 3800$ MeV,
 $\theta_f = 7$ deg,
 $T_N = 140$ MeV,
 $\phi_N = 180$ deg.
 Each line corresponds to a different value of the proton scattering angle θ_N (in degrees).

Actually, these cross section are (proportional to) the **probability distribution function of having an initial neutrino energy for the given muon and proton kinematics.**

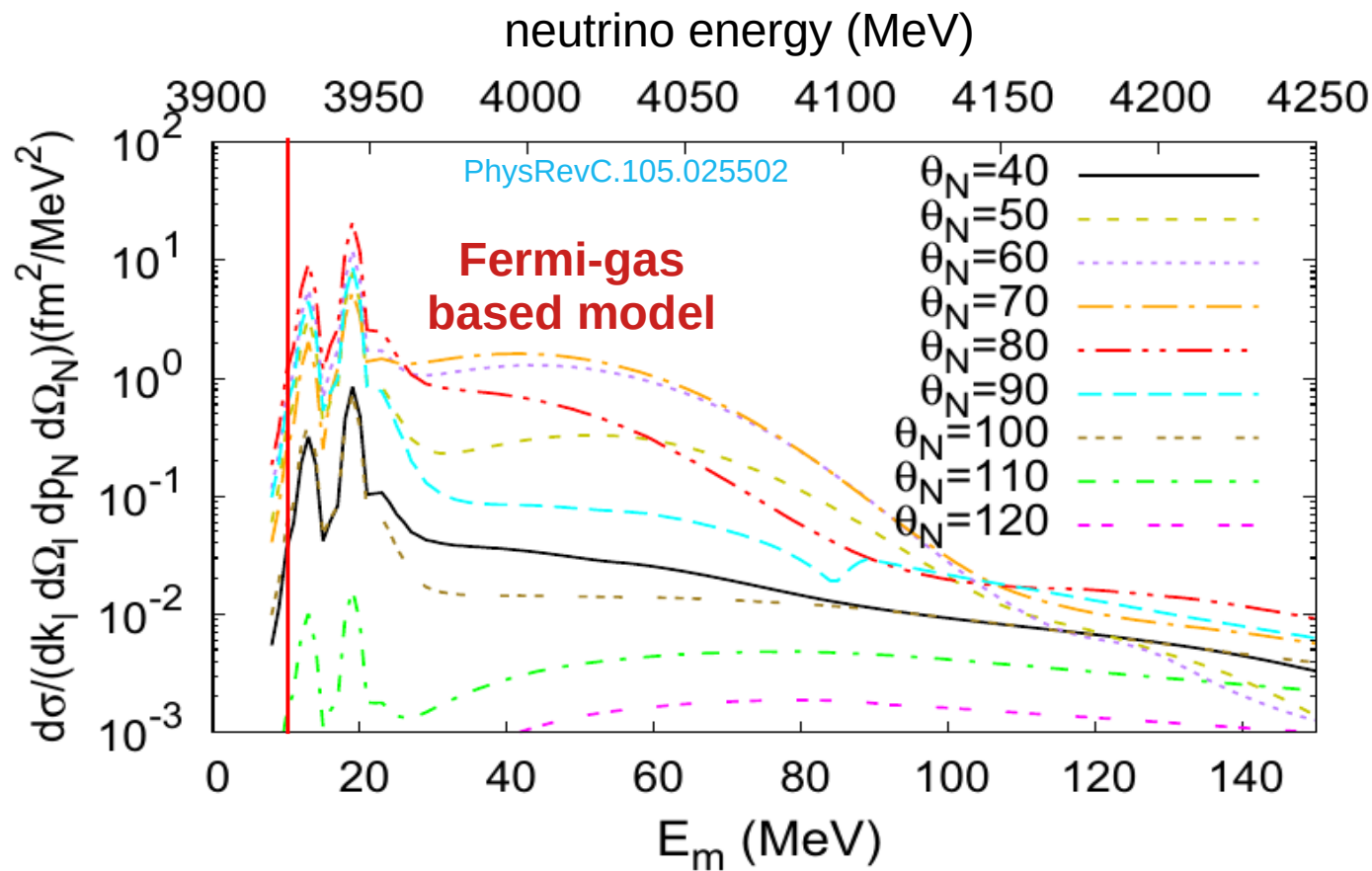


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Obviously, these distributions **do depend** on the nuclear model.

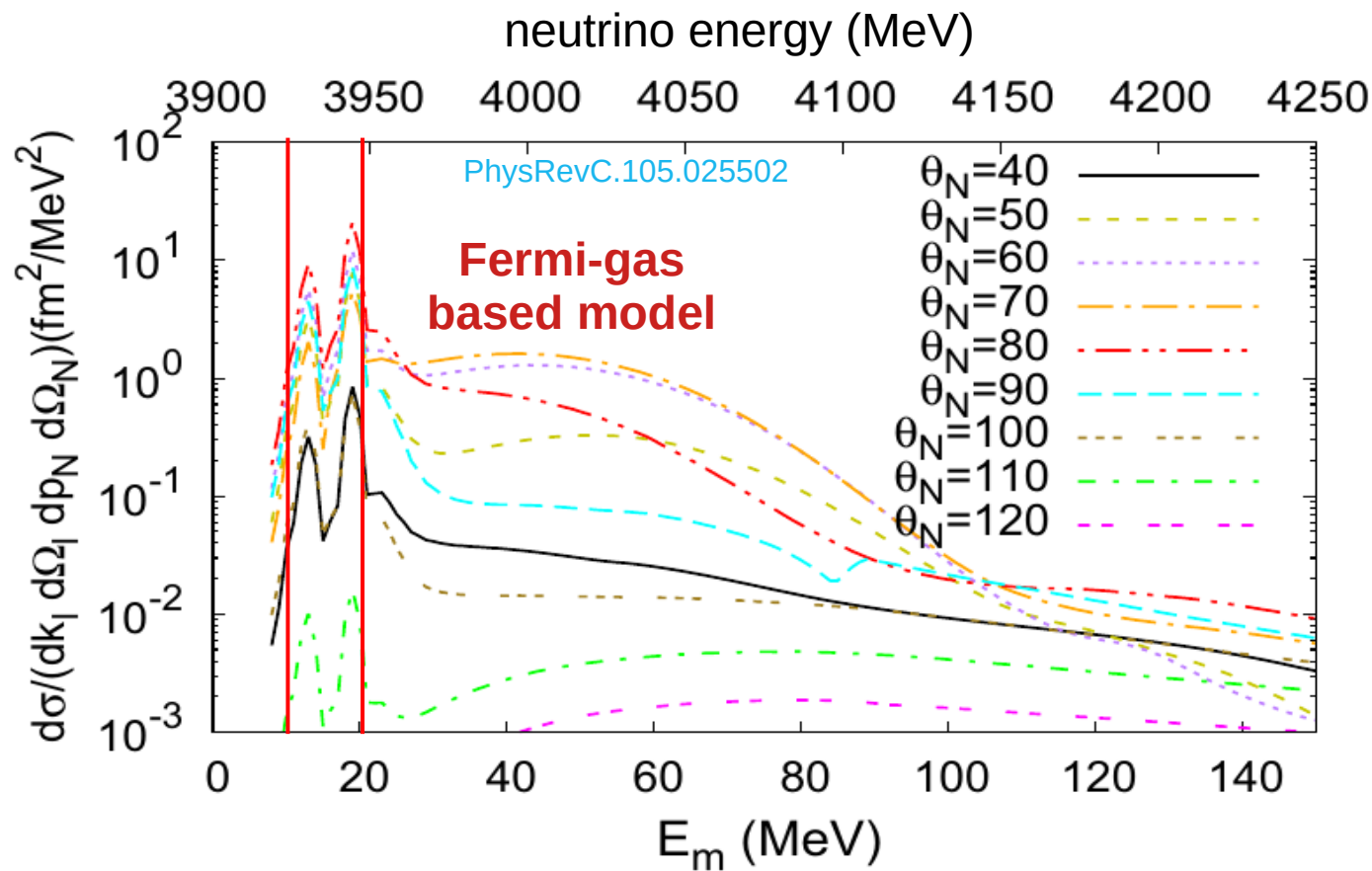


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In Fermi-gas based approaches, each event corresponds to a **unique neutrino energy**. Assuming a binding energy of 24 MeV, one gets:

θ_N	E_i (MeV)		
40	3922		

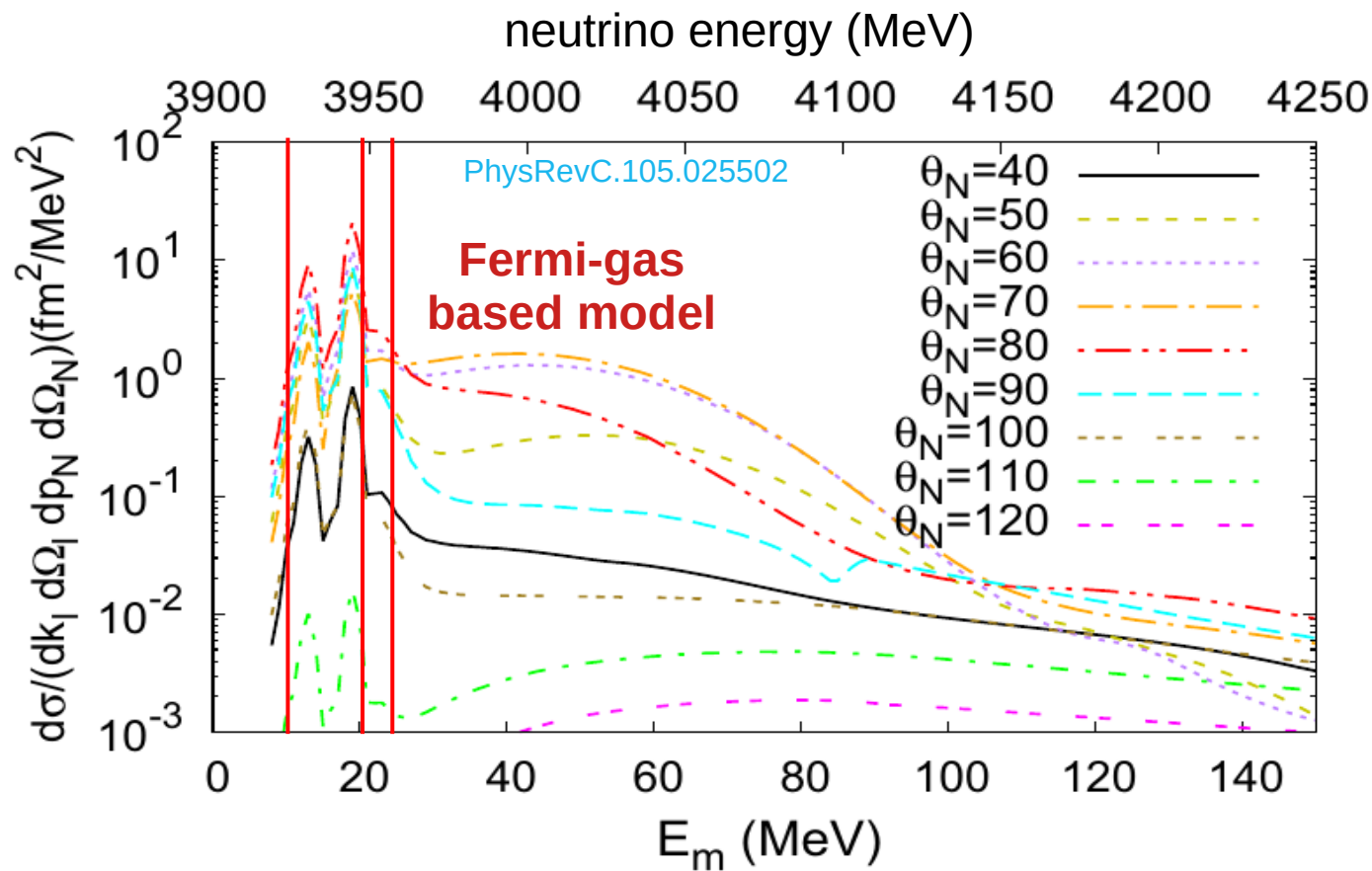


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θ_N	E_i (MeV)		
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50	3948		

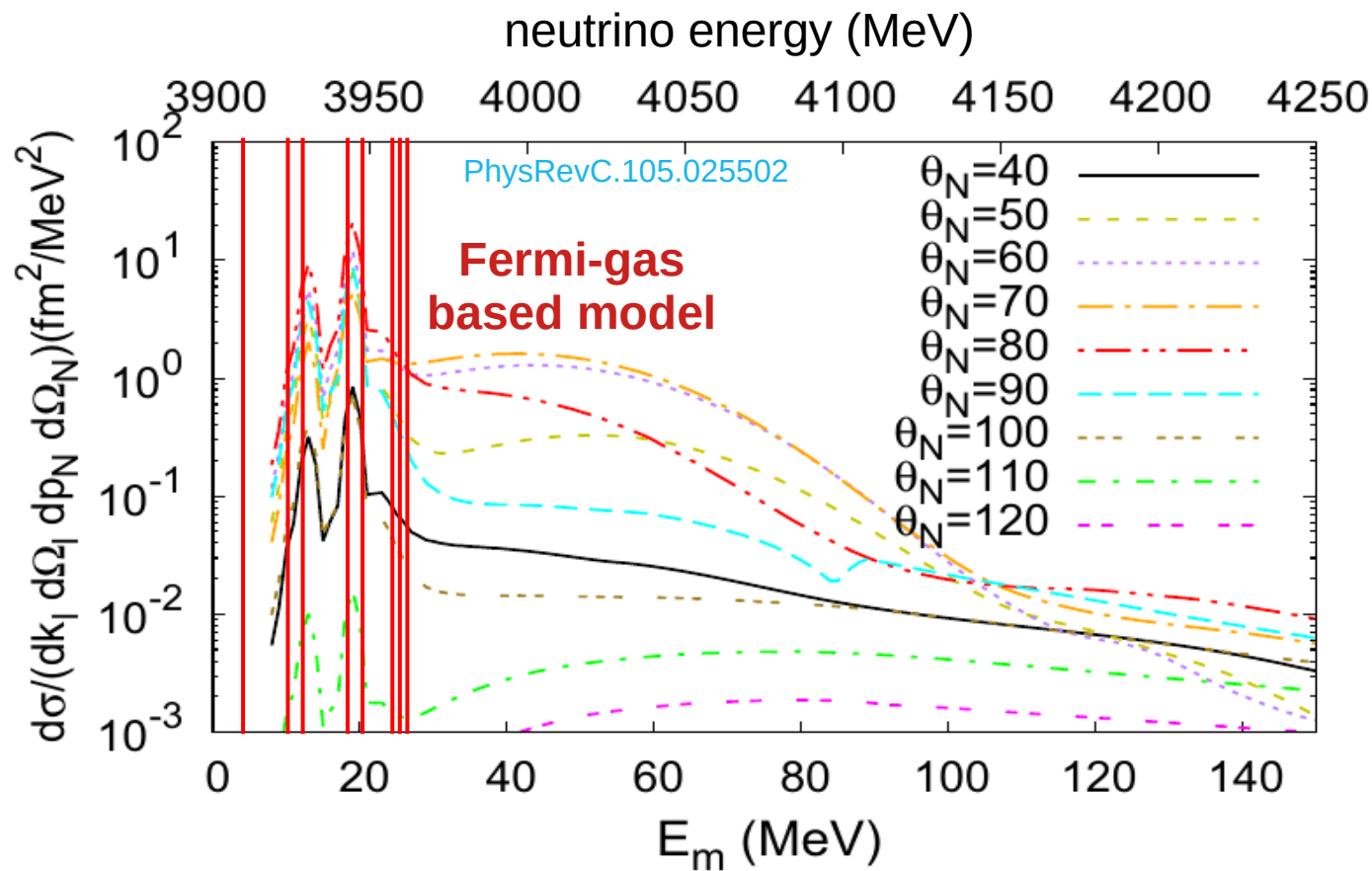


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θ_N	E_i (MeV)		
40	3922		
50	3948		
60	3961		

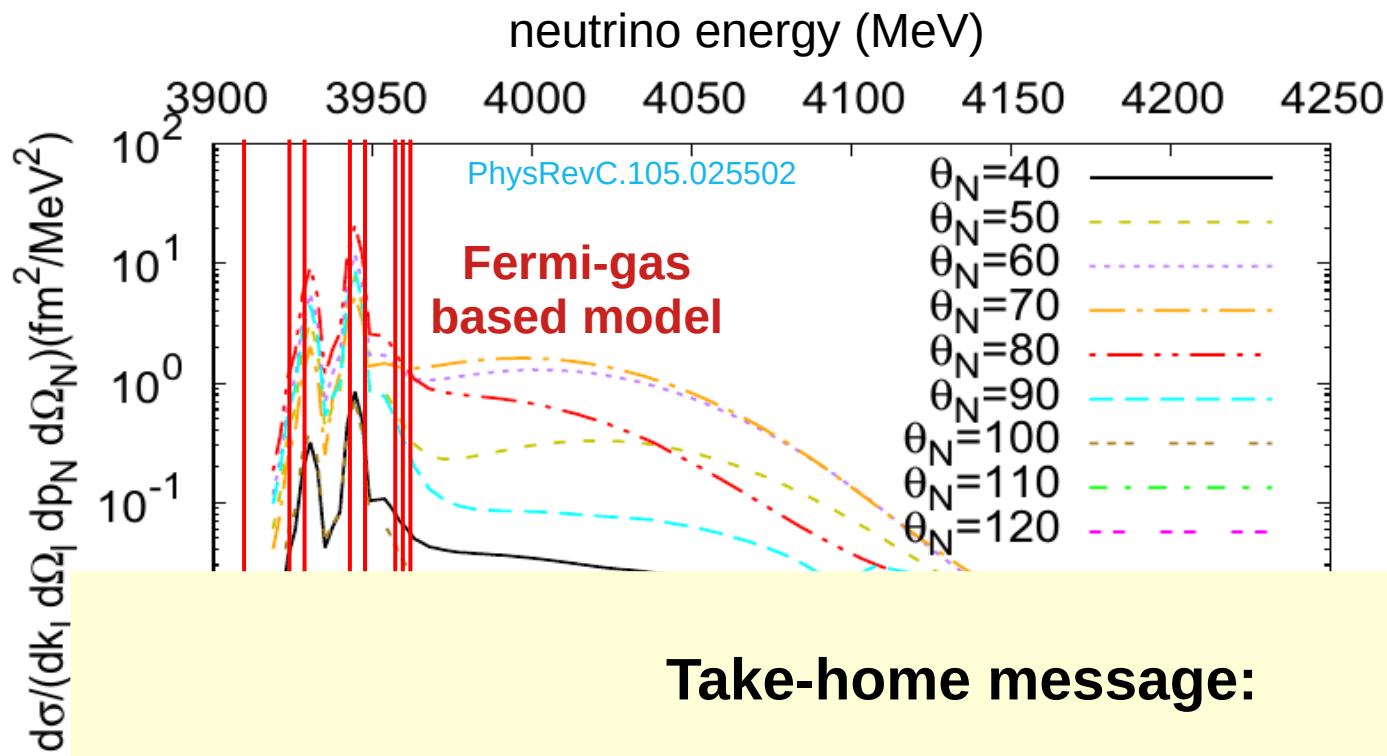


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θ_N	E_i (MeV)	θ_N	E_i (MeV)
40	3922	80	3957
50	3948	90	3945
60	3961	100	3929
70	3963	110	3910



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QE kinematics:
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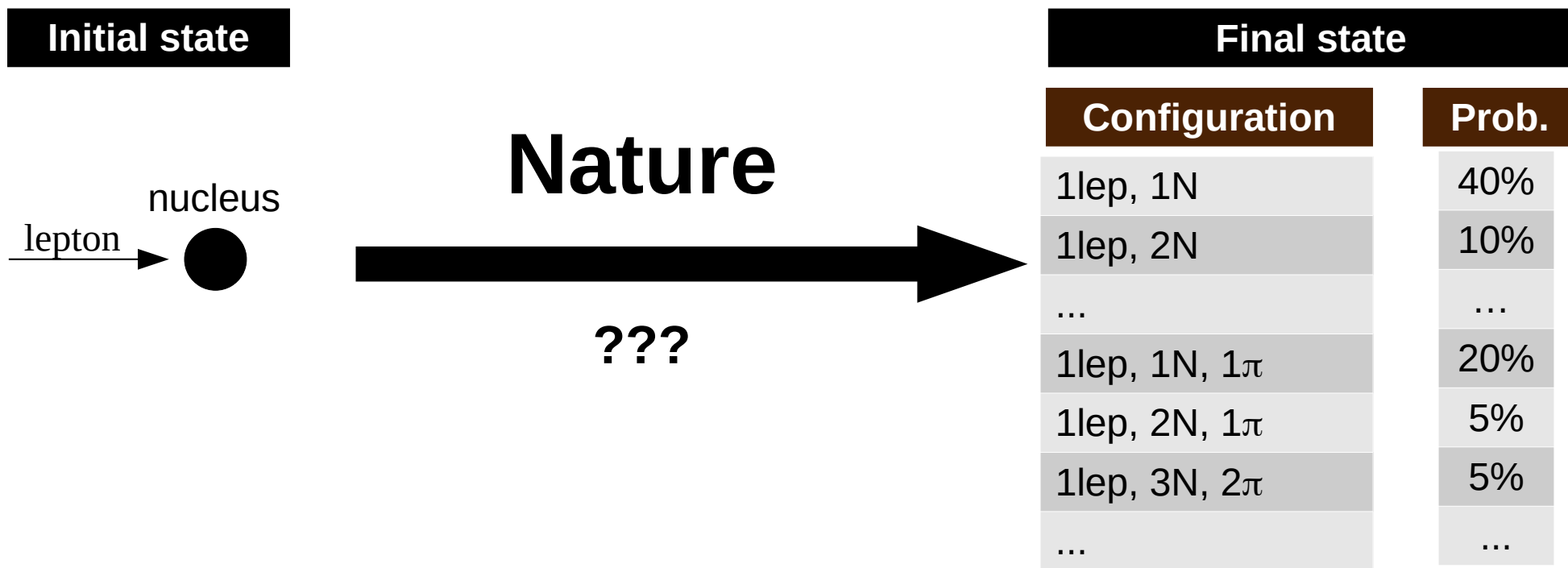
Take-home message:

Neutrino energy reconstruction depends on the theoretical models

The better the nuclear model is, the better the estimate of the neutrino energy.

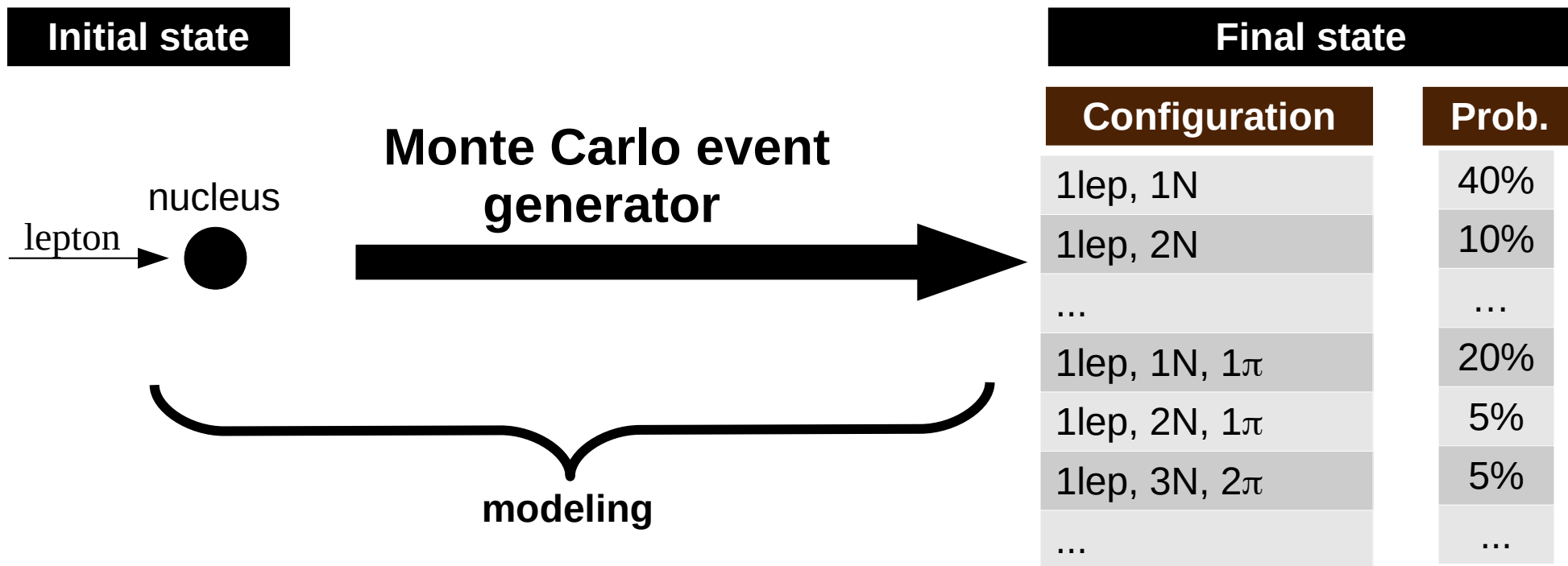
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Monte Carlo neutrino event generators



Nature: there is an initial state and a final state, she does not inform us about what happens in the middle.

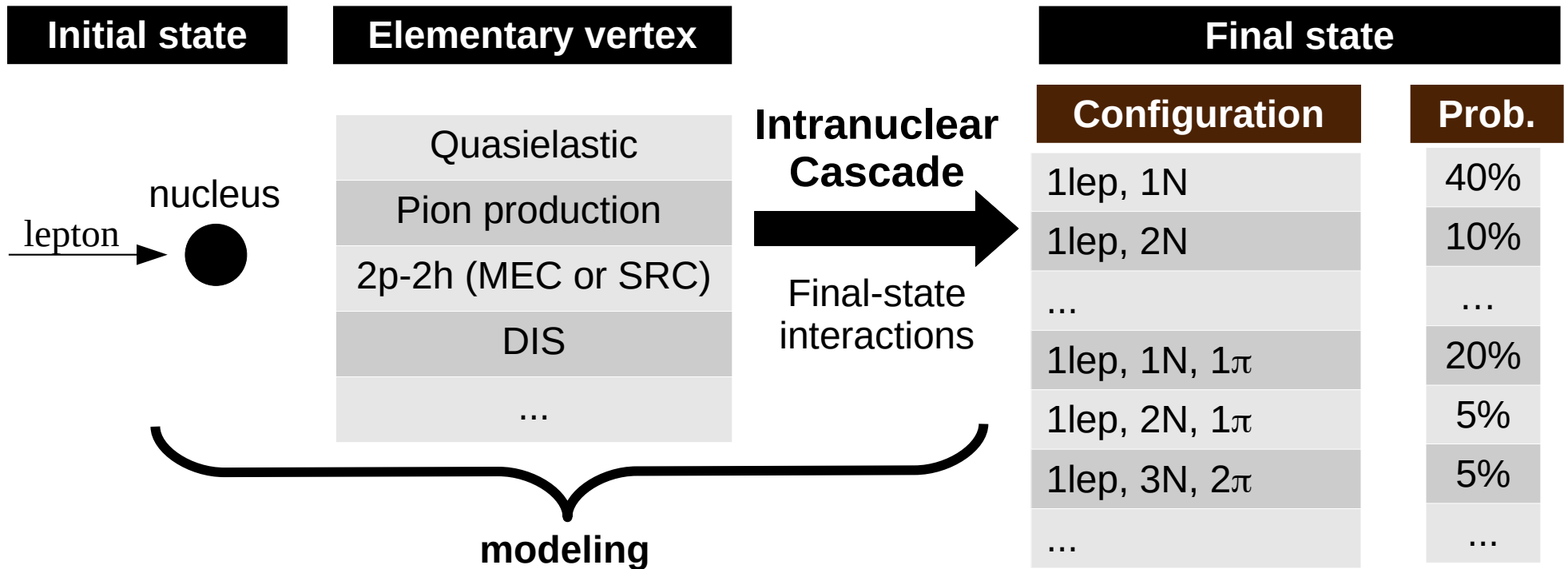
In a scattering experiment, the lepton-nucleus interaction happens many times, so one finds many different final-state configurations, each one with a given probability.



MC event generators...

the goal is to get the same final state configurations as in Nature.

HOW ?



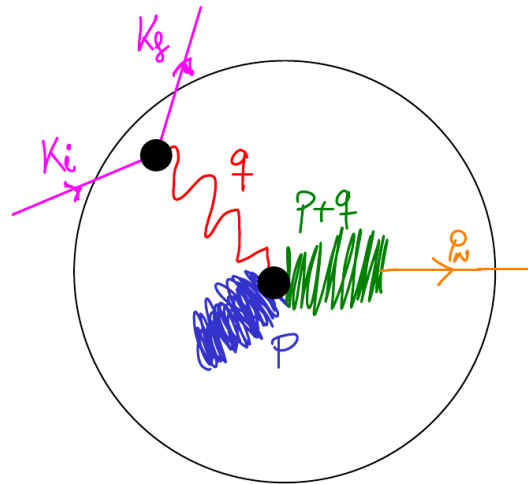
MC event generators...

Two steps:

(1): Reaction channel (elementary vertex).

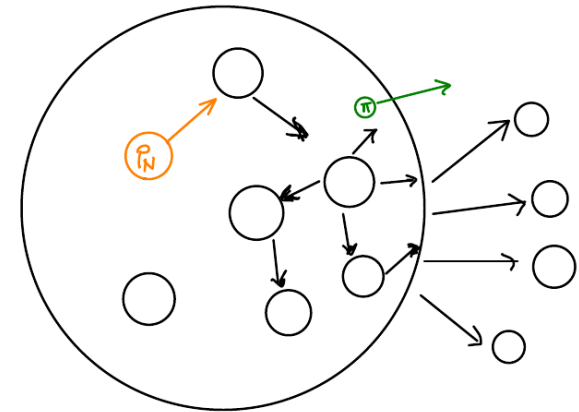
(2): Intranuclear Cascade provides the hadron multiplicity of the final state.

A possible **interpretation** is:
there is a time separation...



This happens first

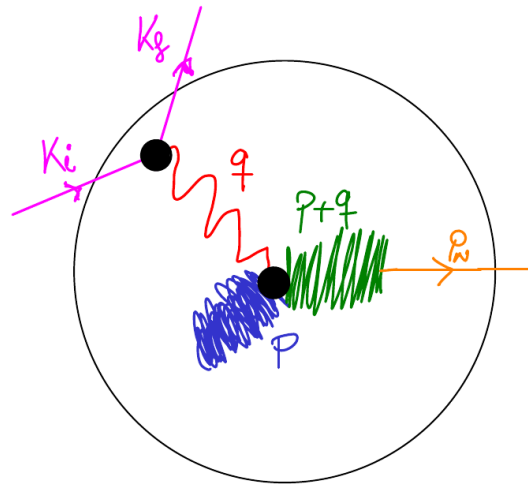
**elementary-vertex
model**



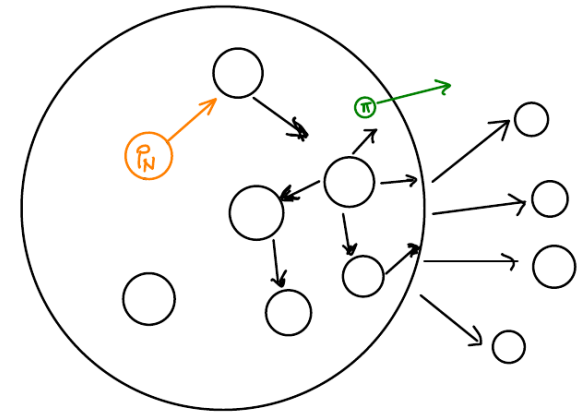
Then, re-scattering can happen

**Intranuclear cascade
(INC) model**

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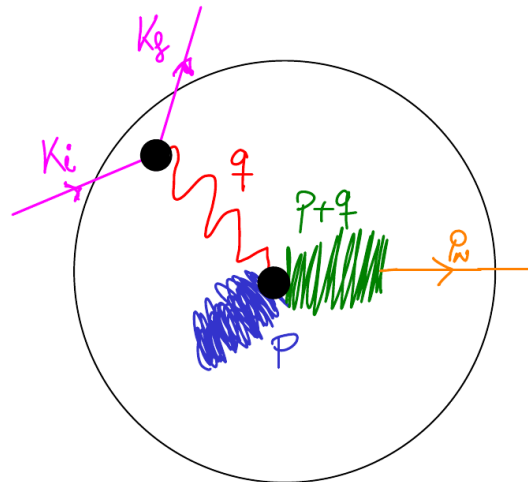
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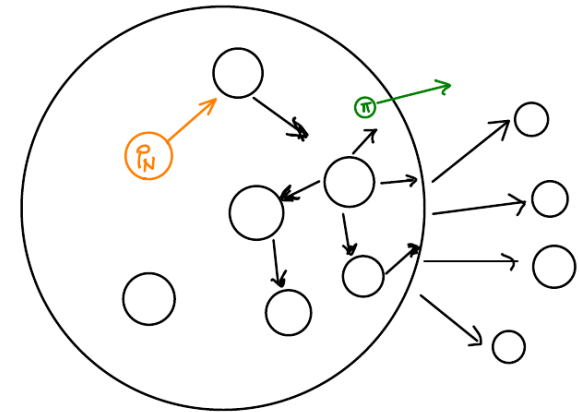
**Is this *two-step* process
what happens in Nature ???**

A possible **interpretation** is:
there is a time separation...



This happens first

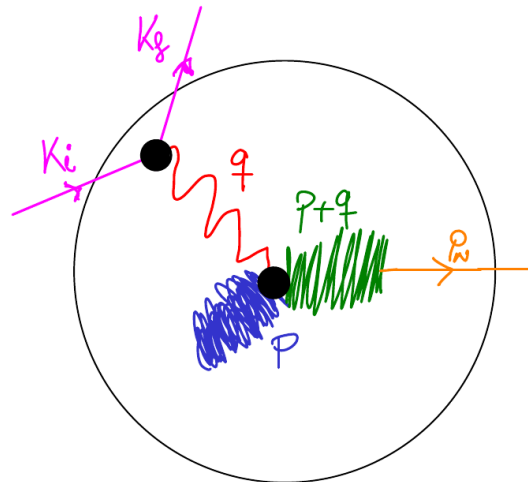
Some time goes by...



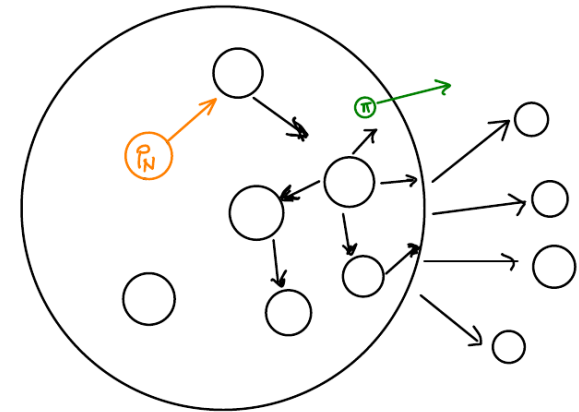
Then, re-scattering can happen

Which ingredients do we include at
each step ???

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This happens first



Then, re-scattering can happen

**Which ingredients do we include at
each step ???**

Inclusive (e,e') cross section

Relativistic Fermi gas

A relativistic gas of non-interacting fermions in the thermodynamic limit.

Set of nuclear effects:

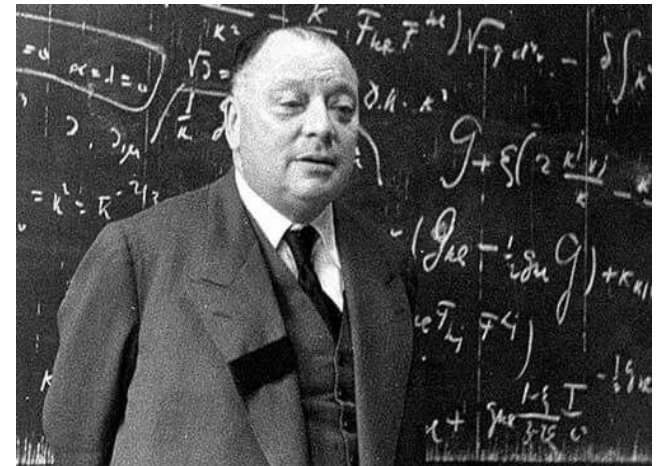
1. Inside a nucleus, the nucleons are moving: **Fermi motion**.
2. **Pauli's exclusion principle**.
3. *Ad hoc* **binding energy**.



Enrico Fermi

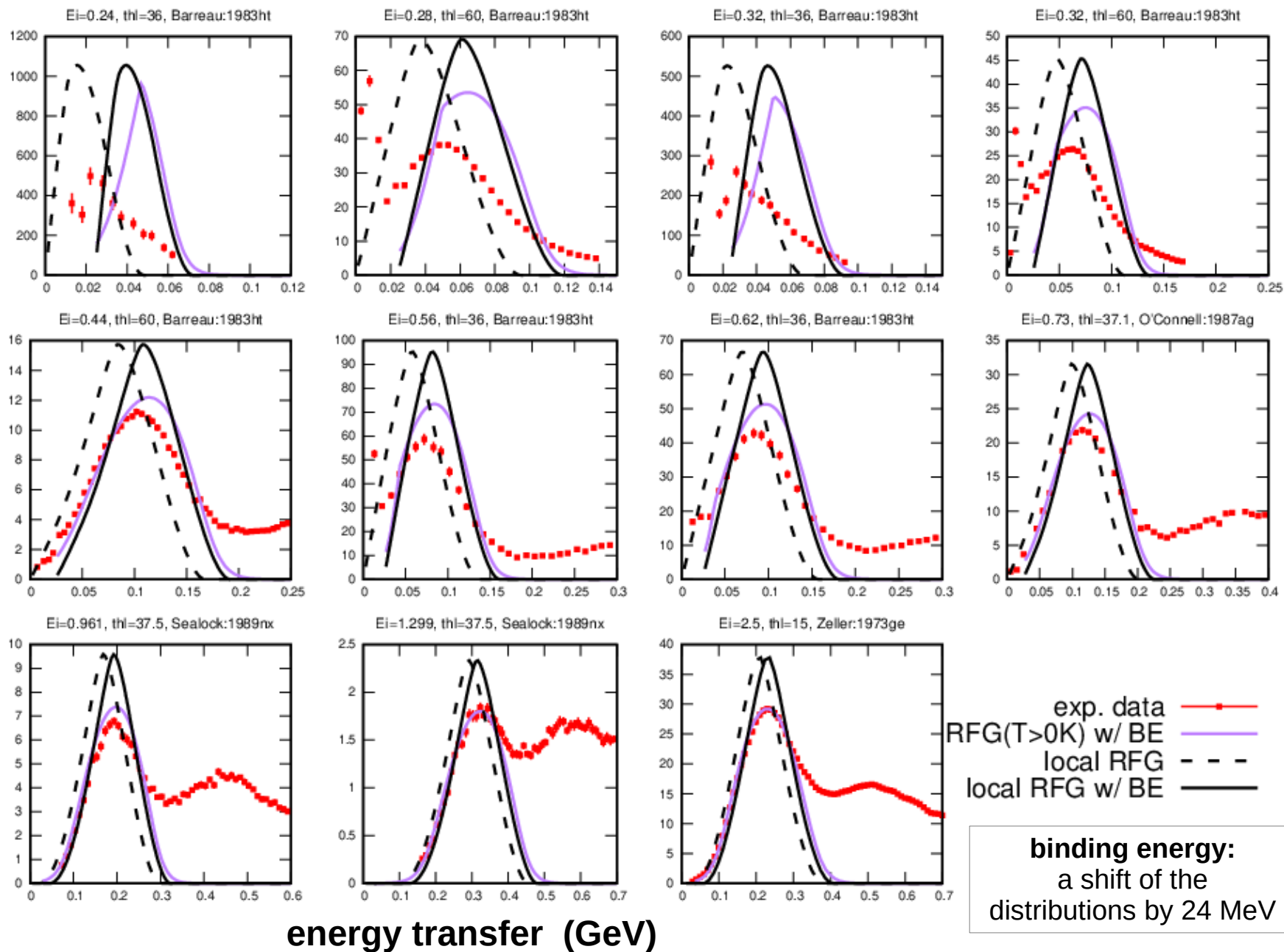


Paul Dirac

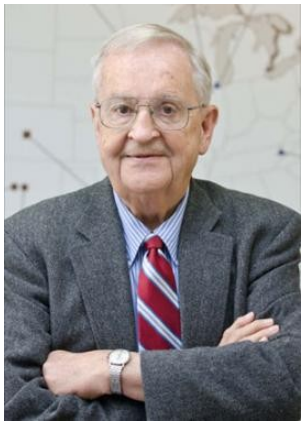


Wolfgang Pauli

$^{12}\text{C}(e,e')$



Independent-particle (relativistic) shell model



J. Dirk Walecka



Brian Serot

Example: a relativistic mean-field model.

Wave equation with strong potentials:

$$[-i\boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + V(r) + \beta (M + S(r))] \Psi(\mathbf{r}) = E\Psi(\mathbf{r})$$

The solutions are energy eigenstates, representing the nucleons in the levels:

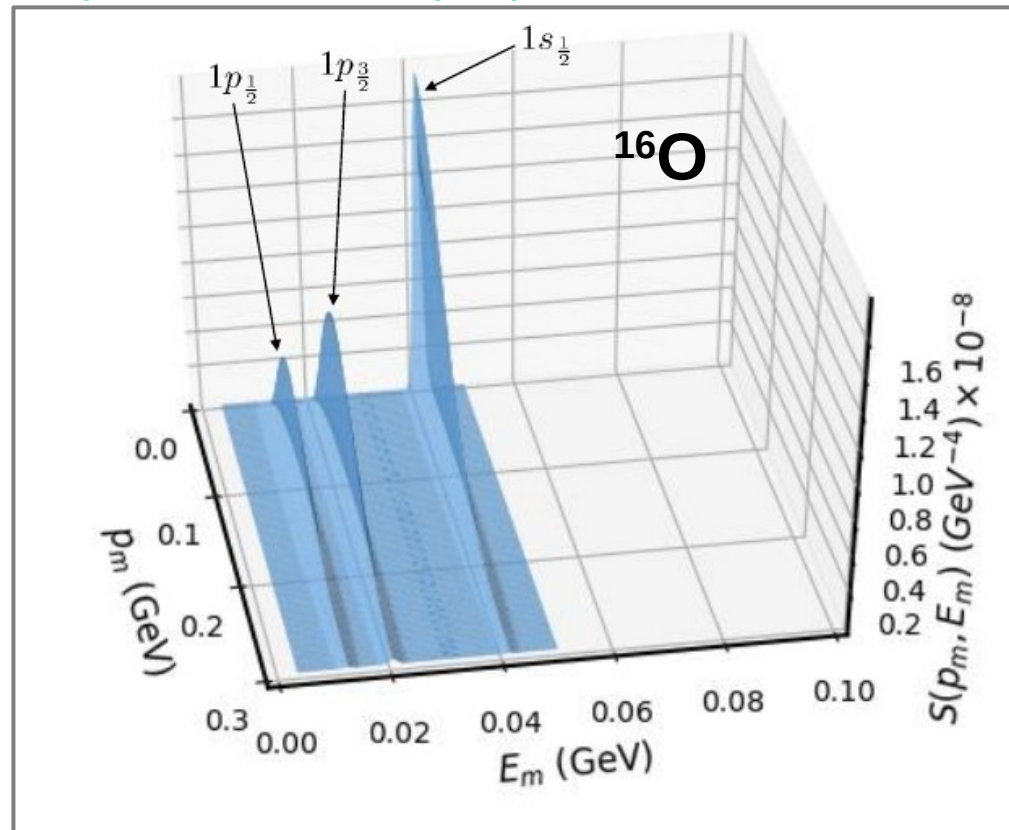
$$\Psi_k^{m_j}(\mathbf{r}) = \begin{pmatrix} g_k(r)\varphi_k^{m_j}(\Omega_r) \\ if_k(r)\varphi_{-k}^{m_j}(\Omega_r) \end{pmatrix},$$

“The **initial-state spectral function (SF)** gives the probability density of knocking out a nucleon from the nucleus, giving to the system an energy E_m and a momentum $\mathbf{p}_m$.”

Spectral function in the IPSM:

$$S(E_m, p_m) = \sum_{\kappa} n_{\kappa}(p_m) \delta(E_m - E_m^{\kappa})$$

Phys. Rev. C 100, 044620 (2019)

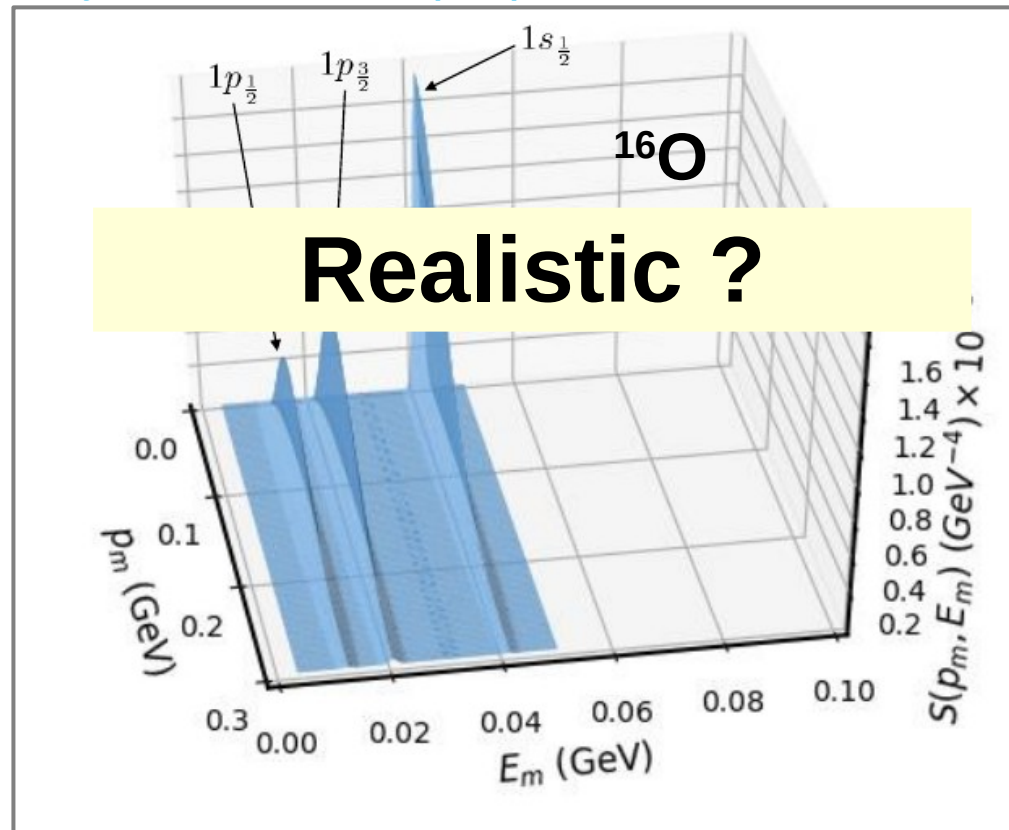


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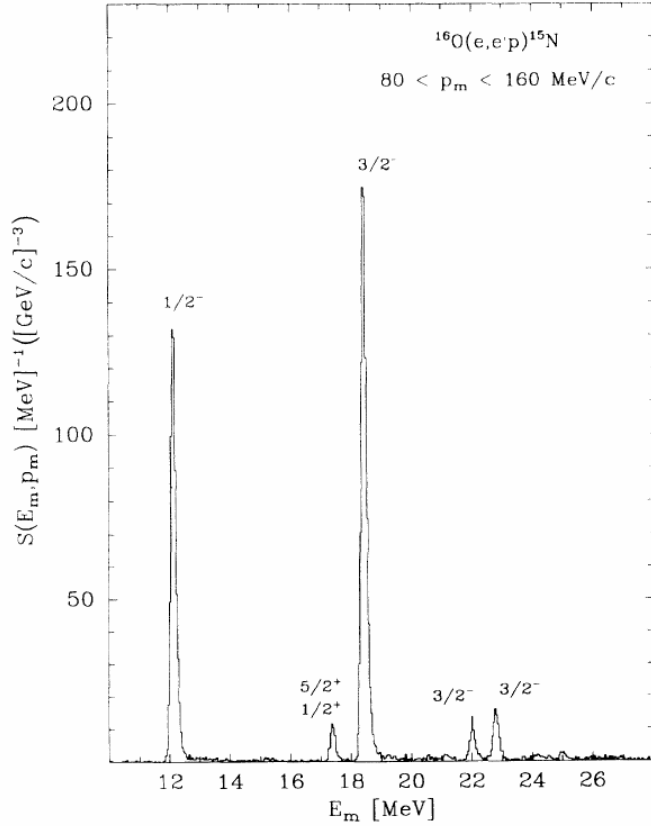
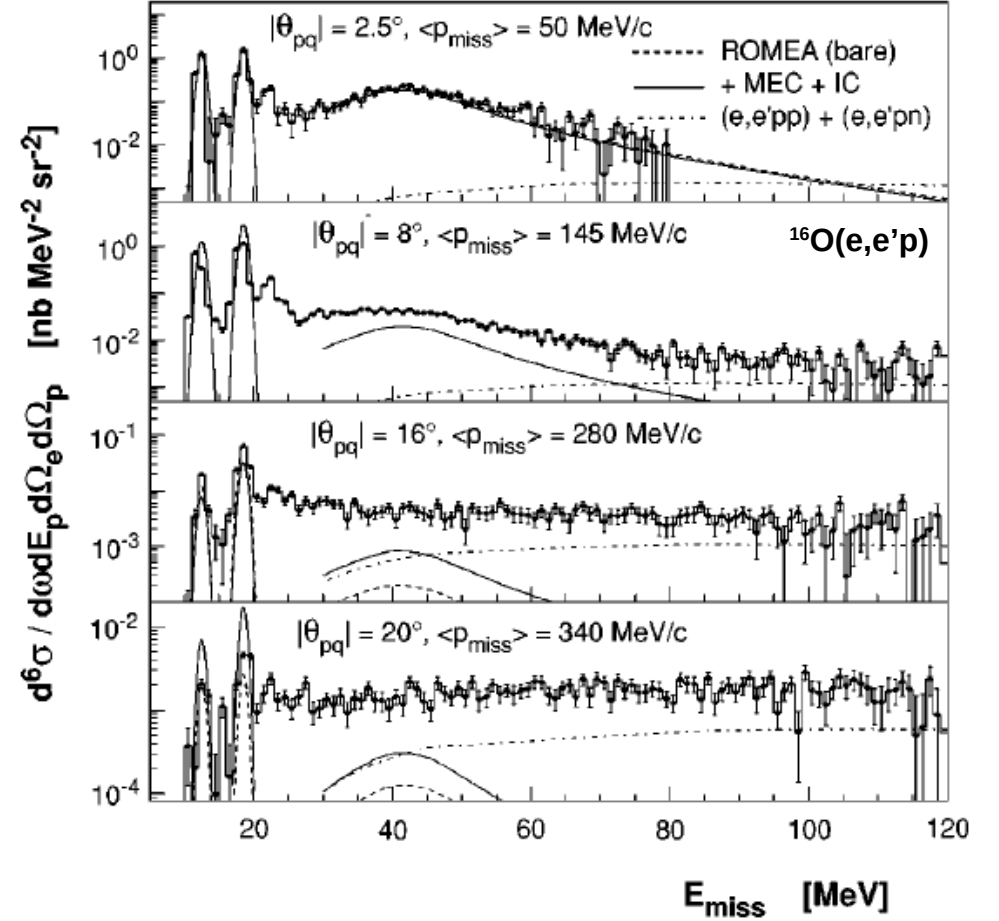


FIG. 1. $^{16}\text{O}(e, e'p)^{15}\text{N}$ missing energy spectrum for the kinematics centered about $p_m = 120 \text{ MeV}/c$.



$$E_m \equiv \omega - T_N - T_B = M_B^* + M - M_A.$$

Effects beyond the independent-particle shell model:

Experimentally one observes:

- + fragmentation of the strength
- + more levels
- + width at the peaks, especially in deep levels
- + spectroscopic factors (< 1)
- + Short Range Correlations: $\sim 20\%$ of the nucleons appear in the high E_m and p_m region.

These effects could be (somehow) included in **a more realistic representation of the spectral function**;

extracted using, e.g., *ab initio* models, and/or experimental data, like (e,e'p).

Plane-wave impulse approximation (PWIA) with a realistic spectral function

$$\frac{d\sigma(k_e)}{d\mathbf{k}'_e d\mathbf{p}_p} = S(E_m, p_m) \sigma_{e-p}^{\text{off-shell}}(\mathbf{k}'_e, \mathbf{p}_p)$$

The spectral function (SF): the Rome approach

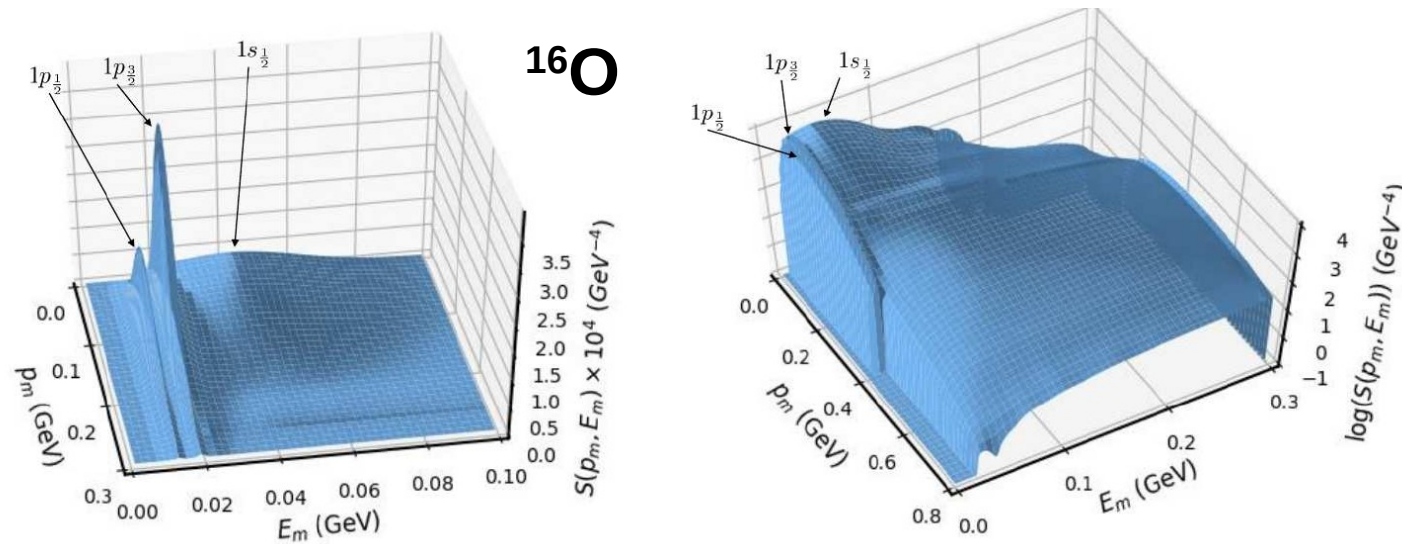
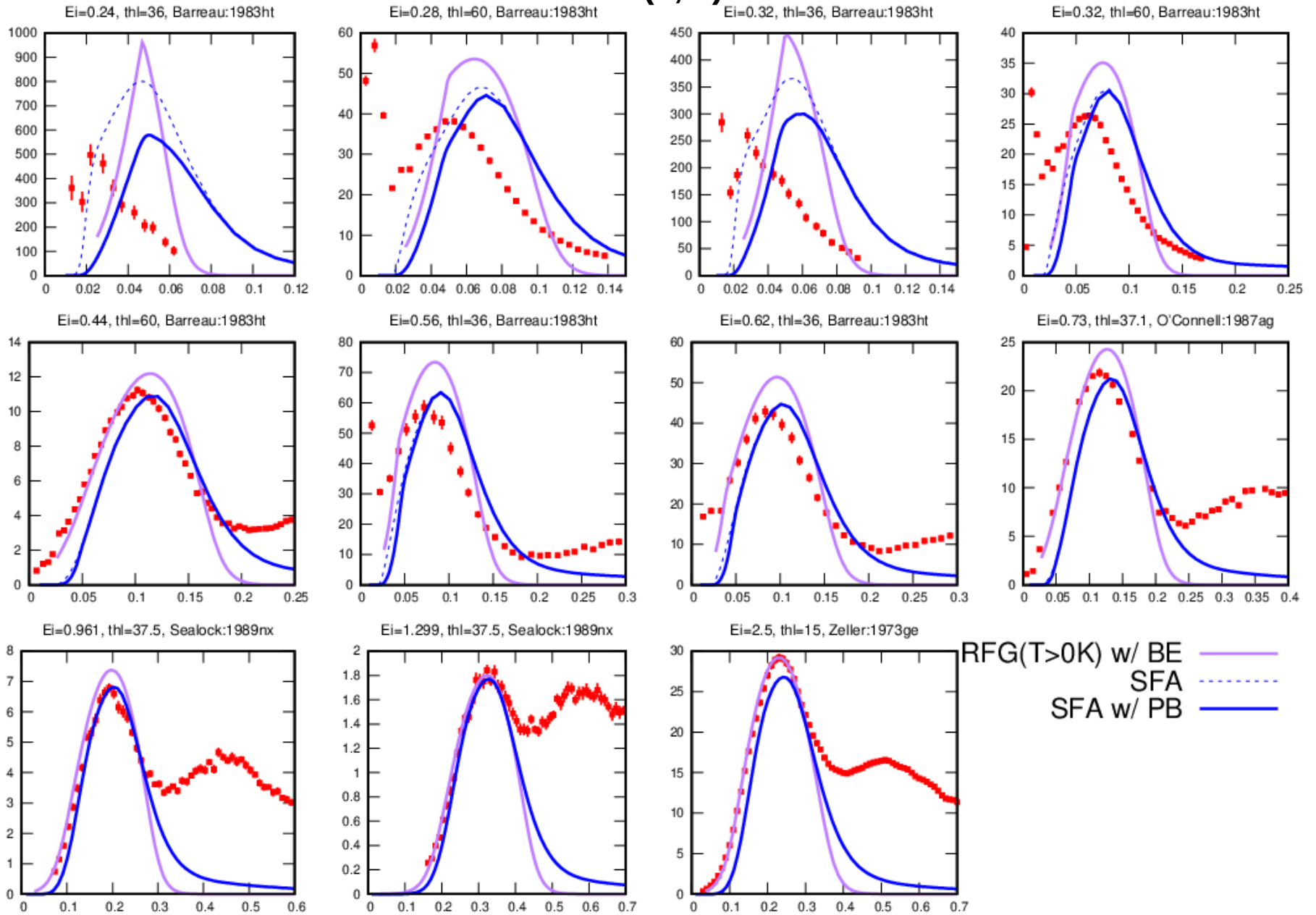


Image from [Phys. Rev. C 100, 044620 \(2019\)](#); spectral function from [Nuclear Physics A 579, 493 \(1994\)](#); [Phys. Rev. D 72, 053005 \(2005\)](#)

$$S(E_m, p) = S_{\text{mean field}}(E_m, p) + S_{\text{SRC}}(E_m, p),$$

$$S_{\text{mean field}}(E_m, p) = \sum_{\kappa} n_{\kappa}(p) \rho(E_m),$$

$^{12}\text{C}(e,e')$



energy transfer (GeV)

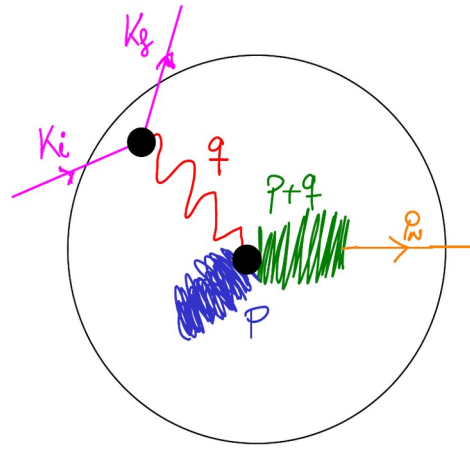
We have already included all the available information about the initial state, from theory and/or experiment.

What is missing?

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What is missing?

A consistent treatment of the initial and final states:

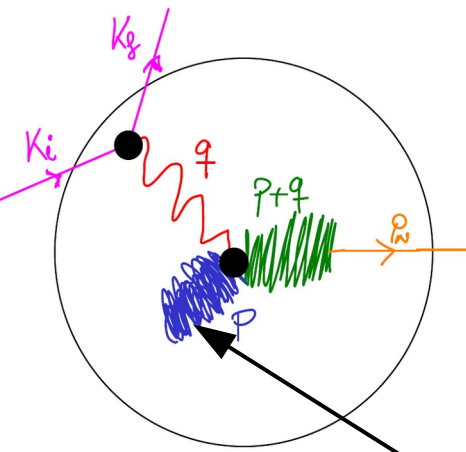


$$J_{had}^{\mu} = \int d\mathbf{p} \, \bar{\Psi}_F(\mathbf{p} + \mathbf{q}, \mathbf{p}_N) \, \mathcal{O}_{\text{one body}}^{\mu} \, \Psi_B(\mathbf{p})$$

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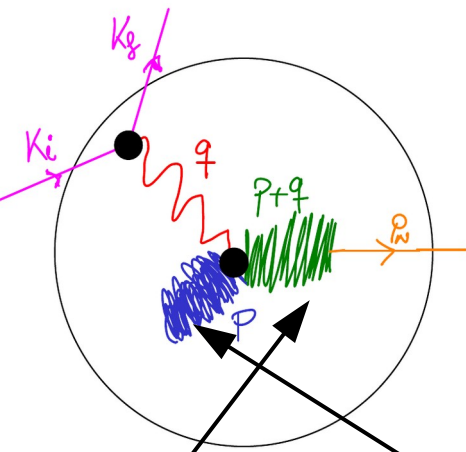
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Bound state from an IPSM

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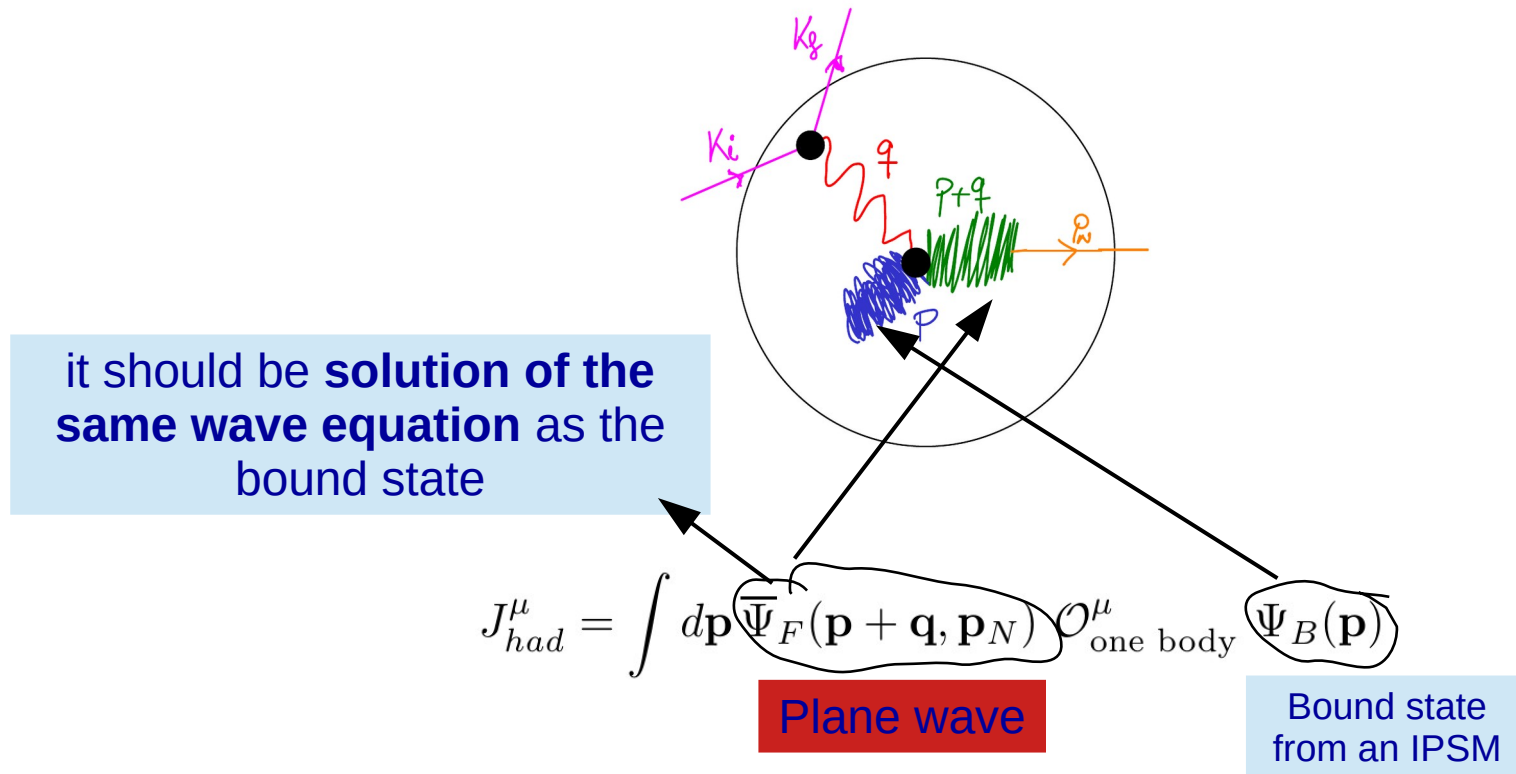


$$J_{had}^{\mu} = \int d\mathbf{p} \underbrace{\bar{\Psi}_F(\mathbf{p} + \mathbf{q}, \mathbf{p}_N)}_{\text{Plane wave}} \mathcal{O}_{\text{one body}}^{\mu} \underbrace{\Psi_B(\mathbf{p})}_{\text{Bound state from an IPSM}}$$

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A consistent treatment of the initial and final states:



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A consistent treatment of the

This ensures:

- + **orthogonality** of the states (Pauli blocking),
- + **vector current conservation**,
- + **distortion** of the final nucleon (elastic FSI) is incorporated

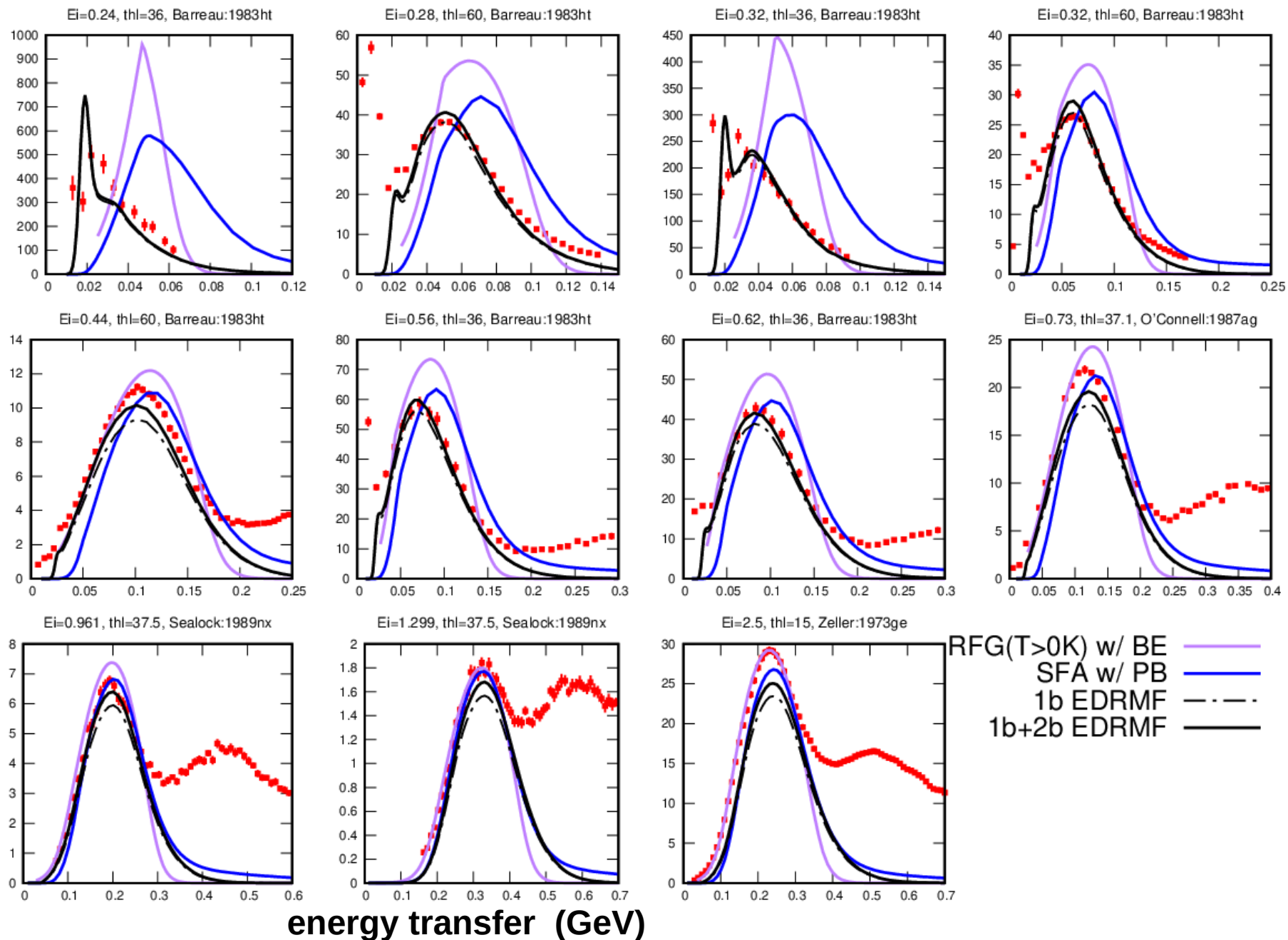
it should be **solution of the same wave equation** as the bound state

$$J_{had}^{\mu} = \int d\mathbf{p} \, \overline{\Psi}_F(\mathbf{p} + \mathbf{q}, \mathbf{p}_N) \mathcal{O}_{\text{one body}}^{\mu} \Psi_B(\mathbf{p})$$

Plane wave

Bound state from an IPSM

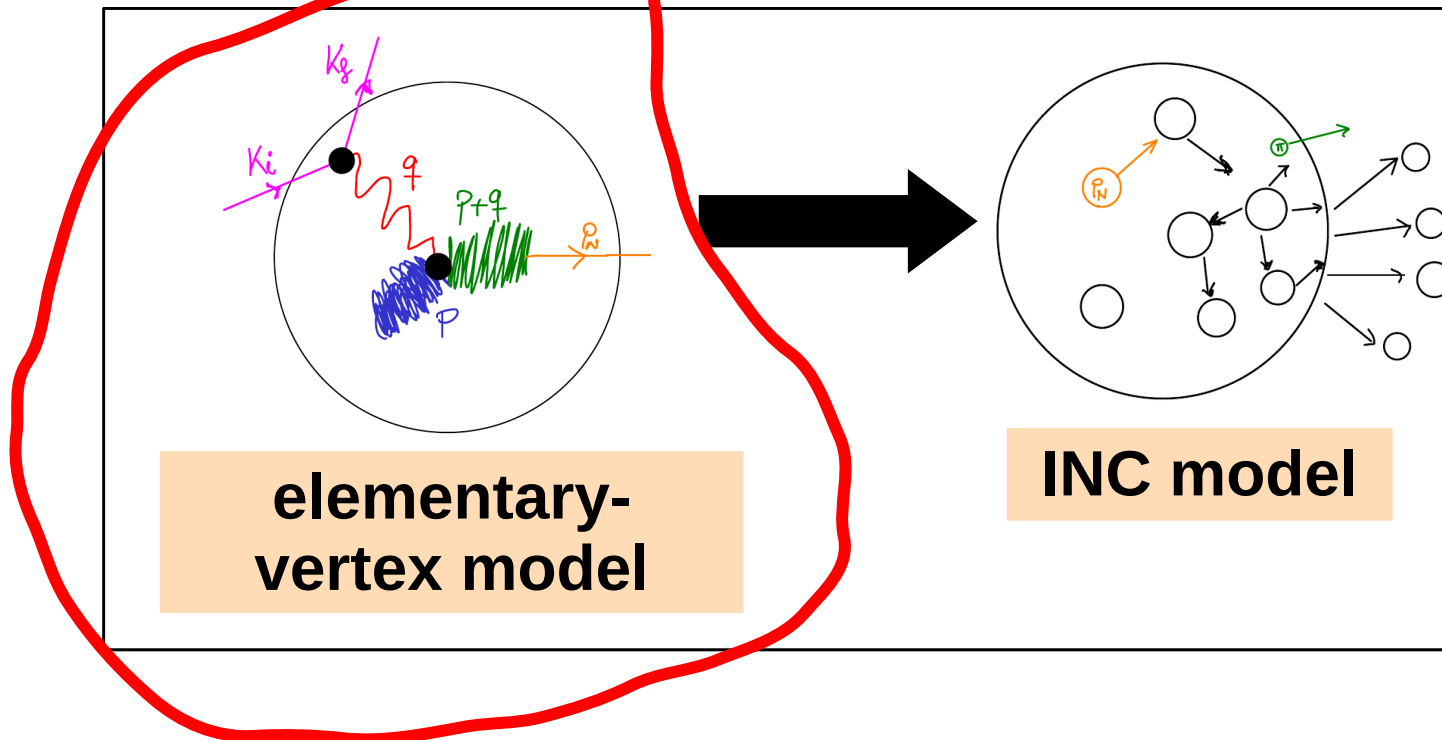
$^{12}\text{C}(e,e')$

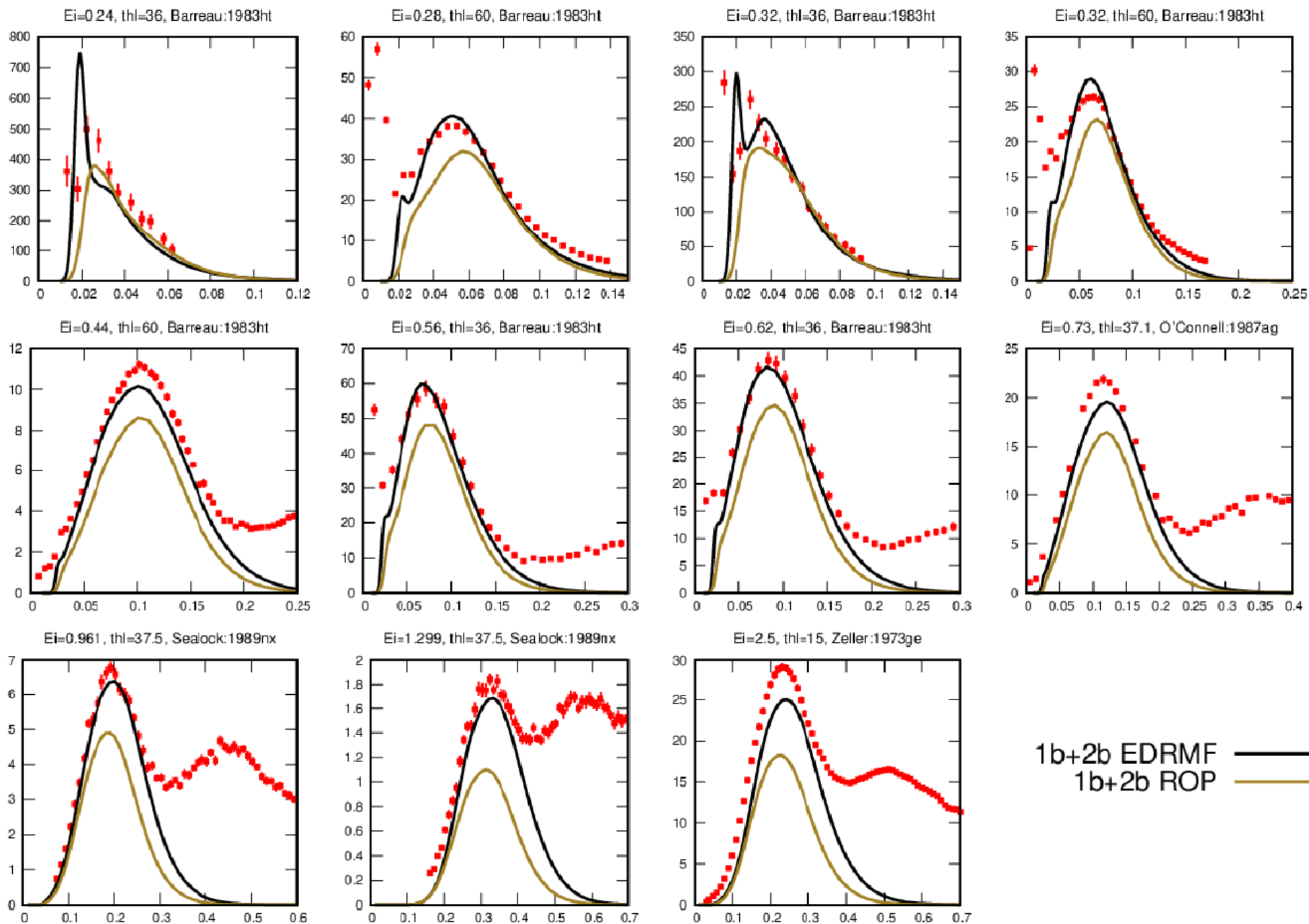


How much of the inclusive strength corresponds to the “true” one-nucleon knockout process?

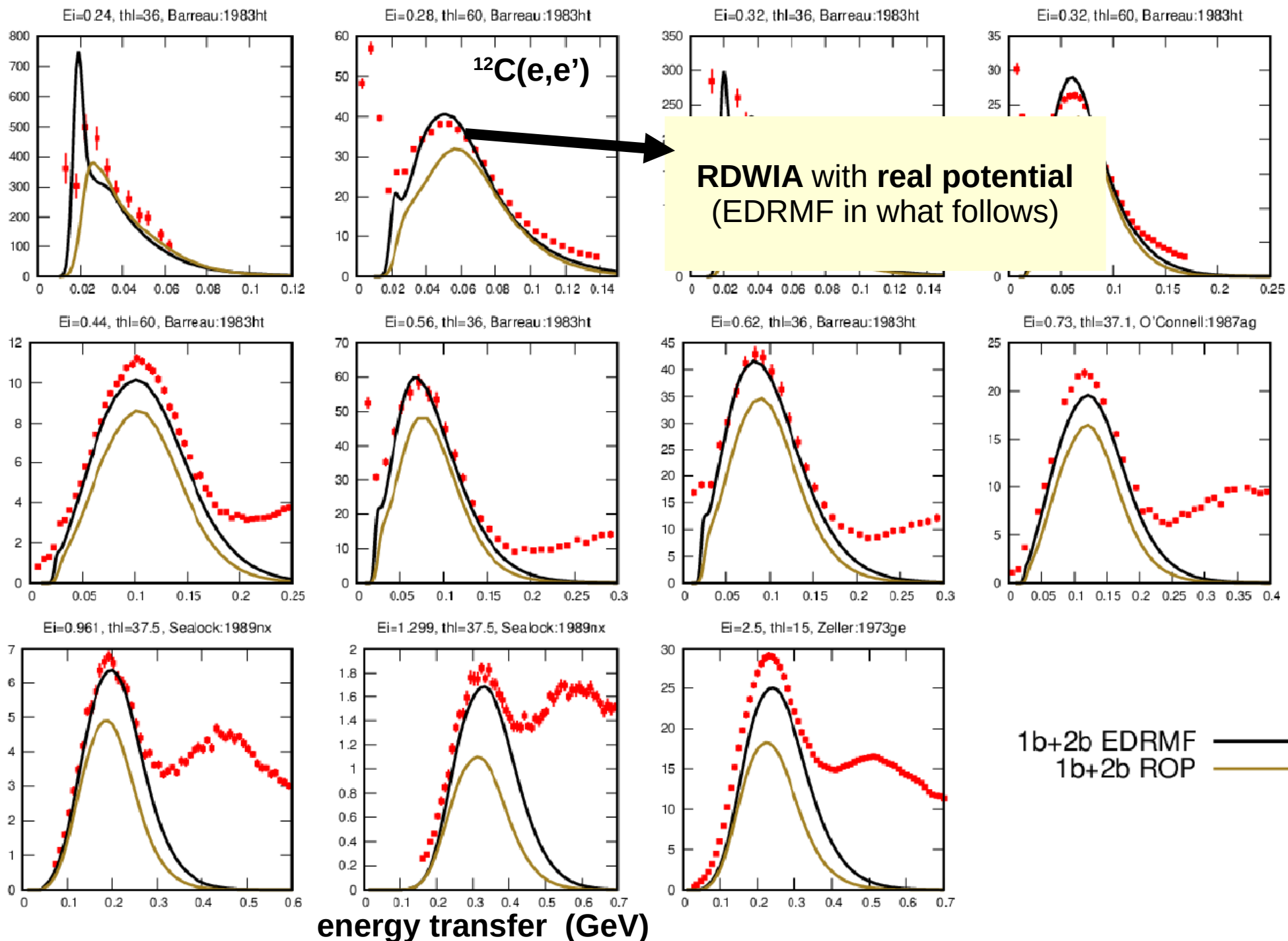
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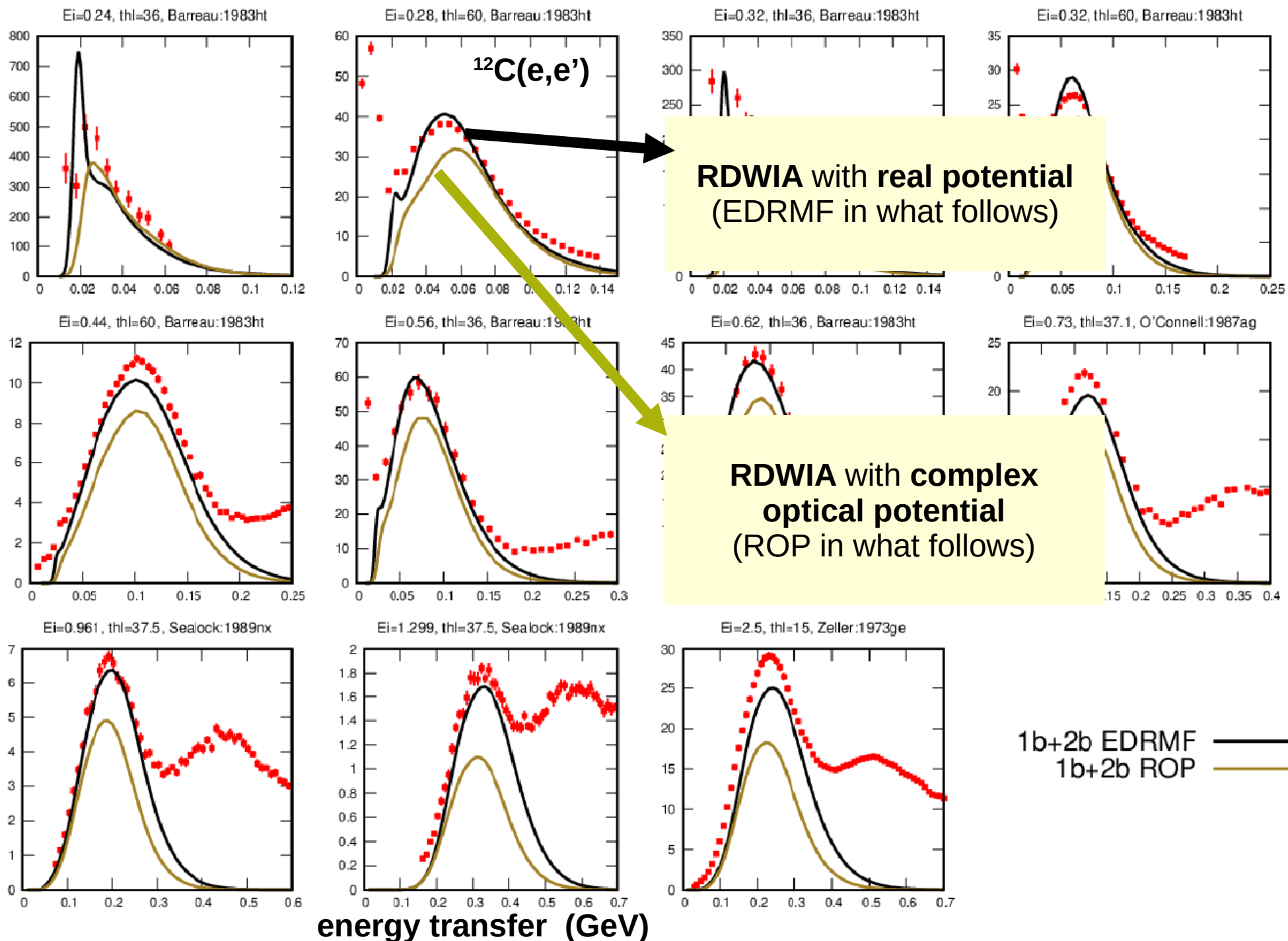
In other words, how often this
is all what happens?





energy transfer (GeV)





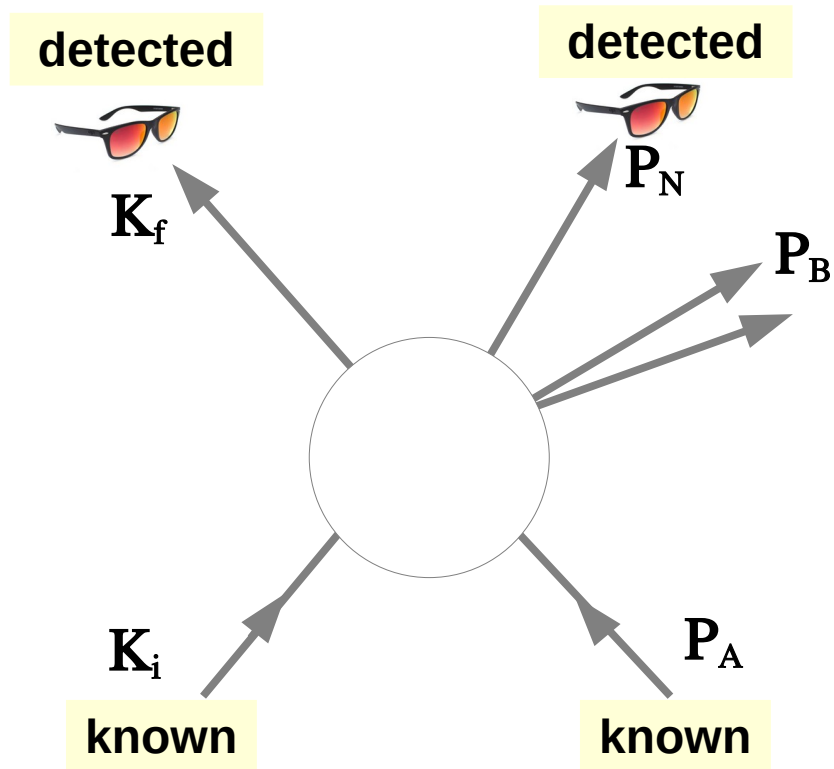
How much of the inclusive strength corresponds to the “true” one-nucleon knockout process?

The RDWIA estimates it using a **complex-optical potential** for the final-nucleon wave function (**ROP** in what follows)

Intranuclear Cascade estimates it as the fraction of events in which the knocked-out nucleon does not interact in its way out
(==> **nuclear Transparency**)

Exclusive (e,e'p) cross section

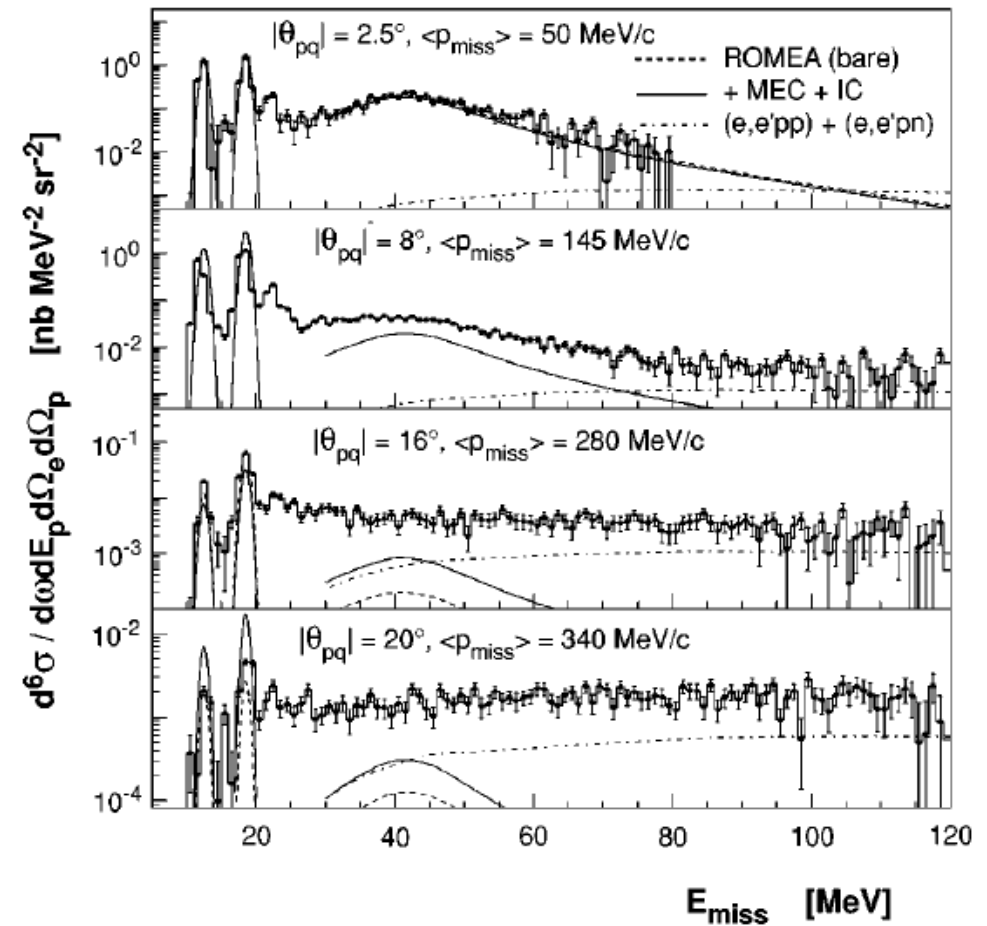
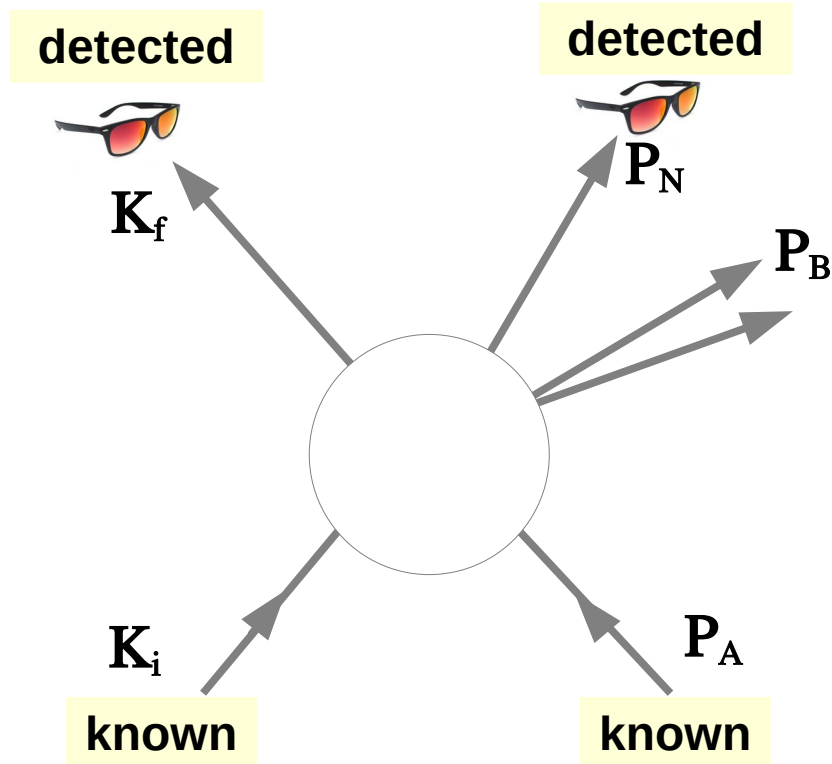
$A(e,e'p)$ experiments



A(e,e'p) experiments

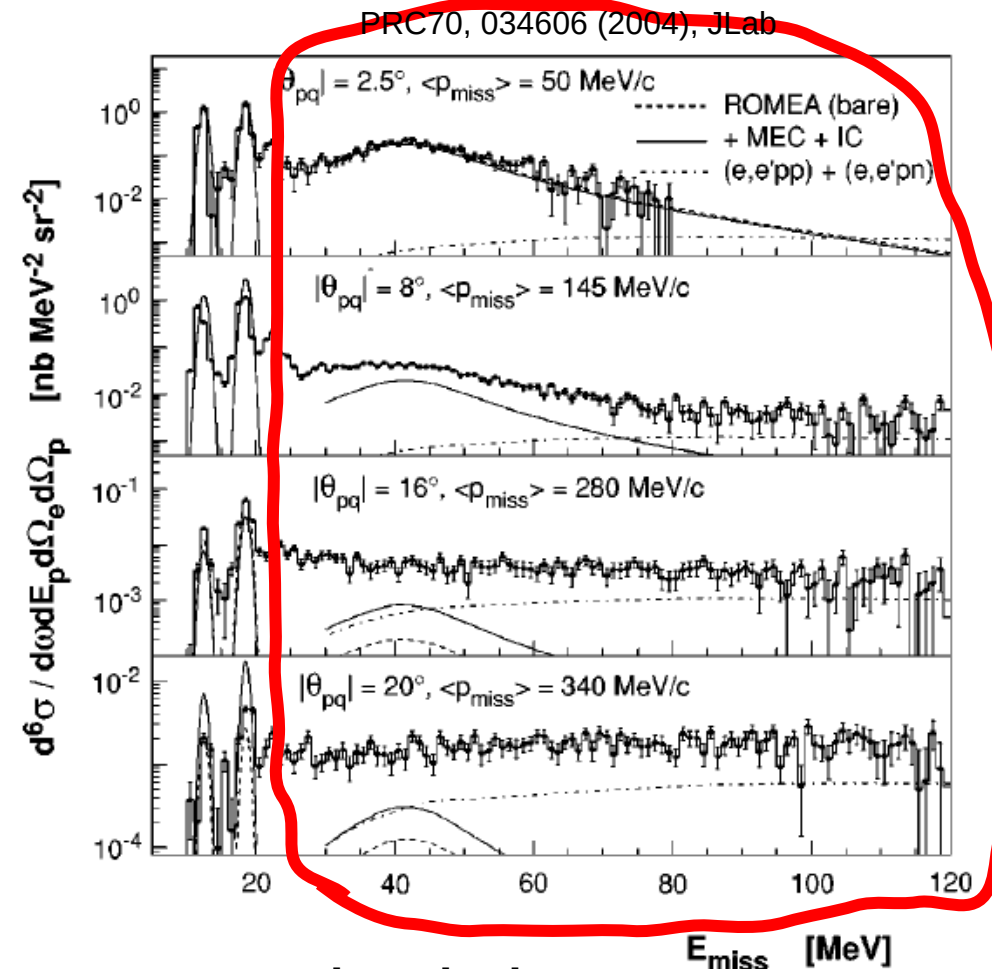
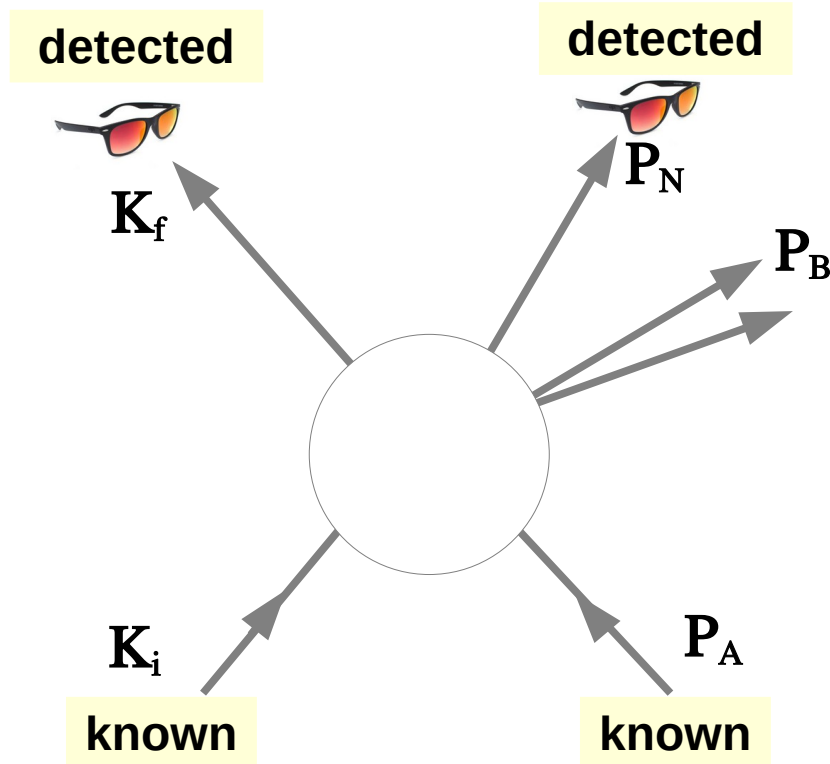
$^{16}\text{O}(e,e'p)$

PRC70, 034606 (2004), JLab



A(e,e'p) experiments

$^{16}\text{O}(e,e'p)$

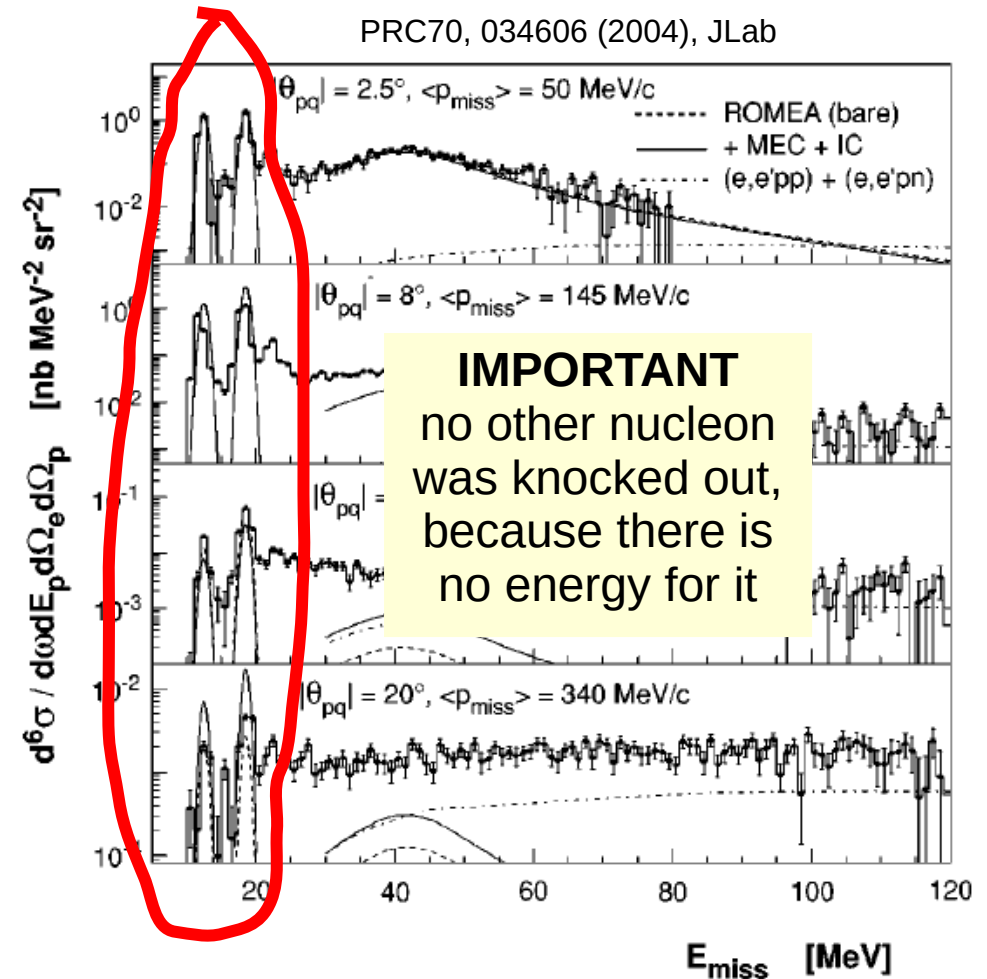
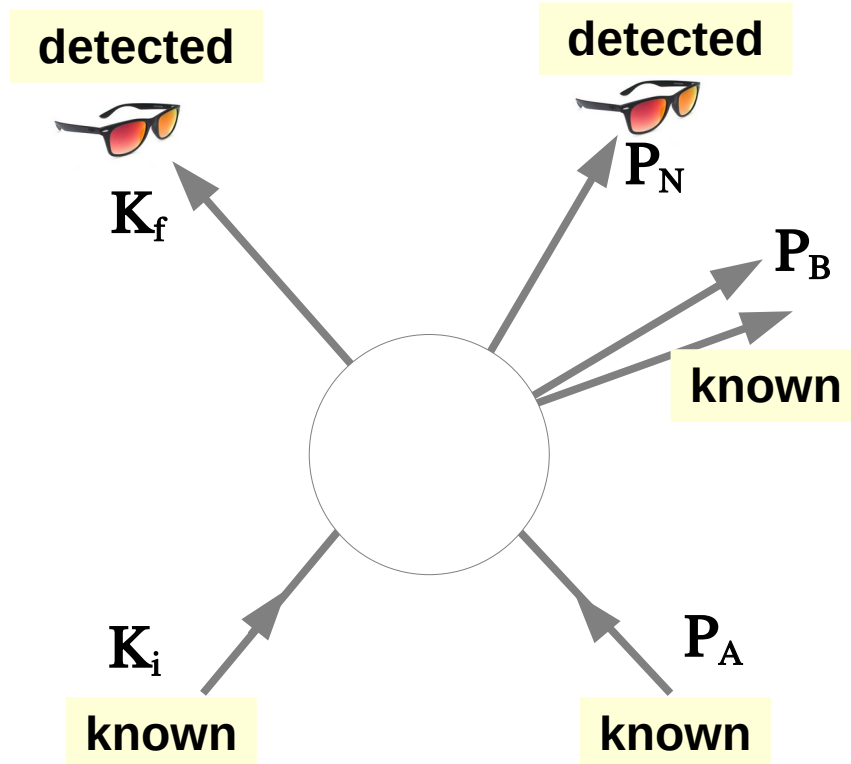


semi-exclusive

We know the energy and momentum of the residual system, but we do **not** know its composition: 1, 2, 3 knockout nucleons...?

A(e,e'p) experiments

$^{16}\text{O}(e,e'p)$



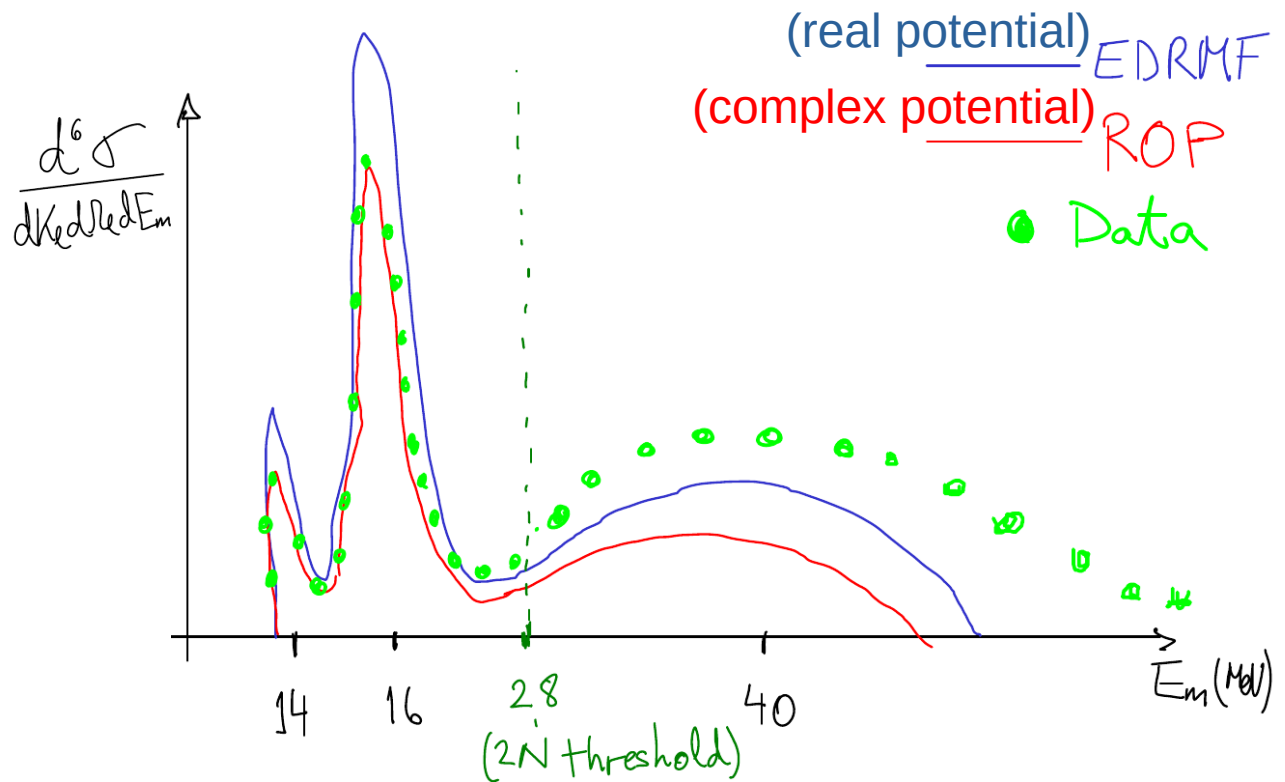
Exclusive

If we stay below the 2N knockout threshold, then we know that the residual system is the $A-1$ nucleus, in the ground or an excited state.

SKETCHING THE IDEA

$^{16}\text{O}(e,e'p)$ data for fixed lepton scattering angle and integrated over the whole solid angle of the nucleon.

What do we expect to see?

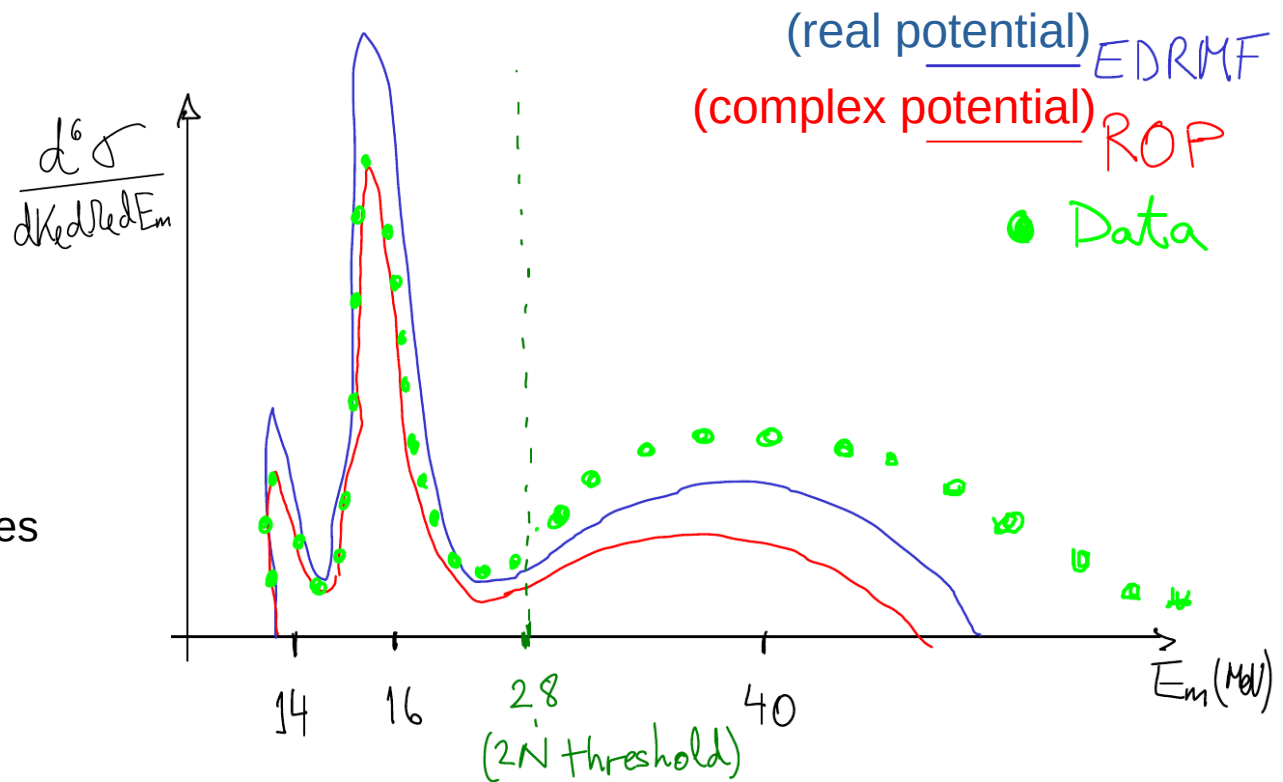


SKETCHING THE IDEA

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What do we expect to see?

1. Around 1p1h threshold: ROP matches the data. EDRMF overestimates them.

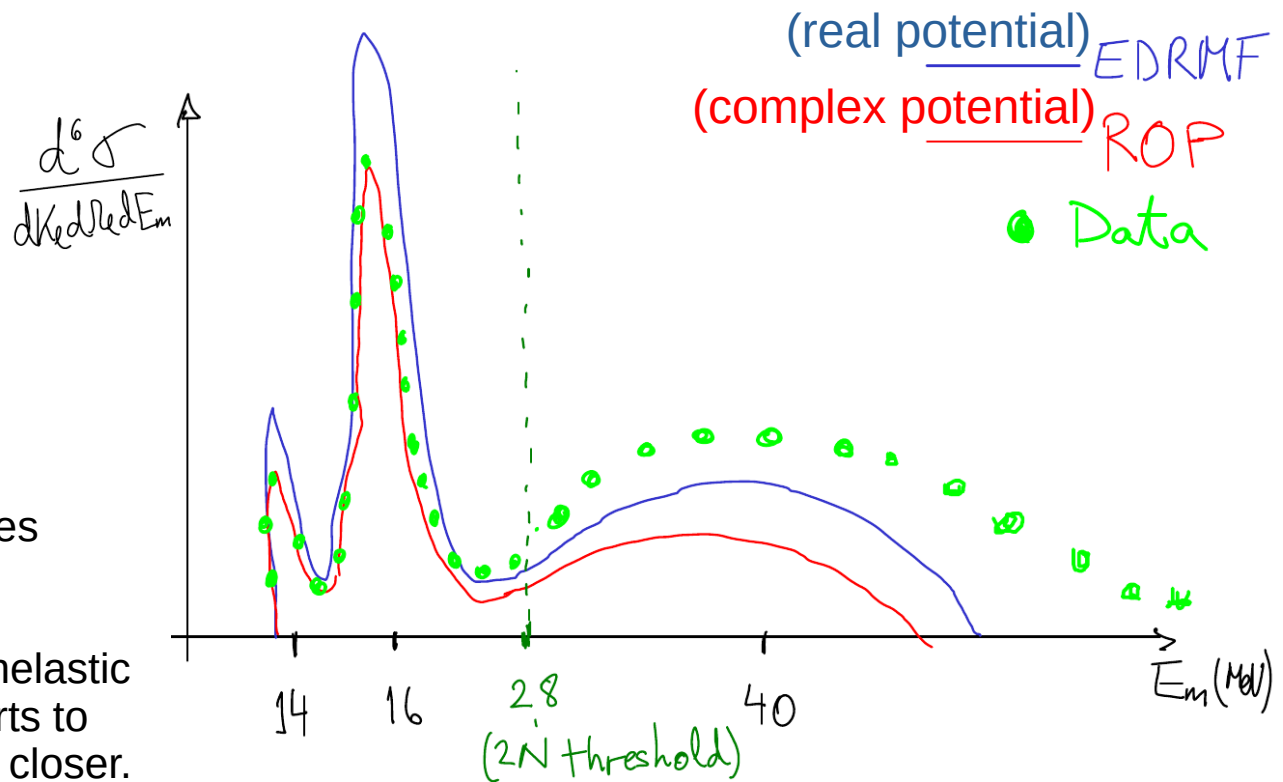


SKETCHING THE IDEA

$^{16}\text{O}(e,e'p)$ data for fixed lepton scattering angle and integrated over the whole solid angle of the nucleon.

What do we expect to see?

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2. A bit beyond 1p1h threshold some inelastic channels have been opened: ROP starts to underestimate data. EDRMF is getting closer.

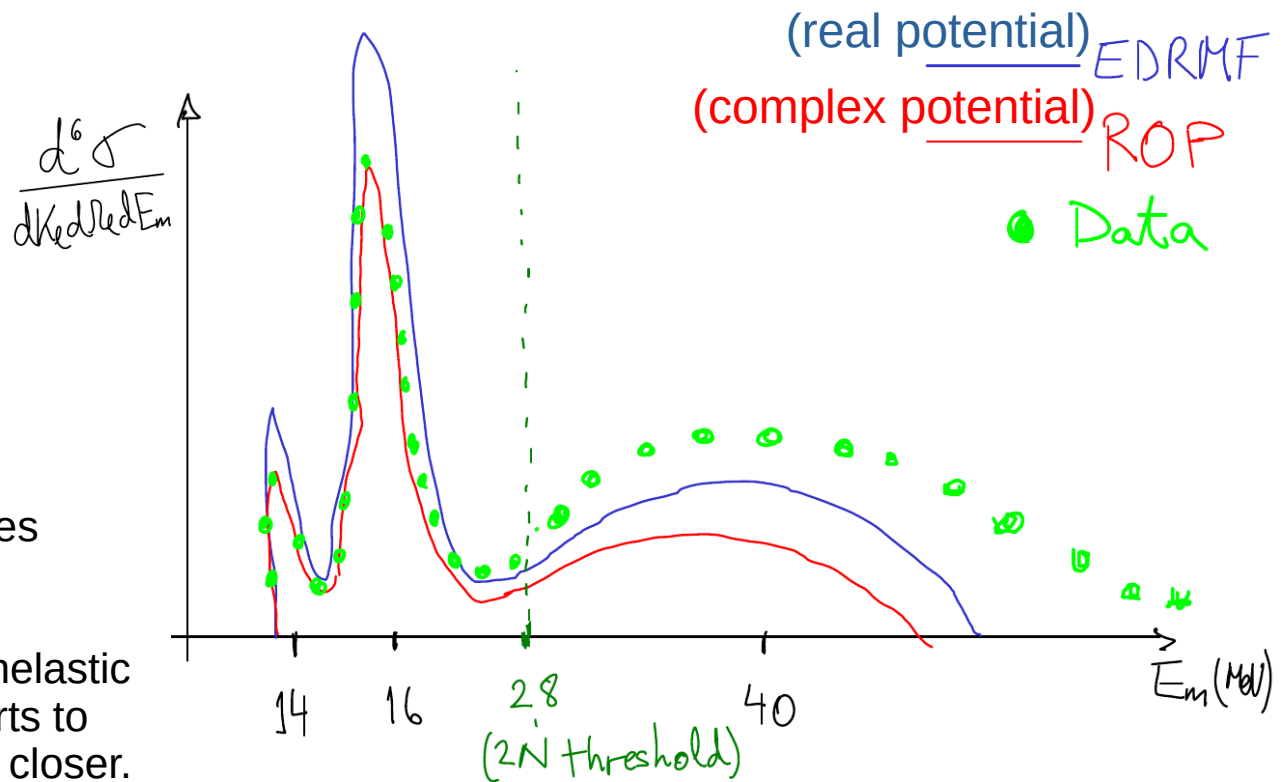


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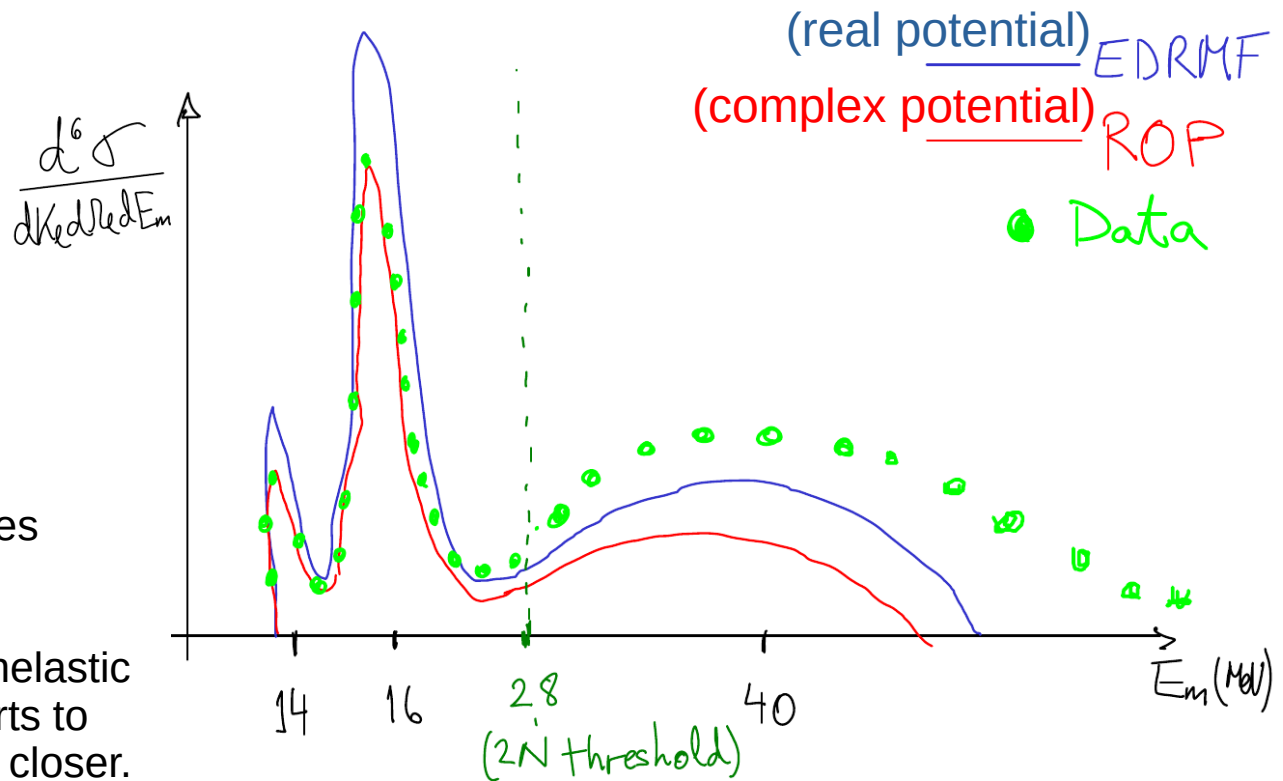


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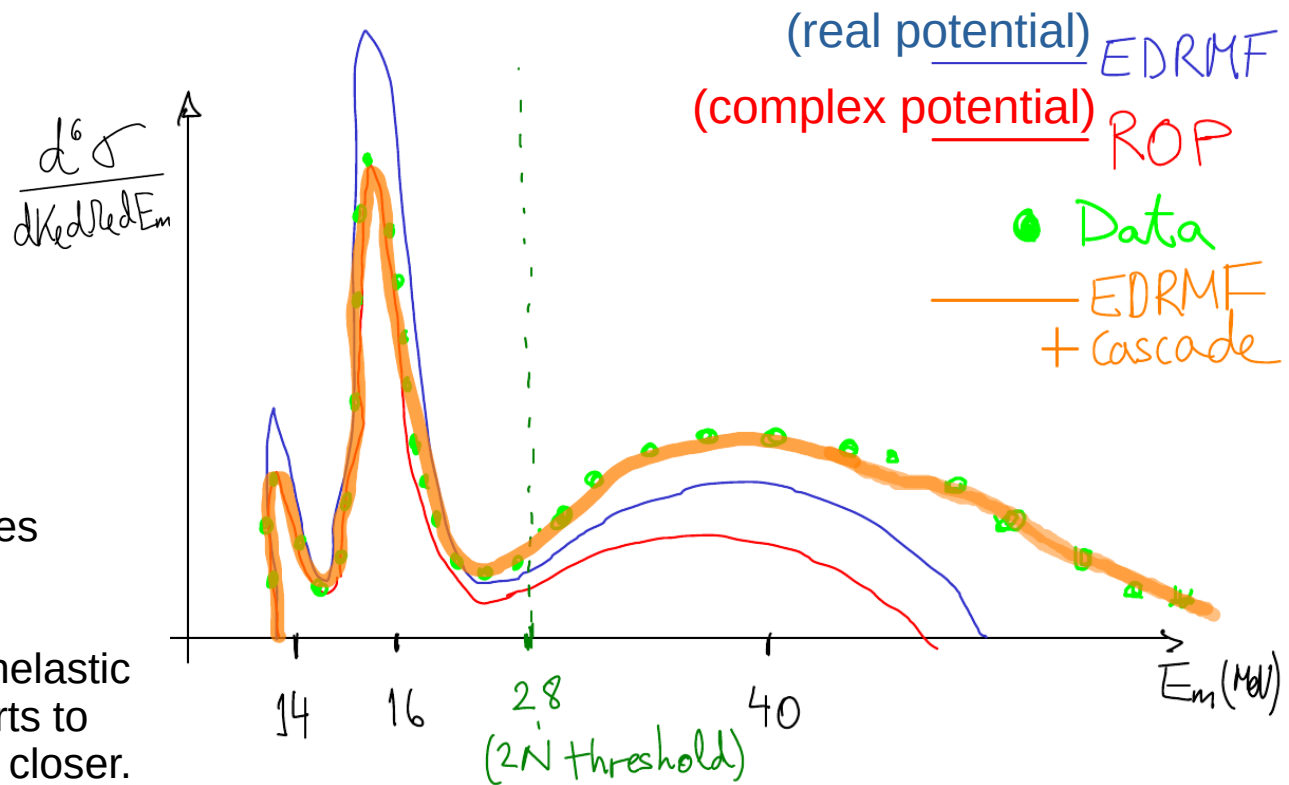


SKETCHING THE IDEA

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5. EDRMF+cascade (hopefully) matches the data nicely.

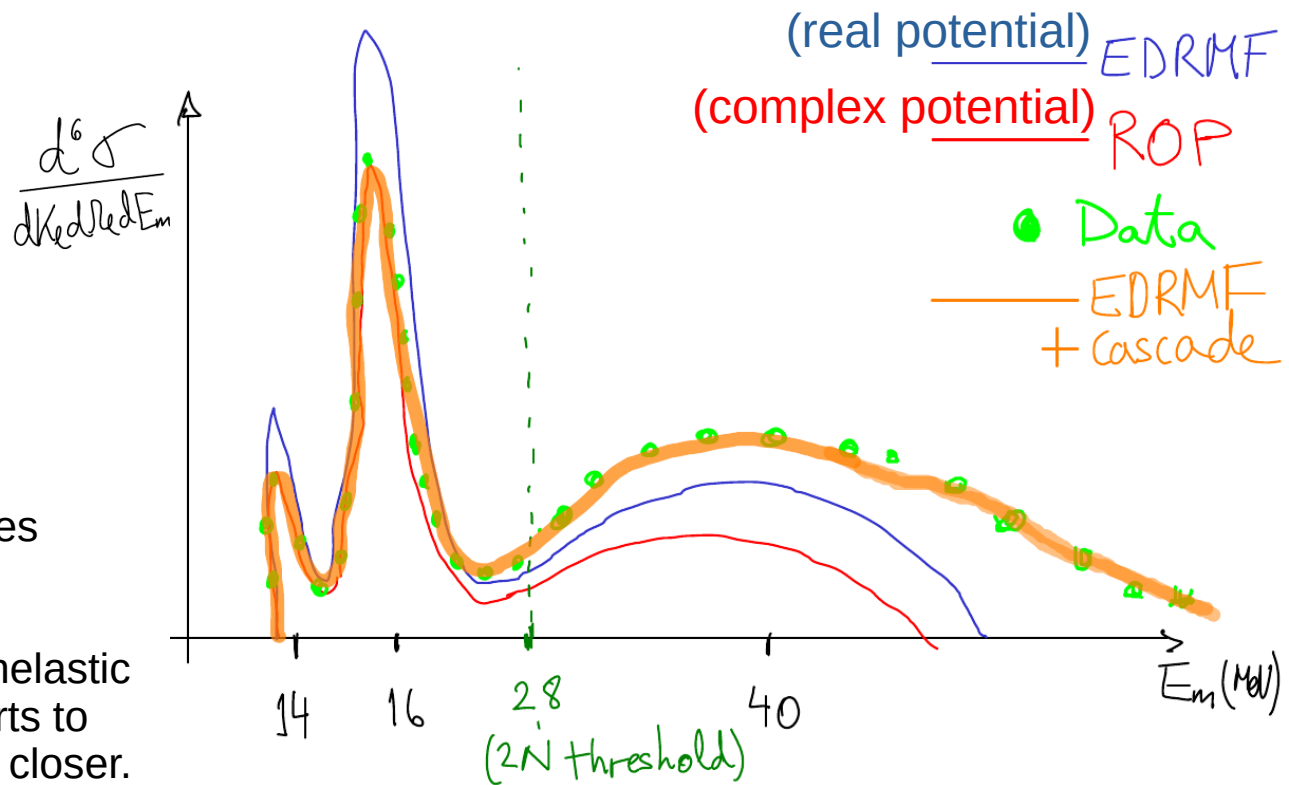


SKETCHING THE IDEA

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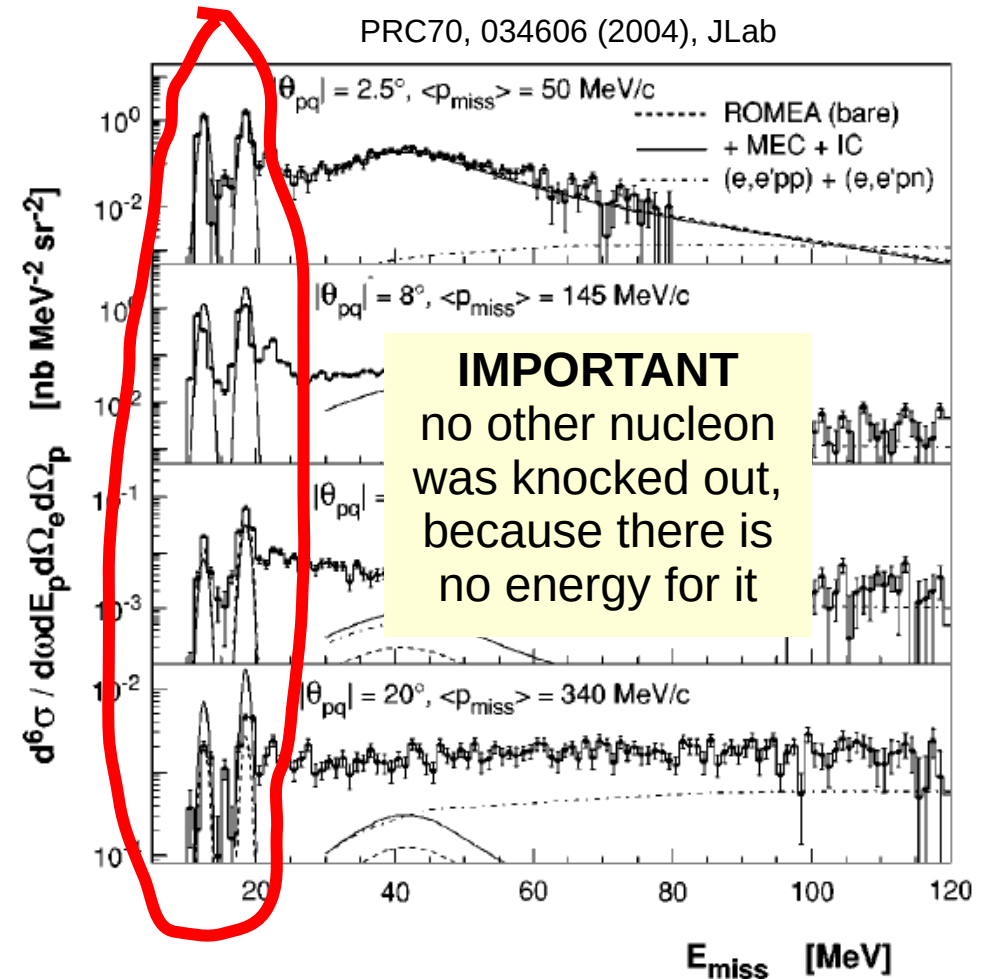
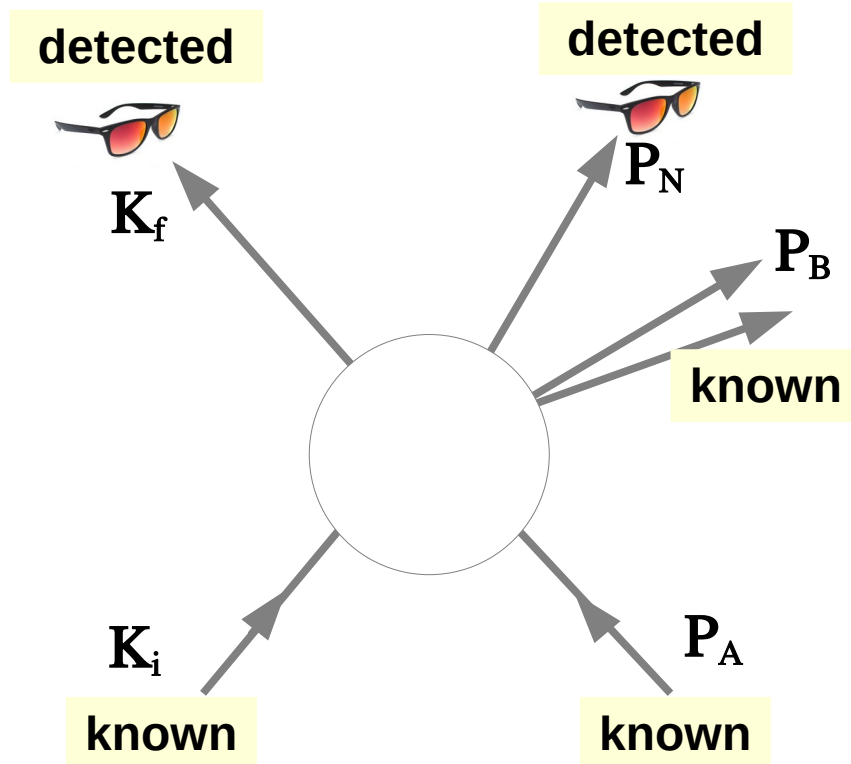
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5. EDRMF+cascade (hopefully) matches the data nicely.



We should work for this to happen

A(e,e'p) experiments

$^{16}\text{O}(e,e'p)$



Exclusive

If we stay below the 2N knockout threshold, then we know that the residual system is the $A-1$ nucleus, in the ground or an excited state.

Exclusive experiments: $^{16}\text{O}(e,e'p)^{15}\text{N}$

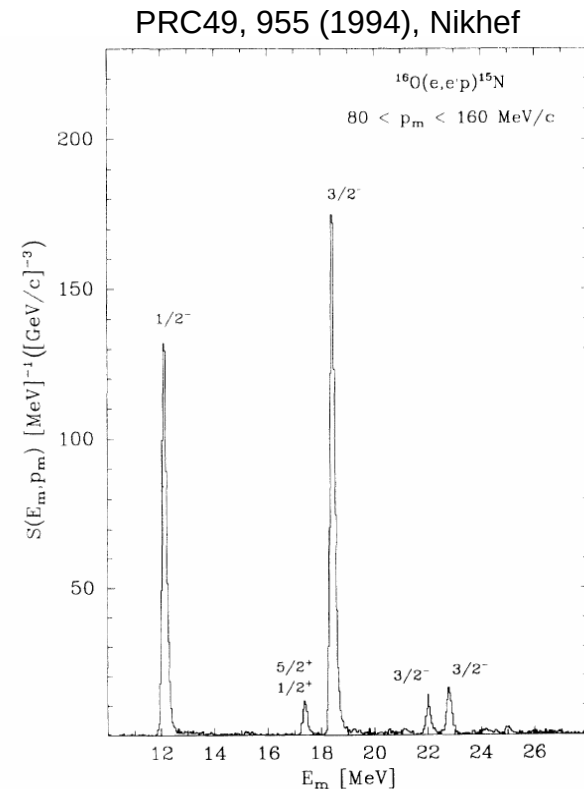
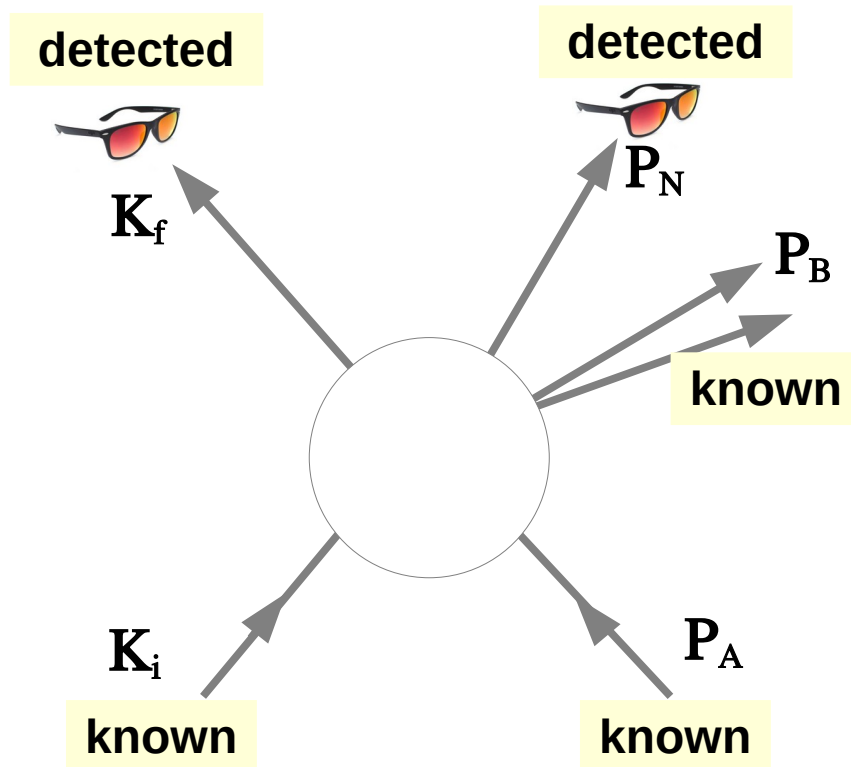


FIG. 1. $^{16}\text{O}(e,e'p)^{15}\text{N}$ missing energy spectrum for the kinematics centered about $p_m = 120 \text{ MeV}/c$.

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If we stay below the 2N knockout threshold, then we know that the residual system is the A-1 nucleus, in the ground or an excited state.

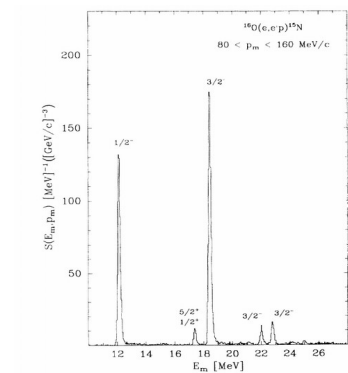


FIG. 1. $^{16}\text{O}(e,e'p)^{15}\text{N}$ missing energy spectrum for the kinematics centered about $p_m = 120 \text{ MeV}/c$.

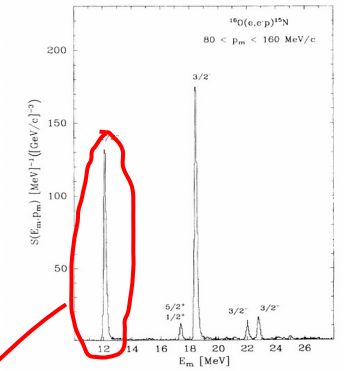
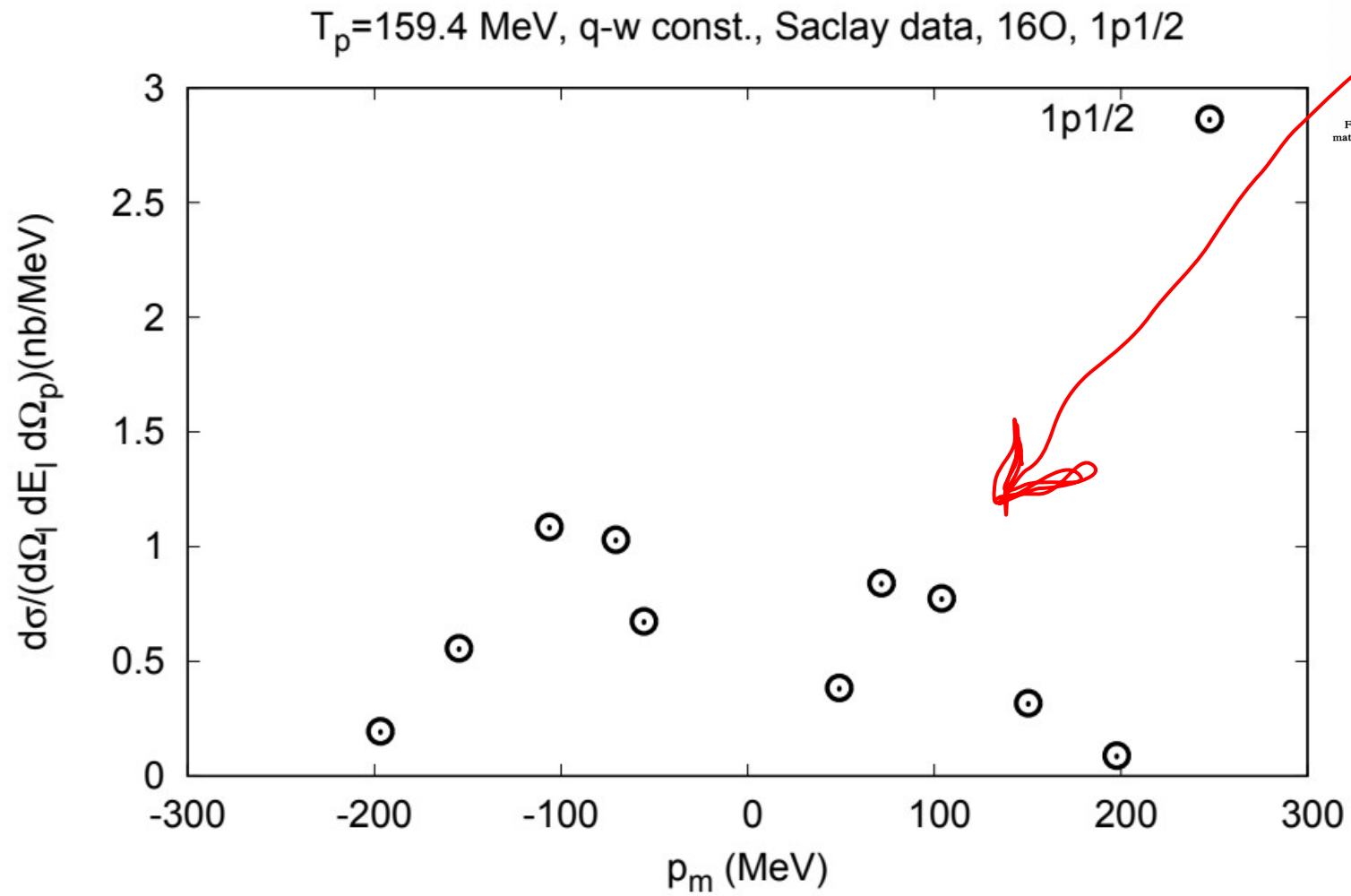
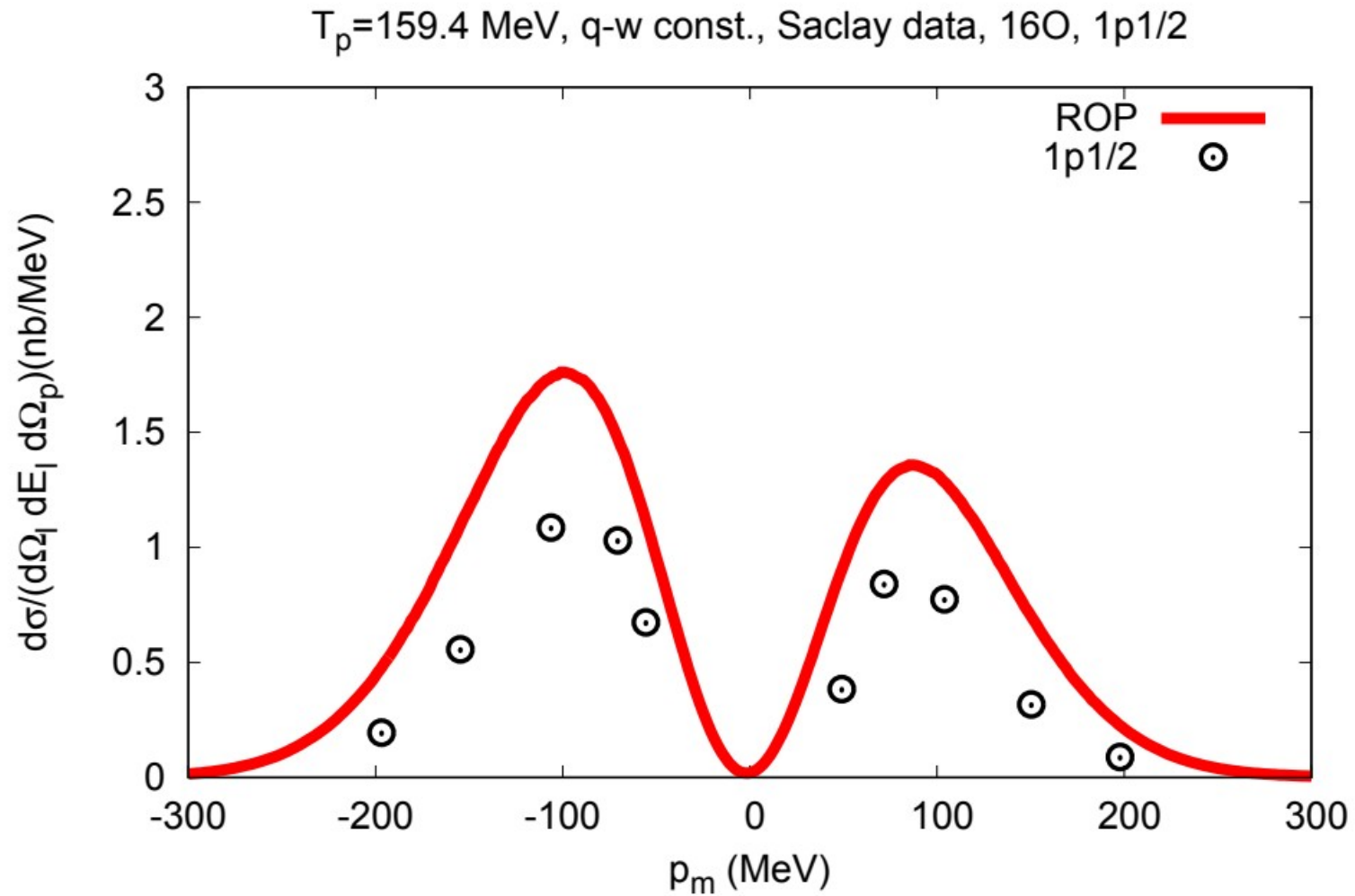


FIG. 1. $^{16}\text{O}(e,e'p)^{15}\text{N}$ missing energy spectrum for the kinematics centered about $p_m = 120$ MeV/c.

data: [PhysRevLett.67.1811.2](https://ui.adsabs.org/abs/1971PhRvLett.67.1811B)

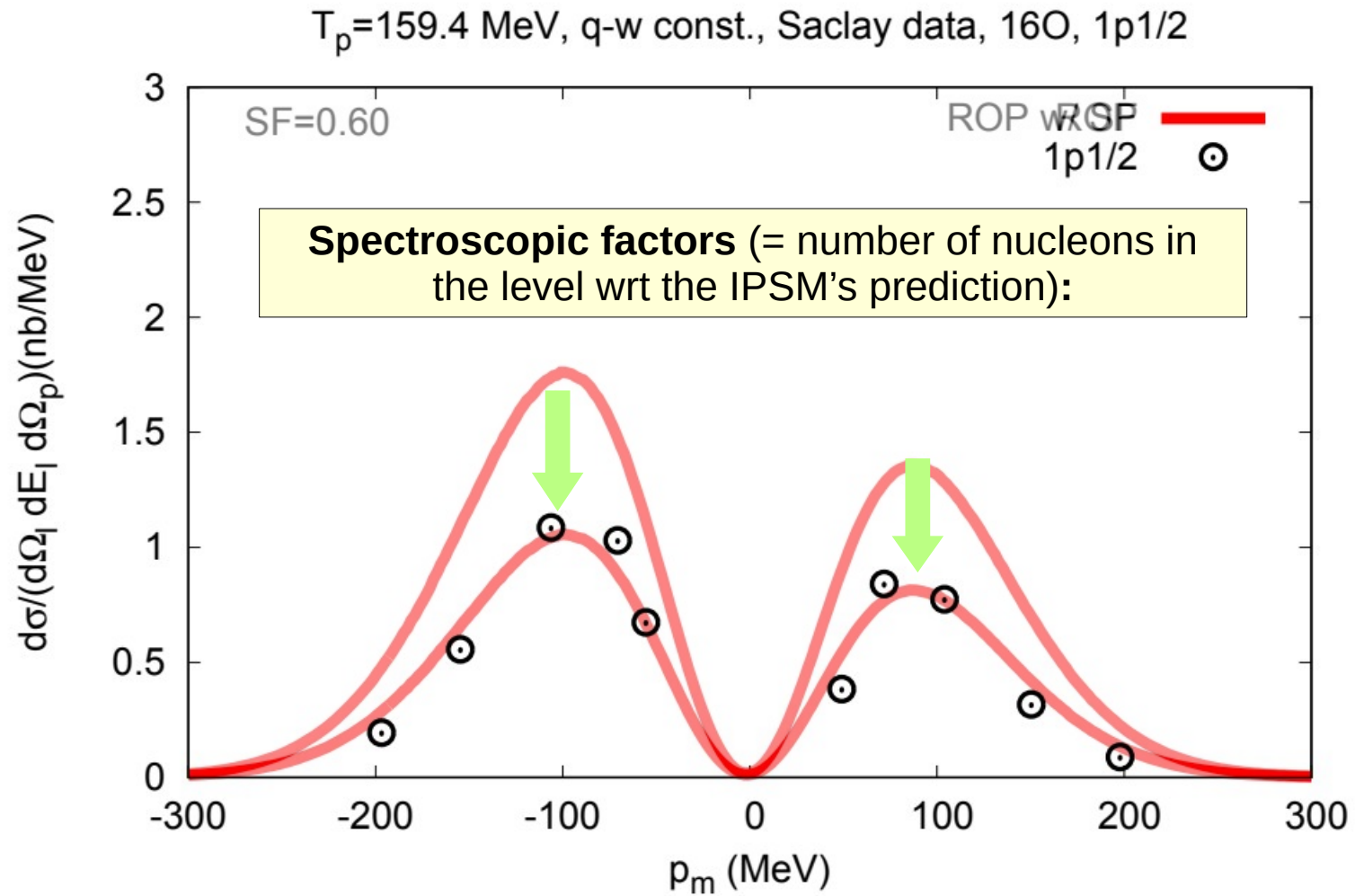
Key:

ROP = IPSM, final nucleon from Dirac equation with relativistic **complex** optical potential



Key:

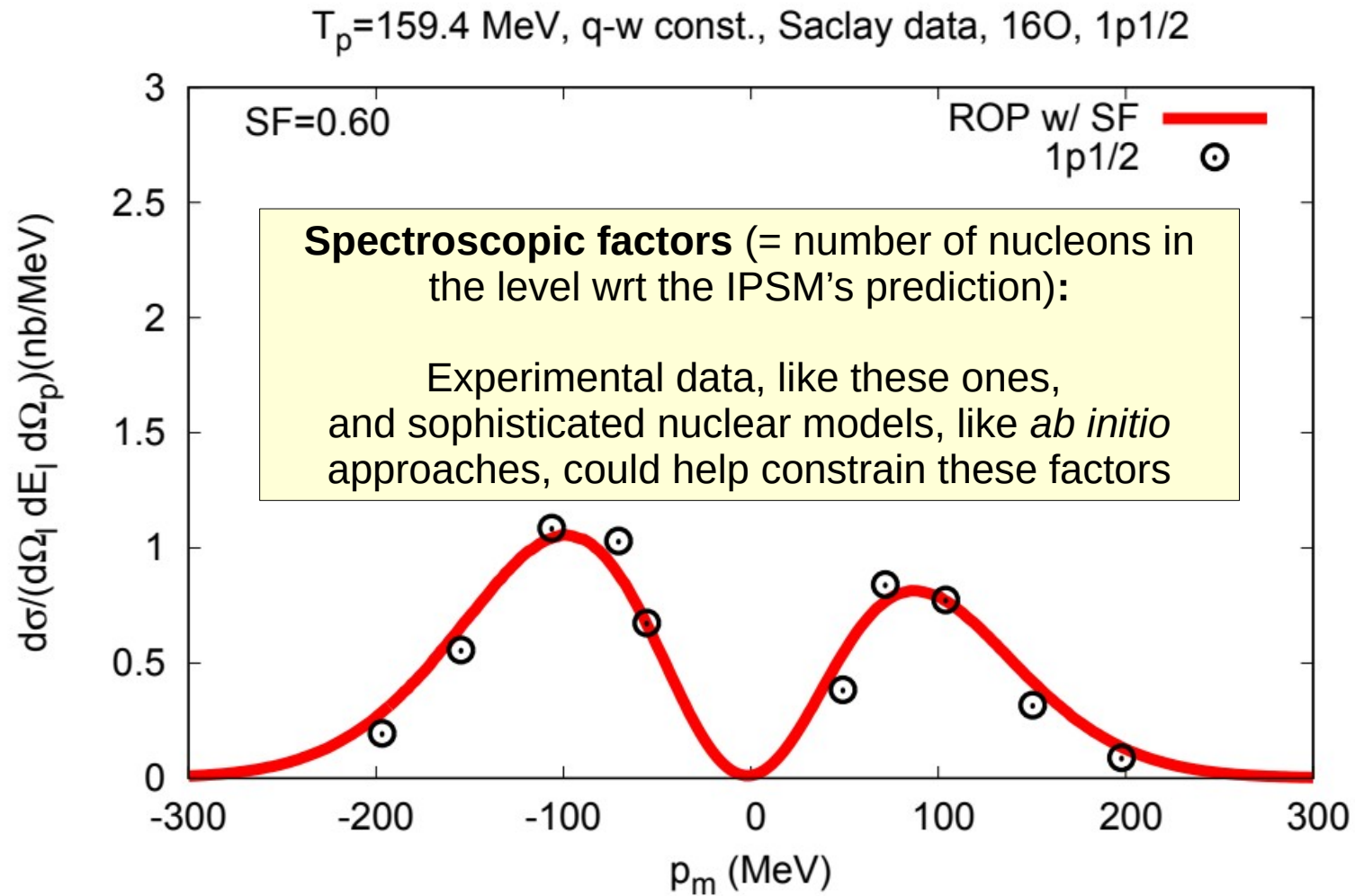
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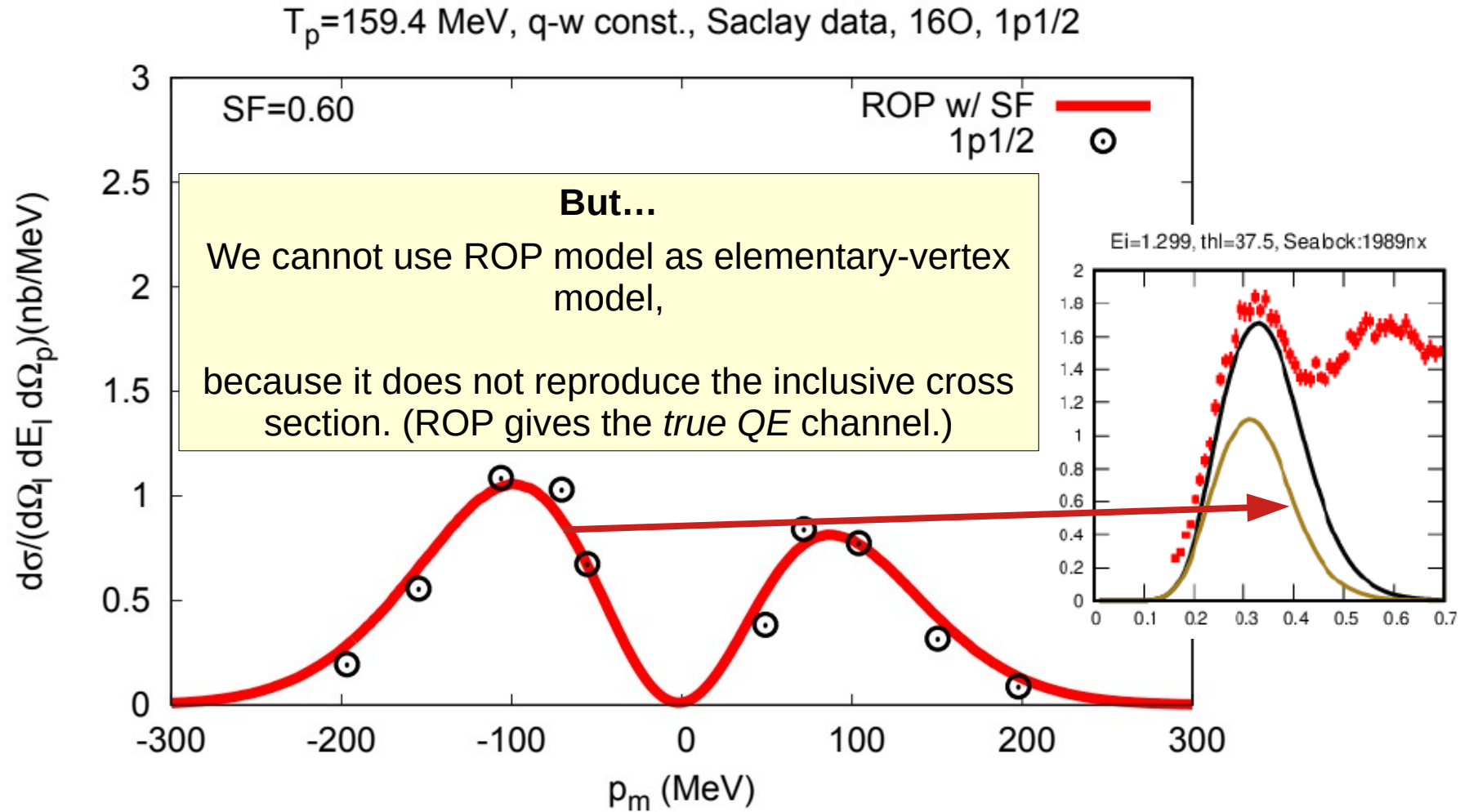
SF = spectroscopic factor



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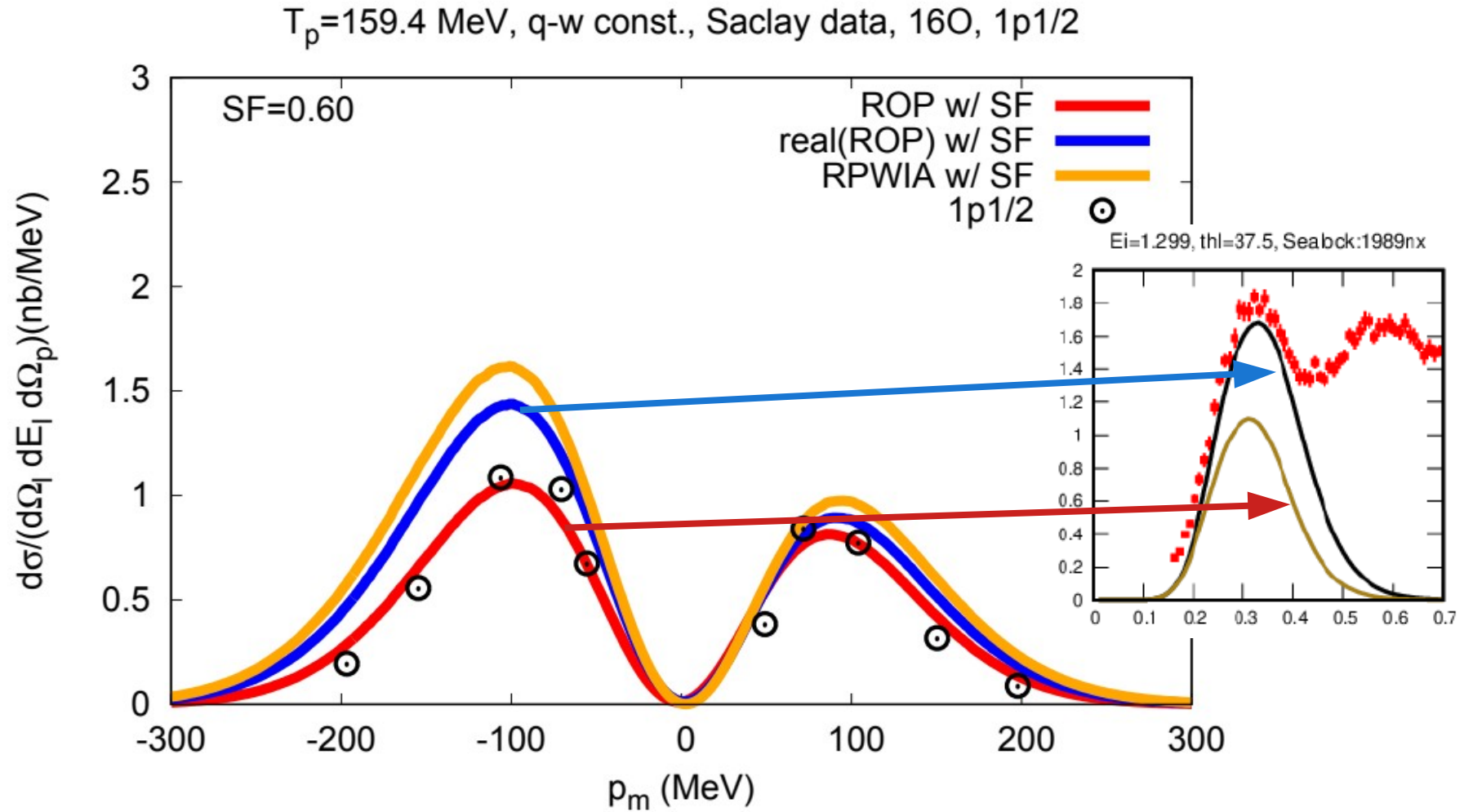
Key:

ROP = IPSM, final nucleon from Dirac equation with relativistic **complex** optical potential

SF = spectroscopic factor

real(ROP) = IPSM, final nucleon from Dirac equation with relativistic **real** optical potential

RPWIA = IPSM, final nucleon is a plane wave



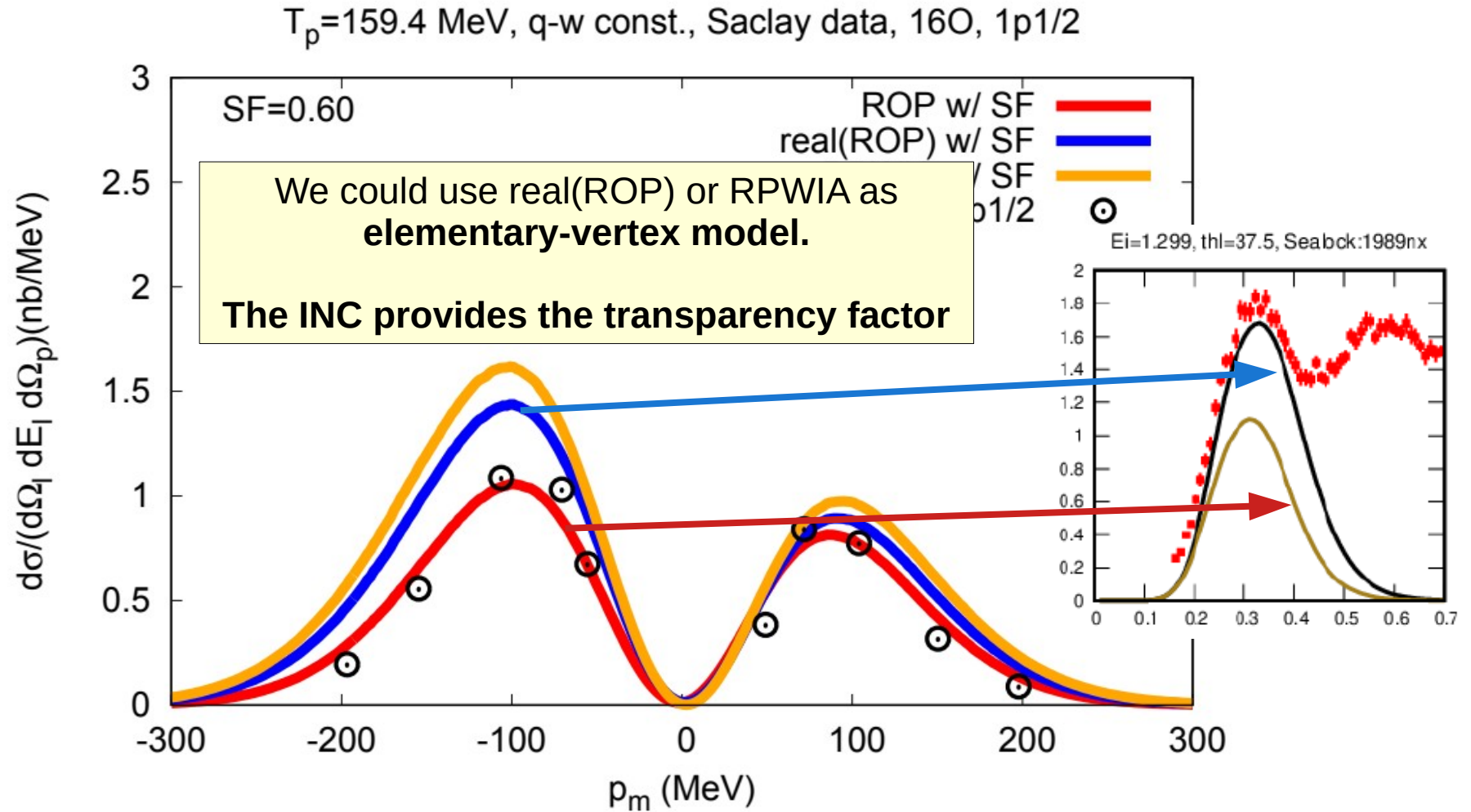
Key:

ROP = IPISM, final nucleon from Dirac equation with relativistic **complex** optical potential

SF = spectroscopic factor

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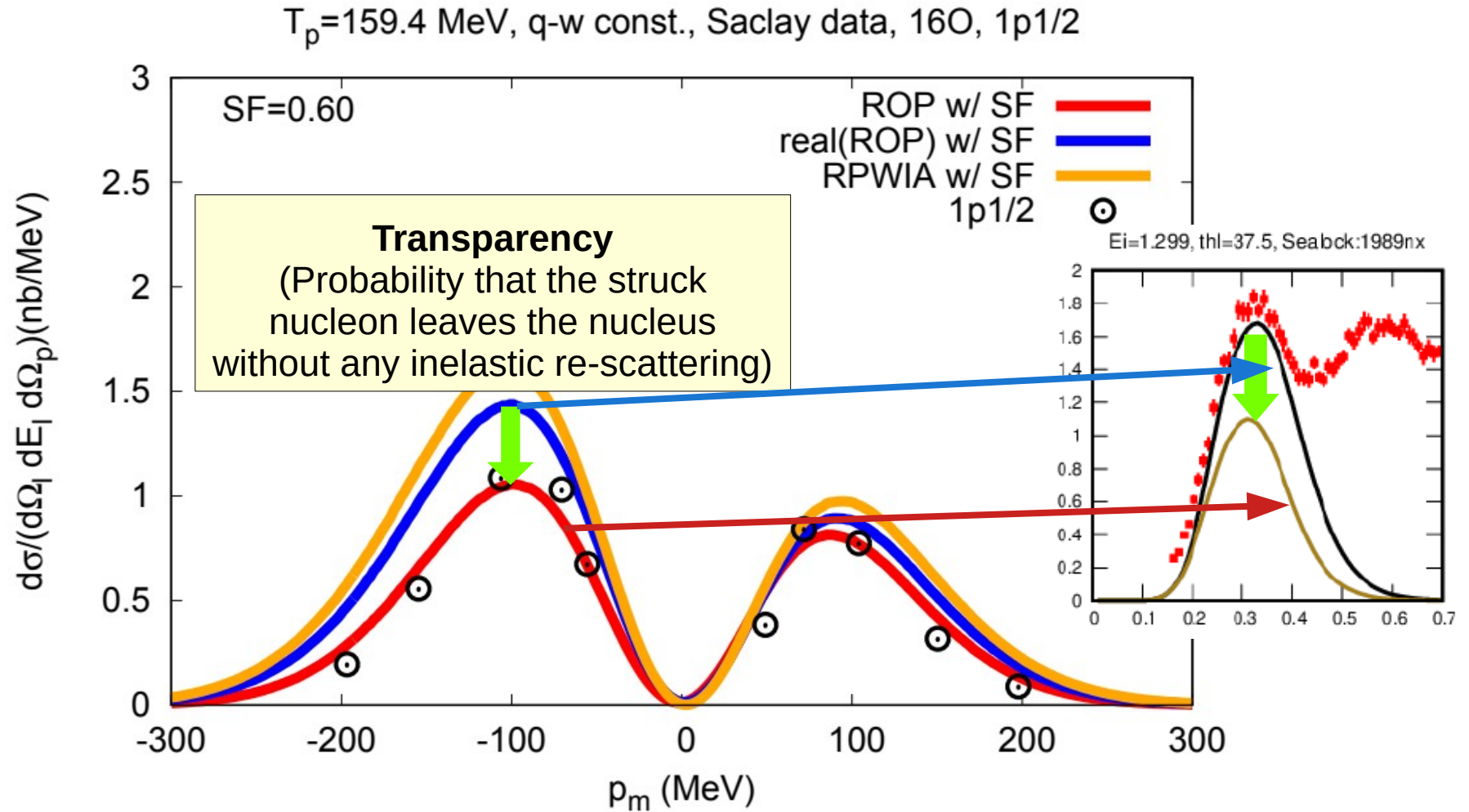
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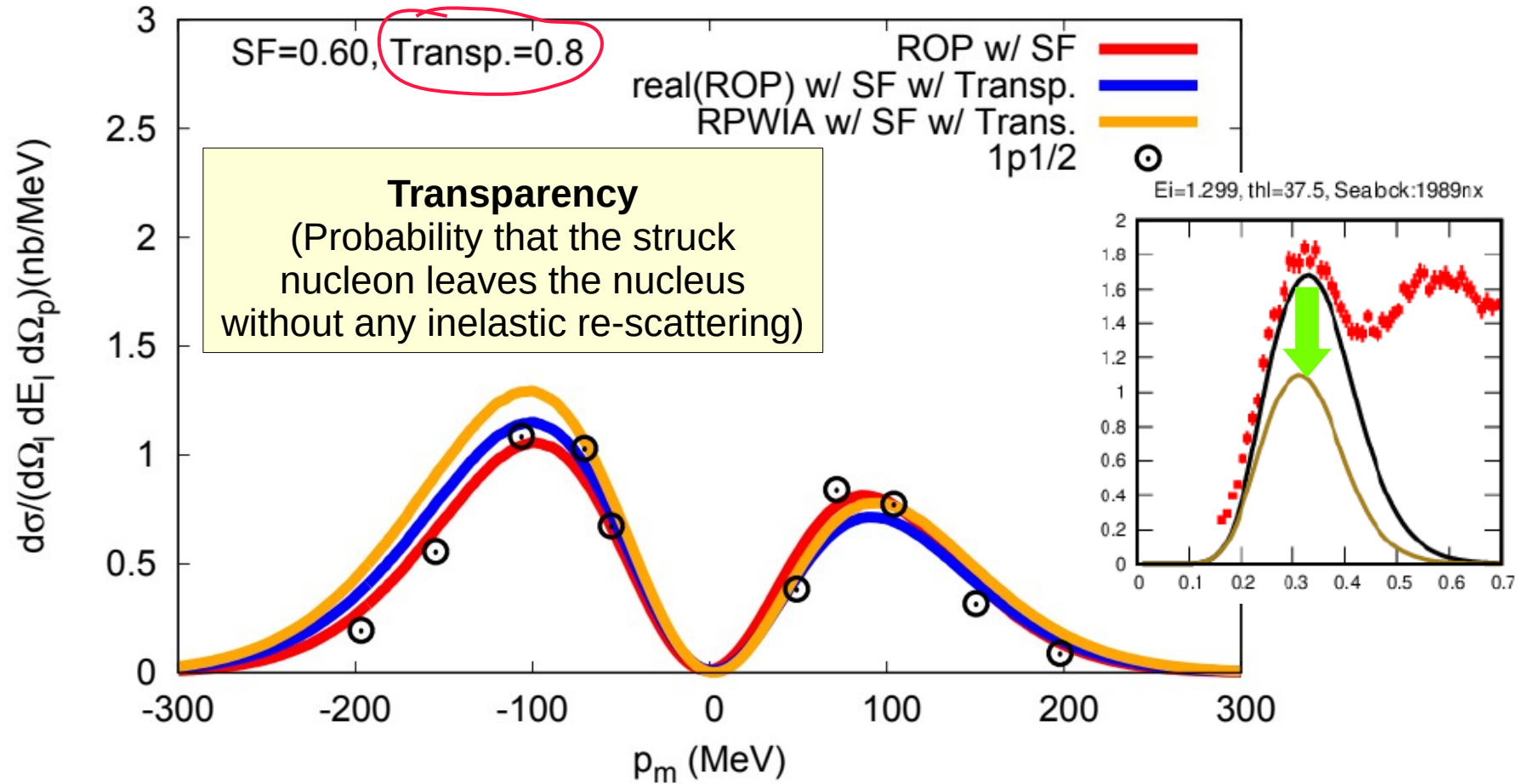
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Transp. = Transparency factor

$T_p = 159.4$ MeV, q -w const., Saclay data, ^{16}O , $1p_{1/2}$



Key:

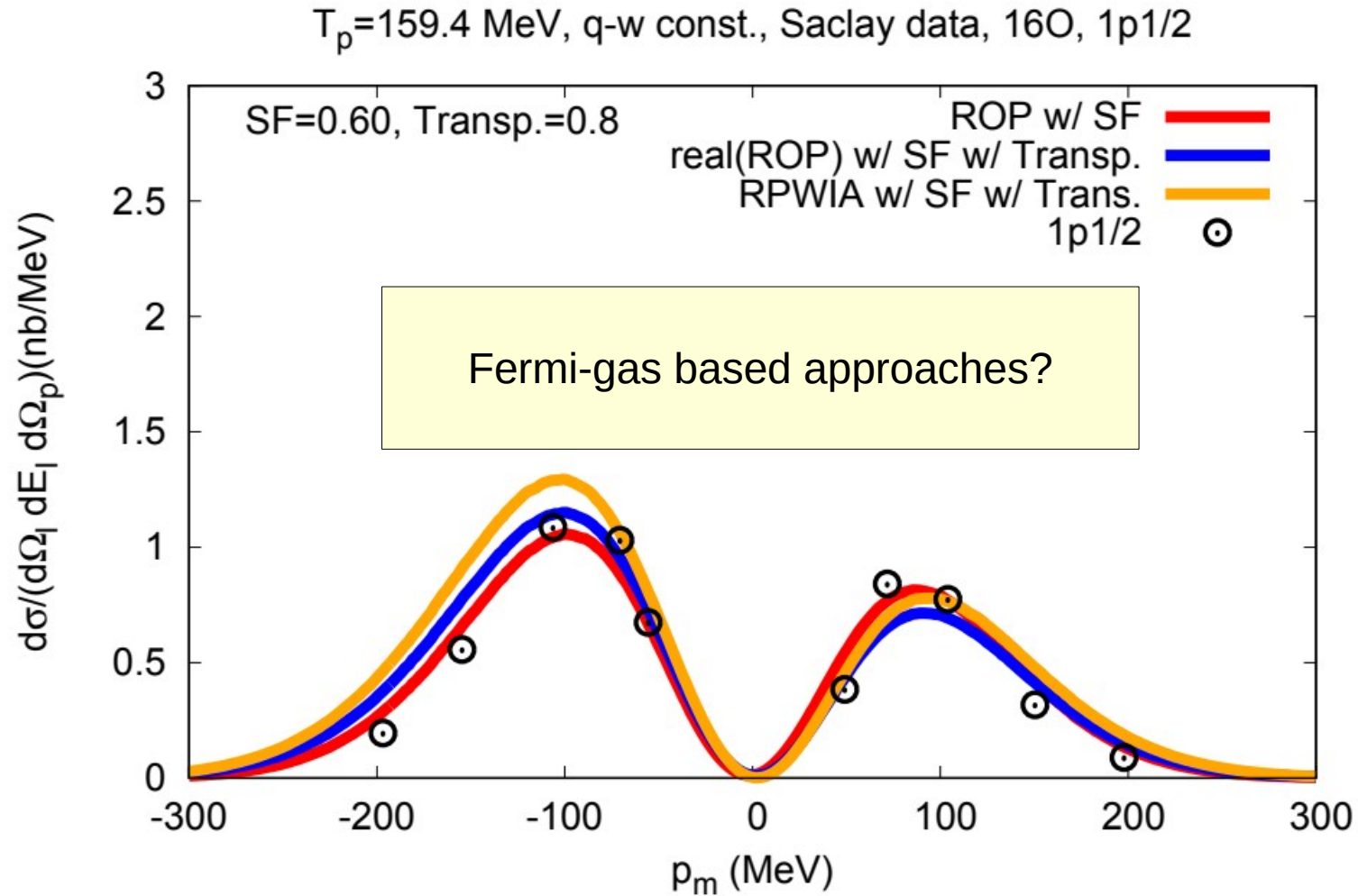
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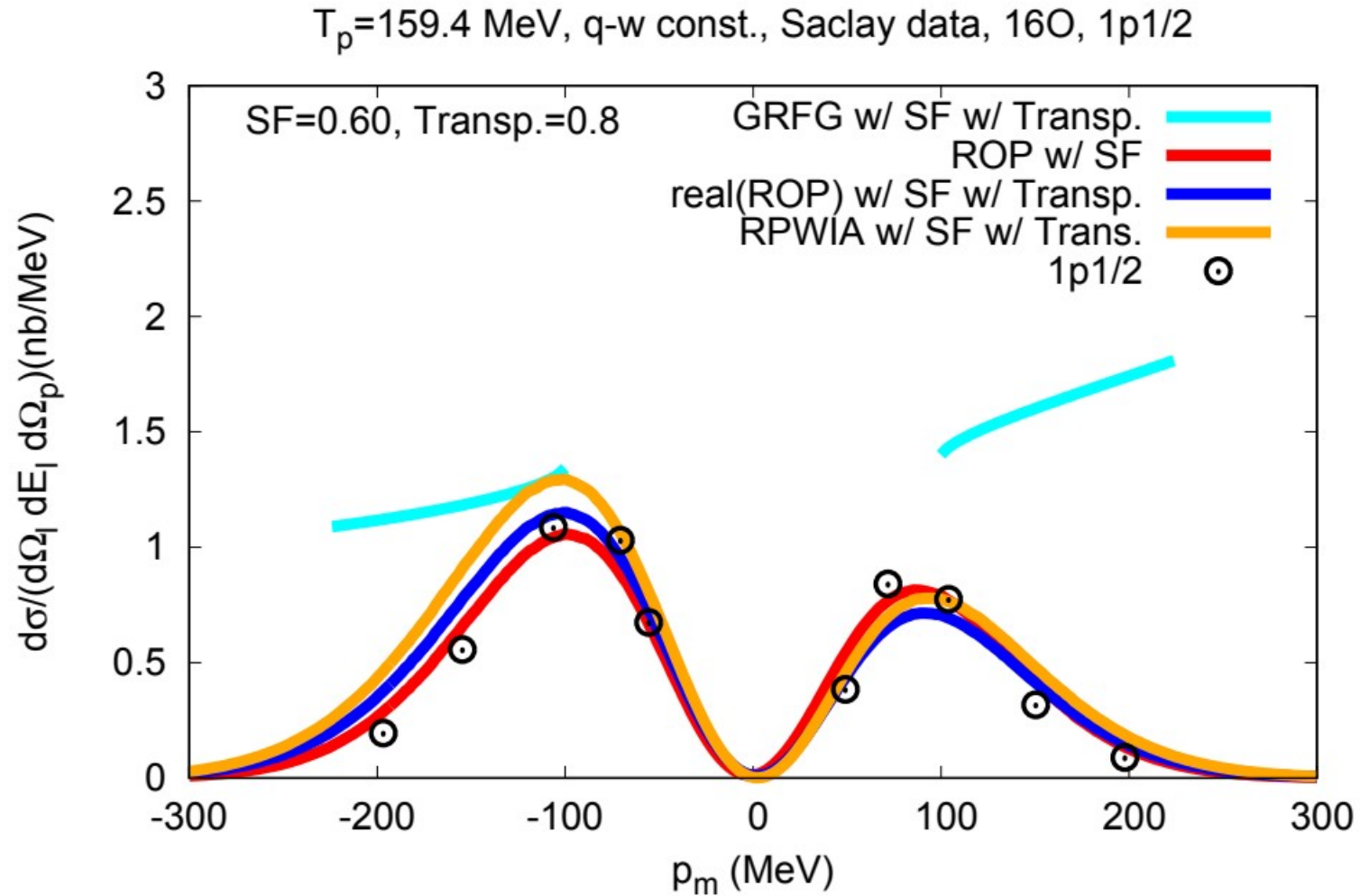
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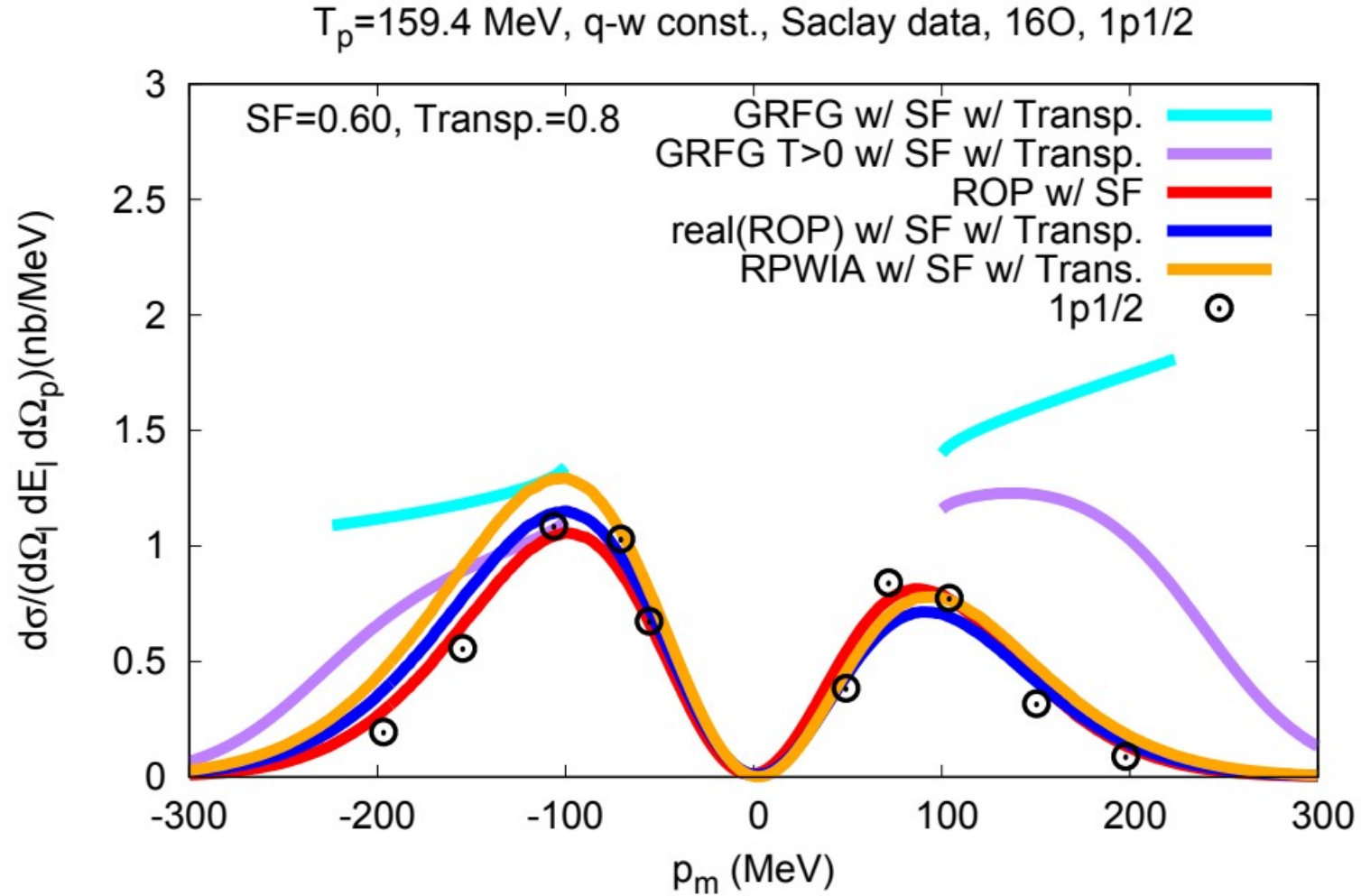
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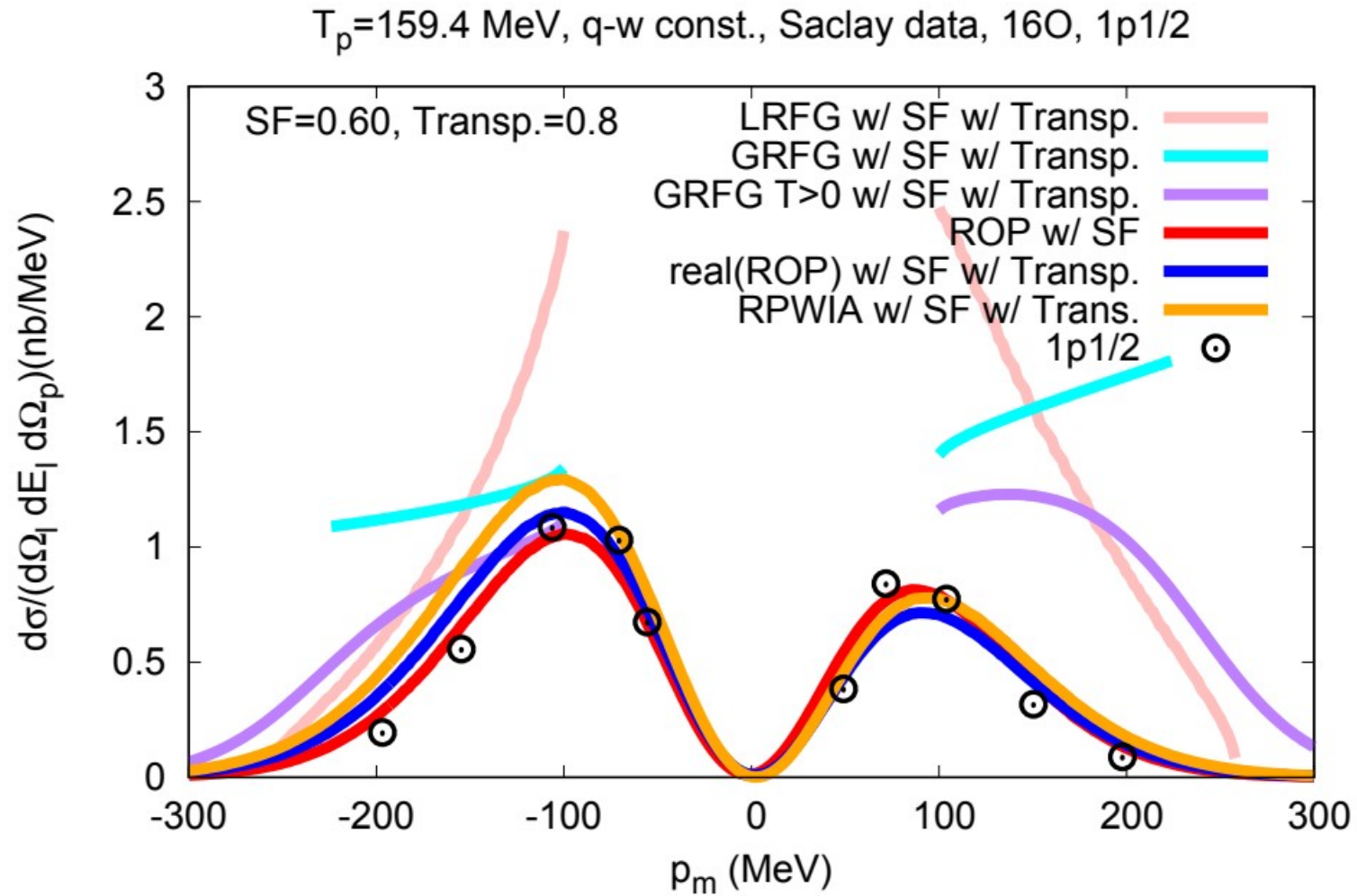
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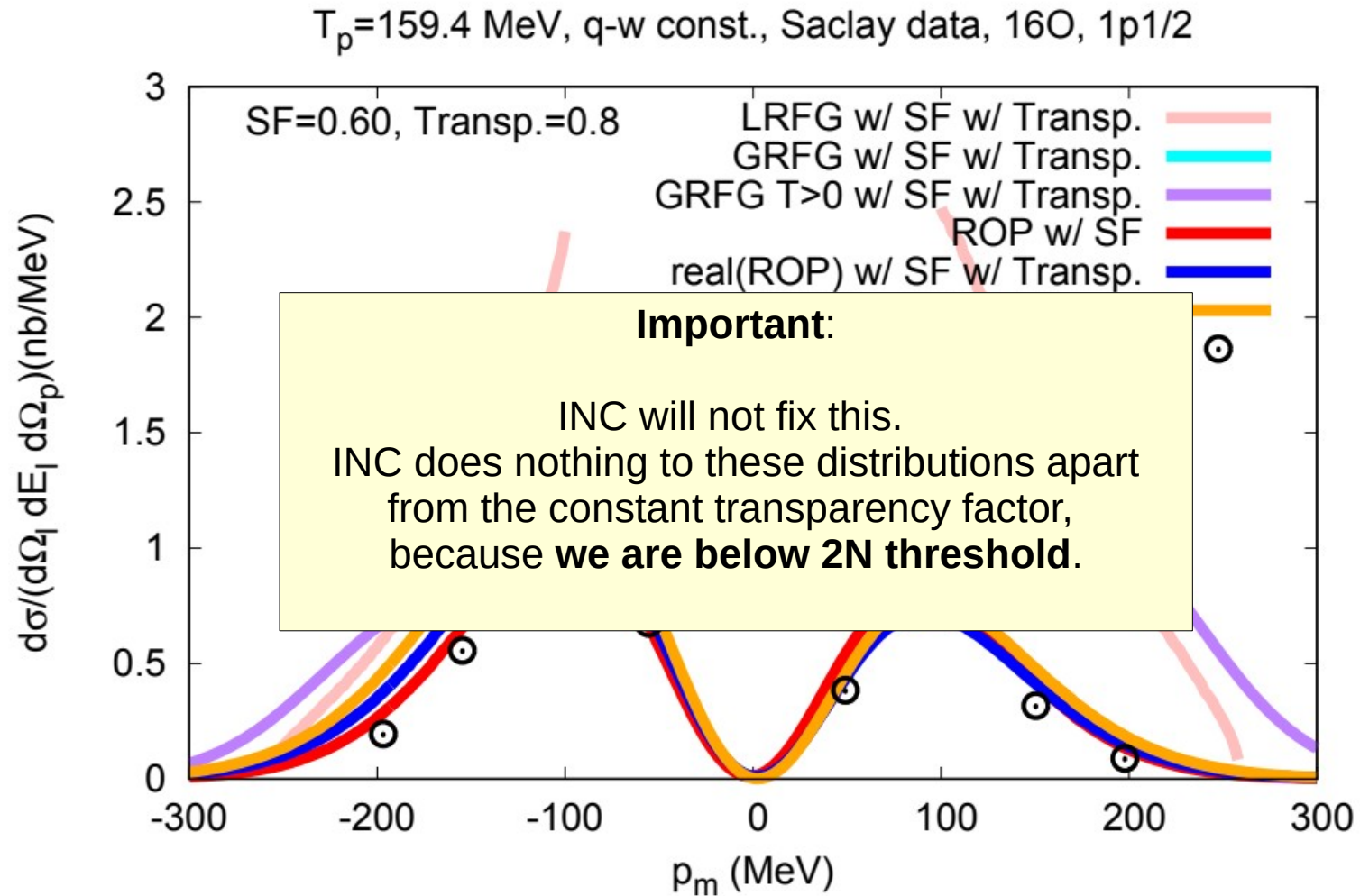
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Exclusive experiments: $^{40}\text{Ca}(e,e'p)^{39}\text{K}$

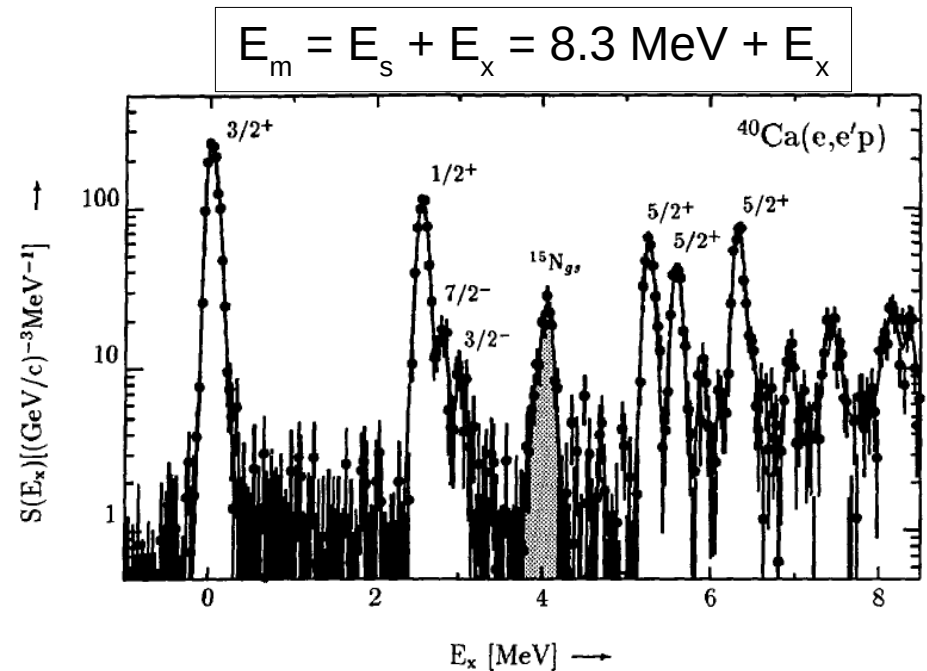
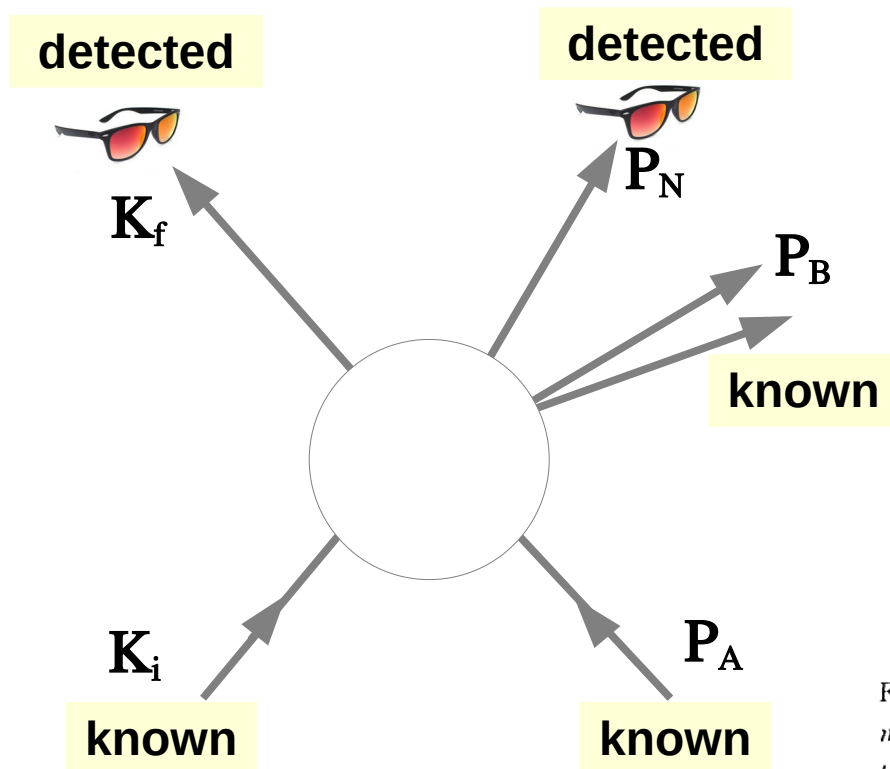
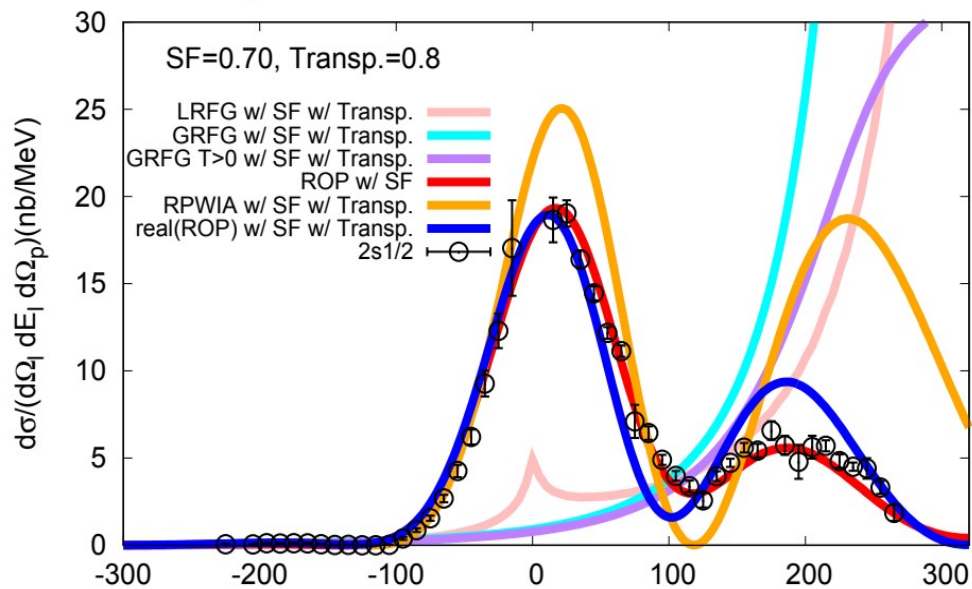


Figure 4.3: A typical excitation-energy spectrum for the reaction $^{40}\text{Ca}(e,e'p)$ at a missing momentum of 100 MeV/c. Some strongly excited states are indicated with their spin and parity. The curve represents a fit to the data. G. Kramer's PhD Thesis (1990) Universiteit van Amsterdam

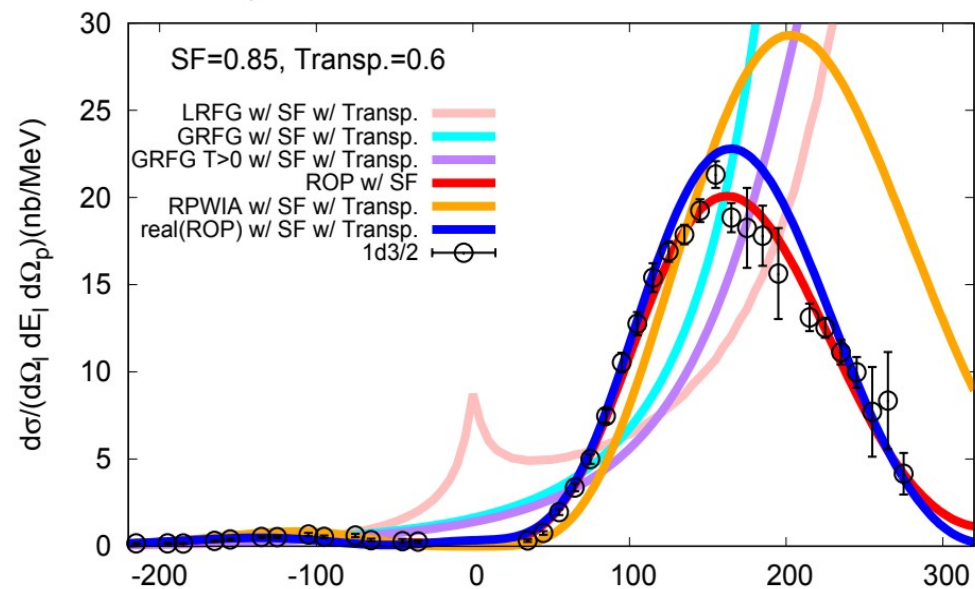
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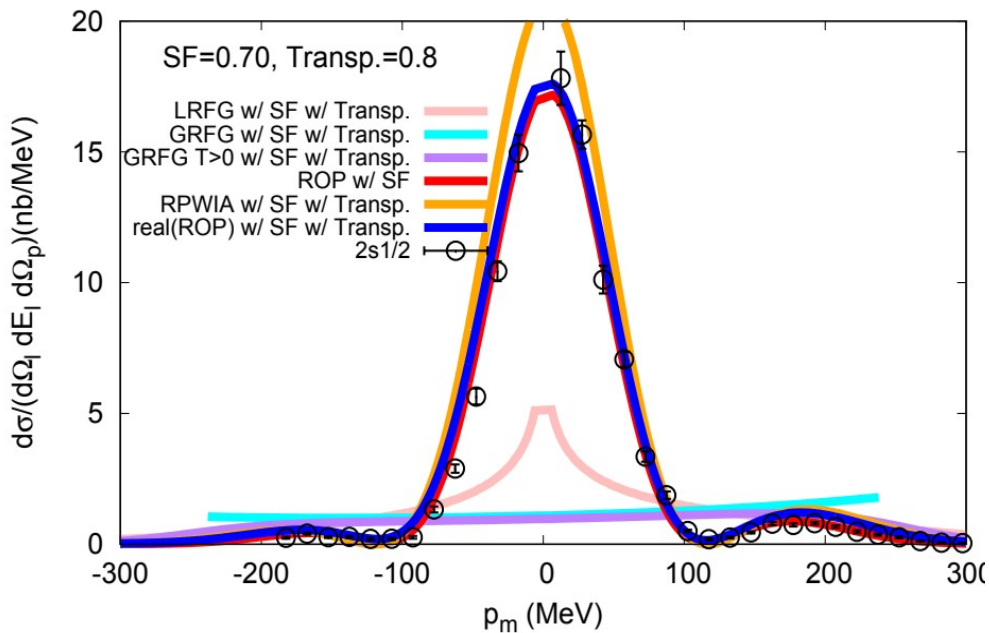
$T_p=100$ MeV, parallel kin, Nikhef data, 40Ca, 2s1/2



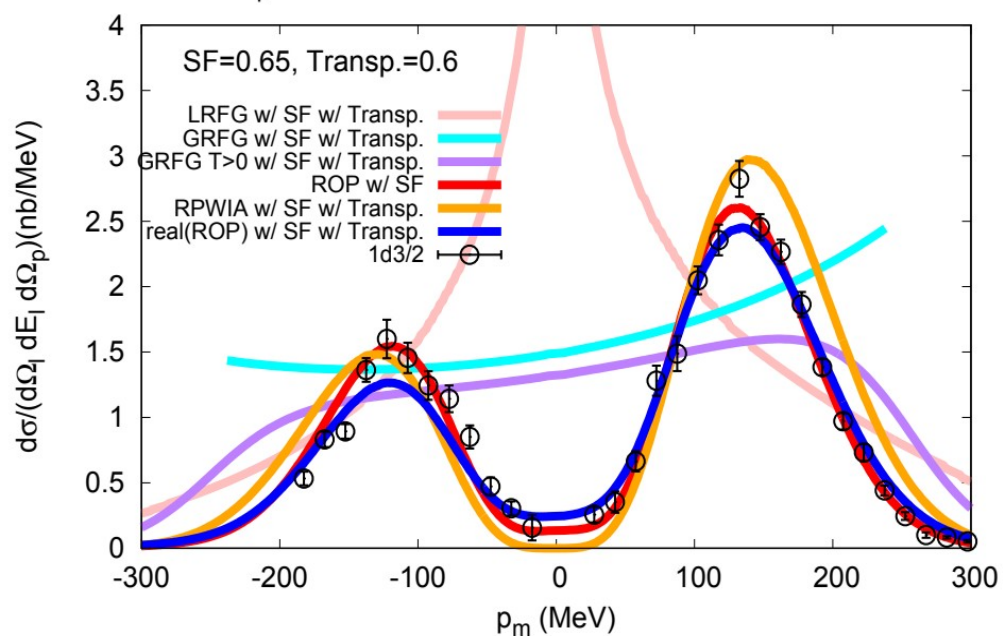
$T_p=100$ MeV, parallel kin, Nikhef data, 40Ca, 1d3/2



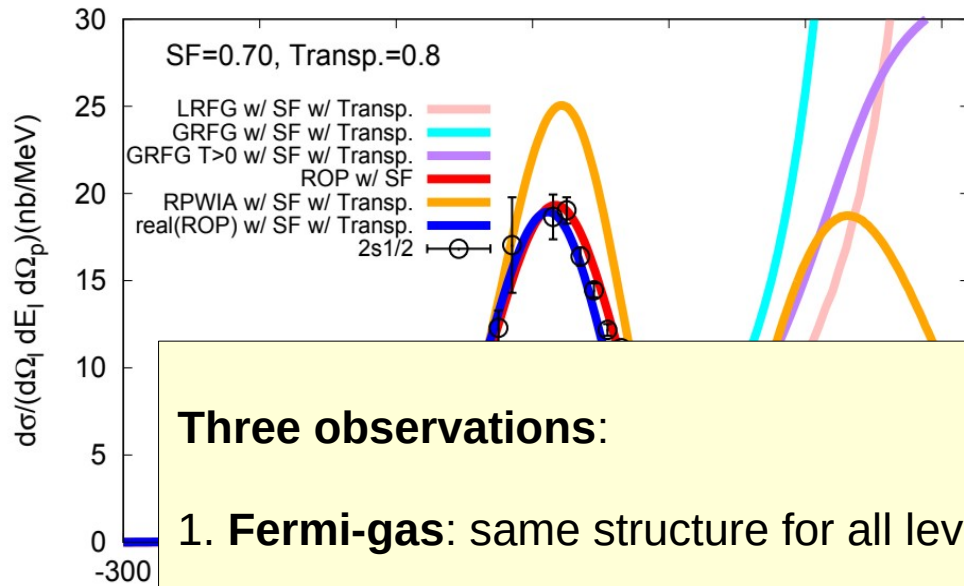
$T_p=100$ MeV, q-w const., Nikhef data, 40Ca, 2s1/2



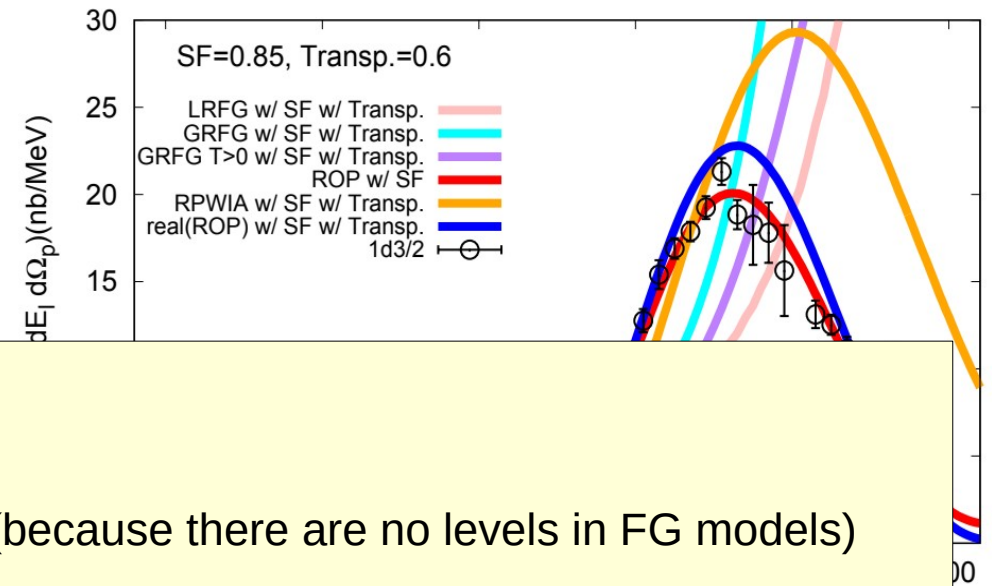
$T_p=100$ MeV, q-w const., Nikhef data, 40Ca, 1d3/2



$T_p=100$ MeV, parallel kin, Nikhef data, 40Ca, 2s1/2

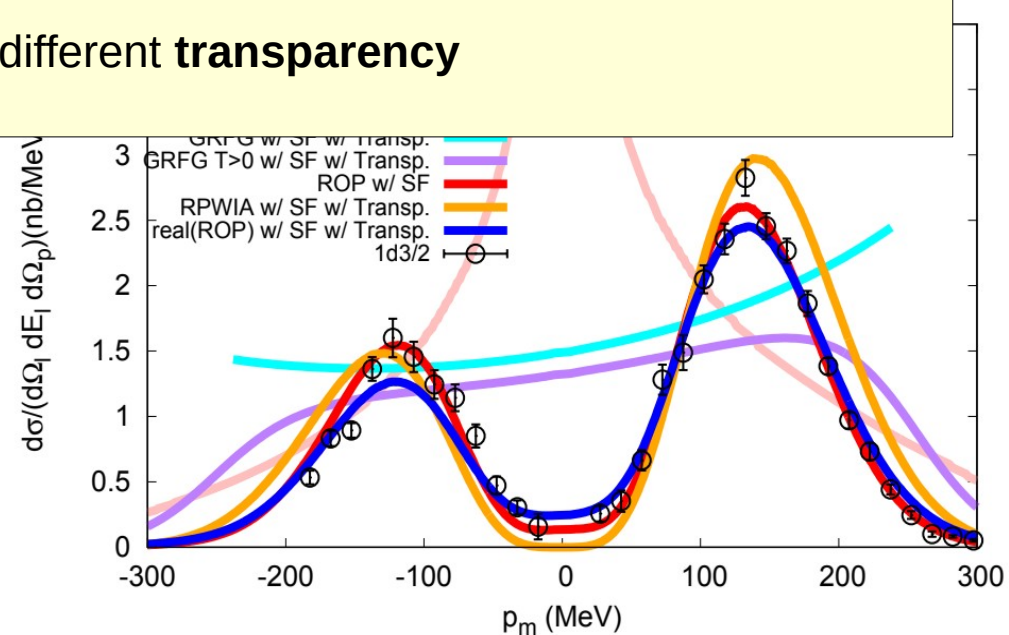
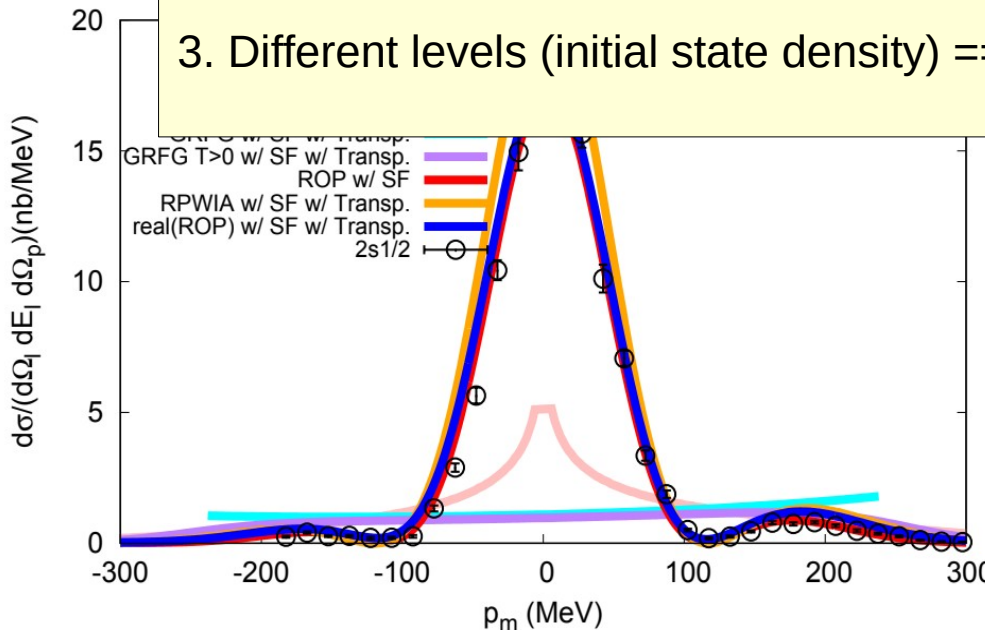


$T_p=100$ MeV, parallel kin, Nikhef data, 40Ca, 1d3/2



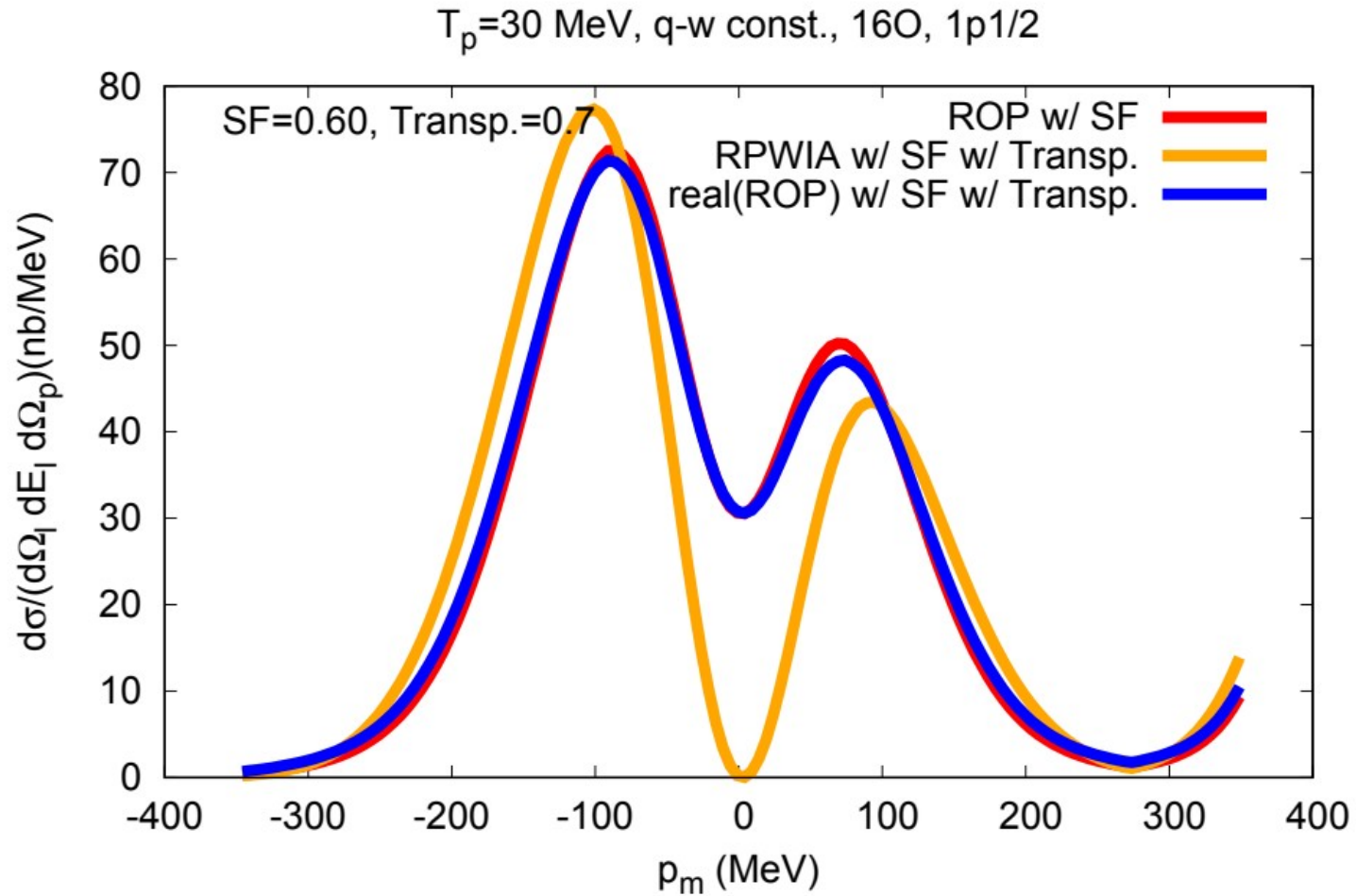
Three observations:

1. **Fermi-gas**: same structure for all levels (because there are no levels in FG models)
2. **'plane waves' x transparency** is superior to **'Fermi gas' x transparency**, but not as good as **'distorted waves' x transparency**
3. Different levels (initial state density) ==> different **transparency**



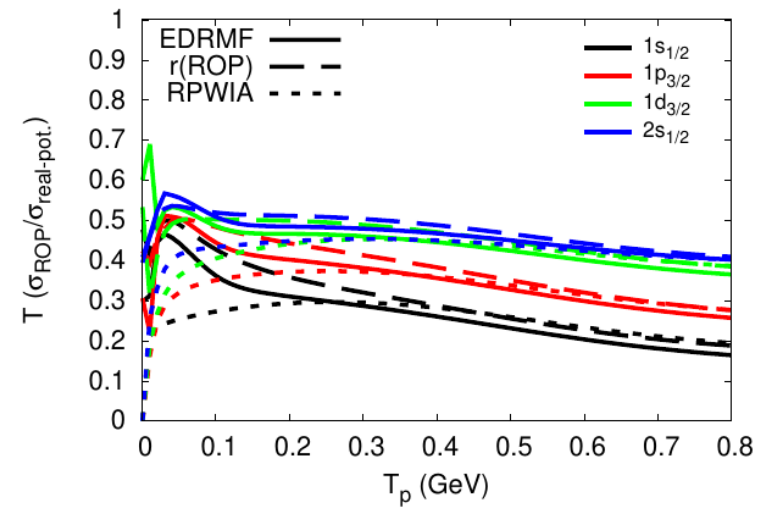
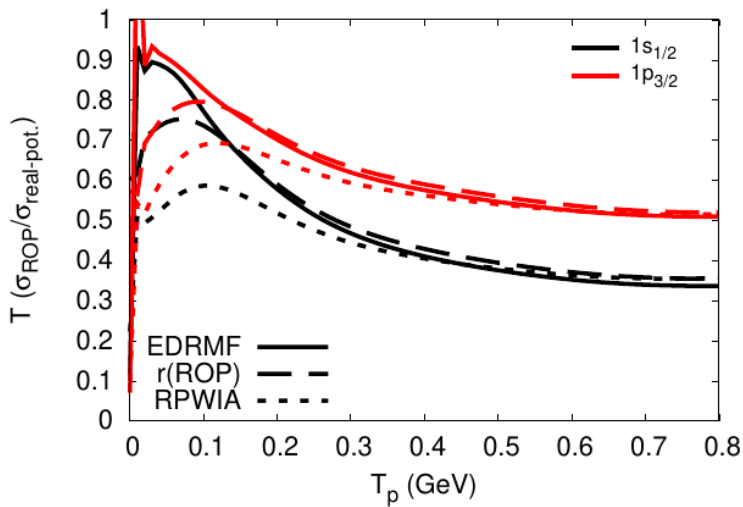
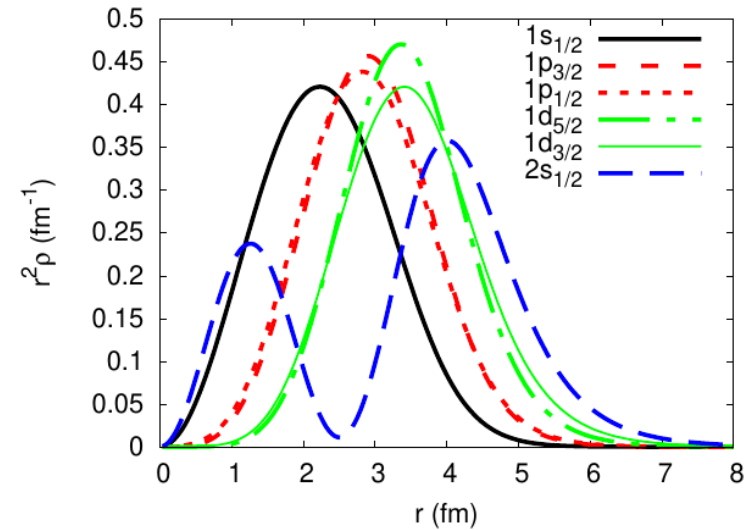
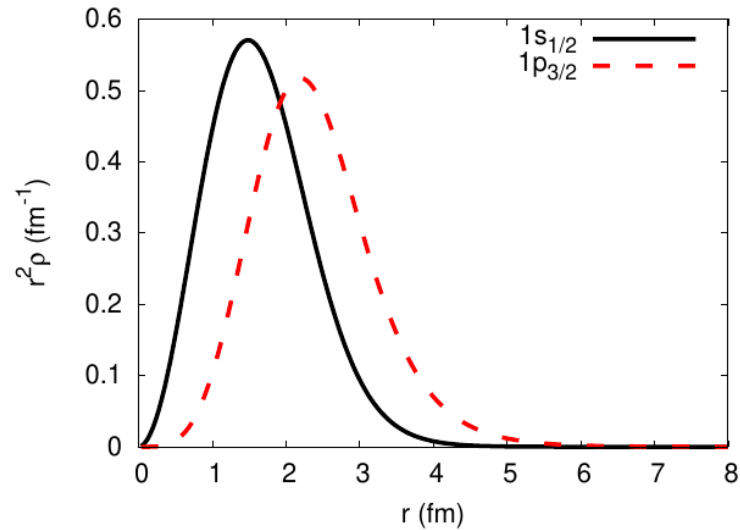
2. 'plane waves' x transparency is superior to 'Fermi gas' x transparency, but not as good as 'distorted waves' x transparency

Clearer for lower nucleon kinetic energies



3. Different levels ==> different transparency

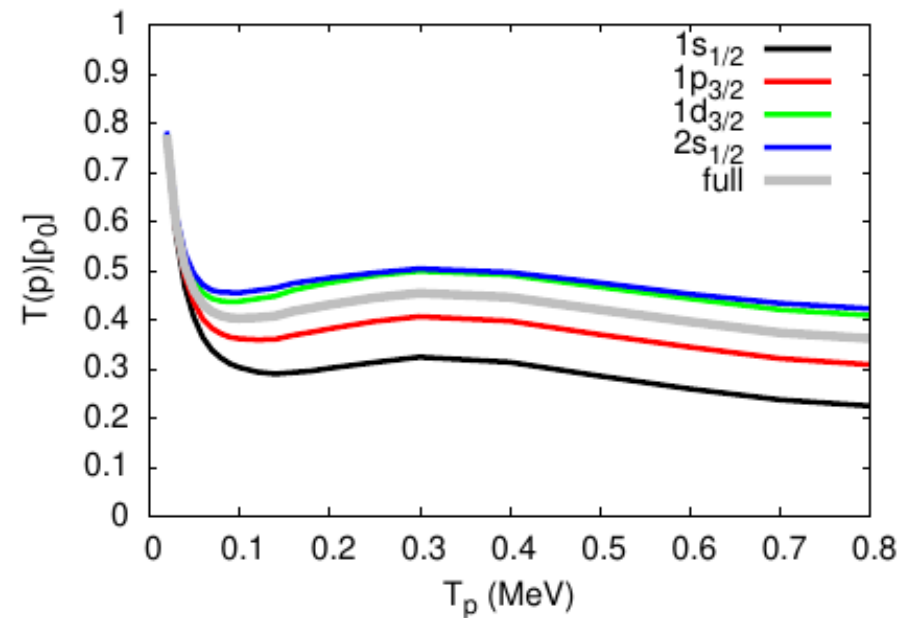
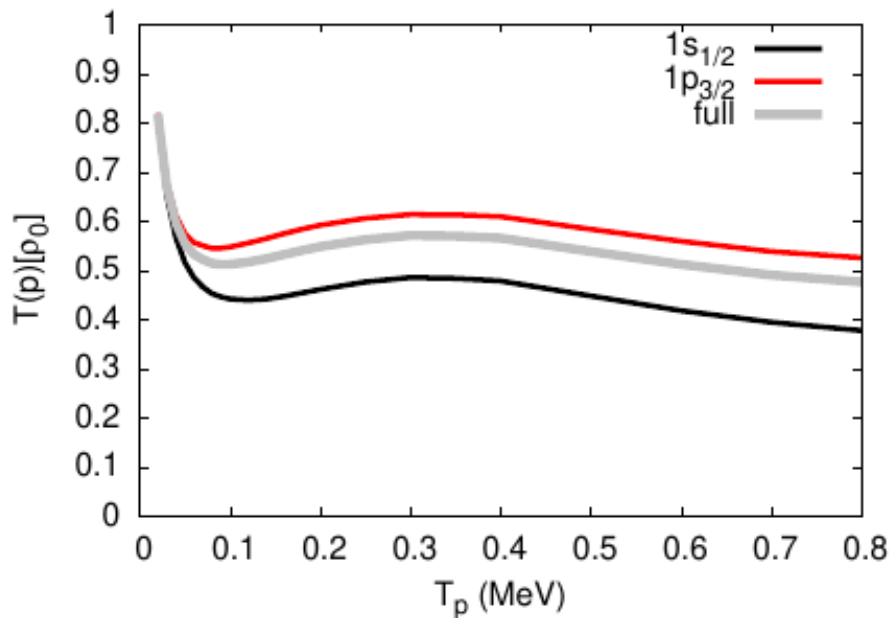
in other words, the transparency depends on where (which level) the nucleon originated;
in other words, the transparency depends on the density.

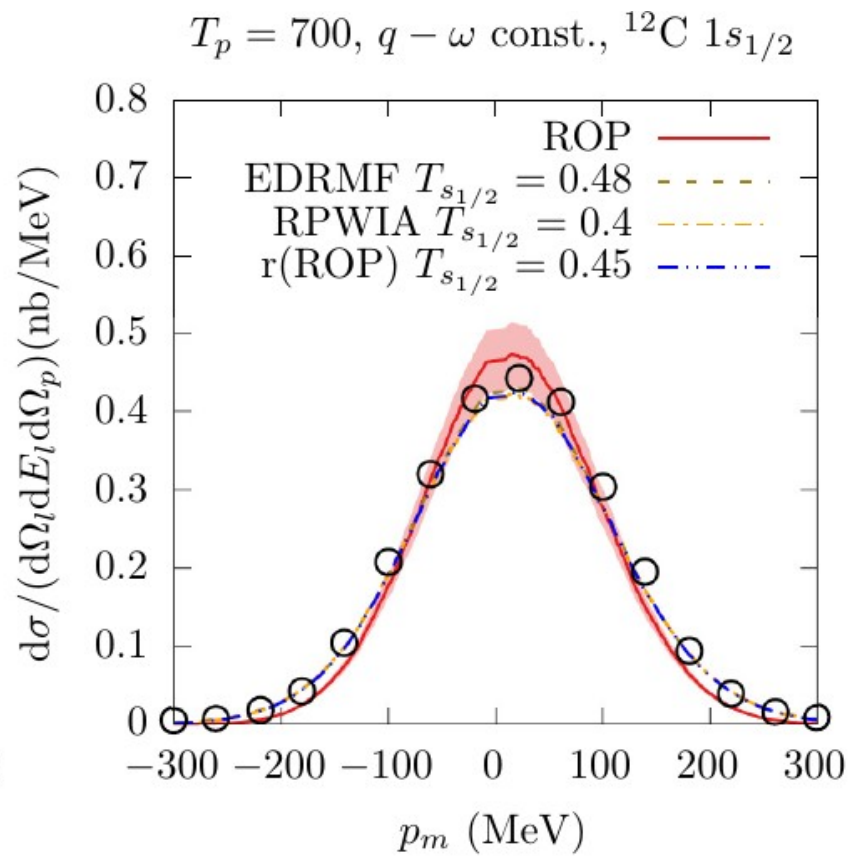
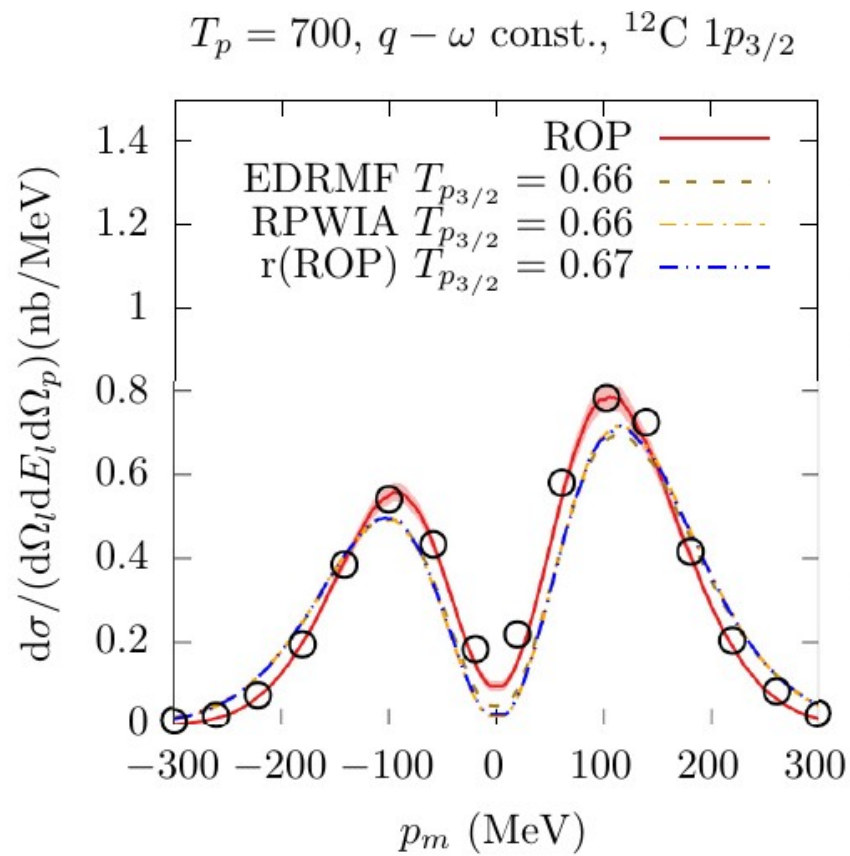


3. Different levels ==> different transparency

in other words, the transparency depends on where (which level) the nucleon originated;
in other words, the transparency depends on the density.

**This feature could be implemented
easily in a MC event generator!!!**





TAKE-HOME MESSAGES FROM THIS PRESENTATION

- 1) Nuclear models should be benchmarked using electron scattering data
- 2) Nuclei have shells
- 3) Elastic FSI (or nucleon distortion) **belong to the elementary-vertex model**
==> Avoid them in the INC (for the struck nucleon)
(Intranuclear cascade (INC) does NOT affect the inclusive (e,e') cross section
==> A realistic model for the elementary vertex is needed .)
- 4) The elementary-vertex model should provide information on the hadron(s) produced in the “hard interaction”, those will be propagated using the INC.
- 5) Exclusive (e,e'p) scattering data are useful to improve:
 - + elementary-vertex models,
 - + and a basic feature of the INC: the **transparency**.
- 6) Nuclear transparency depends on the level from which the knockout nucleon originated. This feature could be easily implemented in event generators.
- 7) “Distorted-wave model (with real potentials) + INC” is a feasible approach, and it **improves** current modeling of **nuclear effects in event generators**.

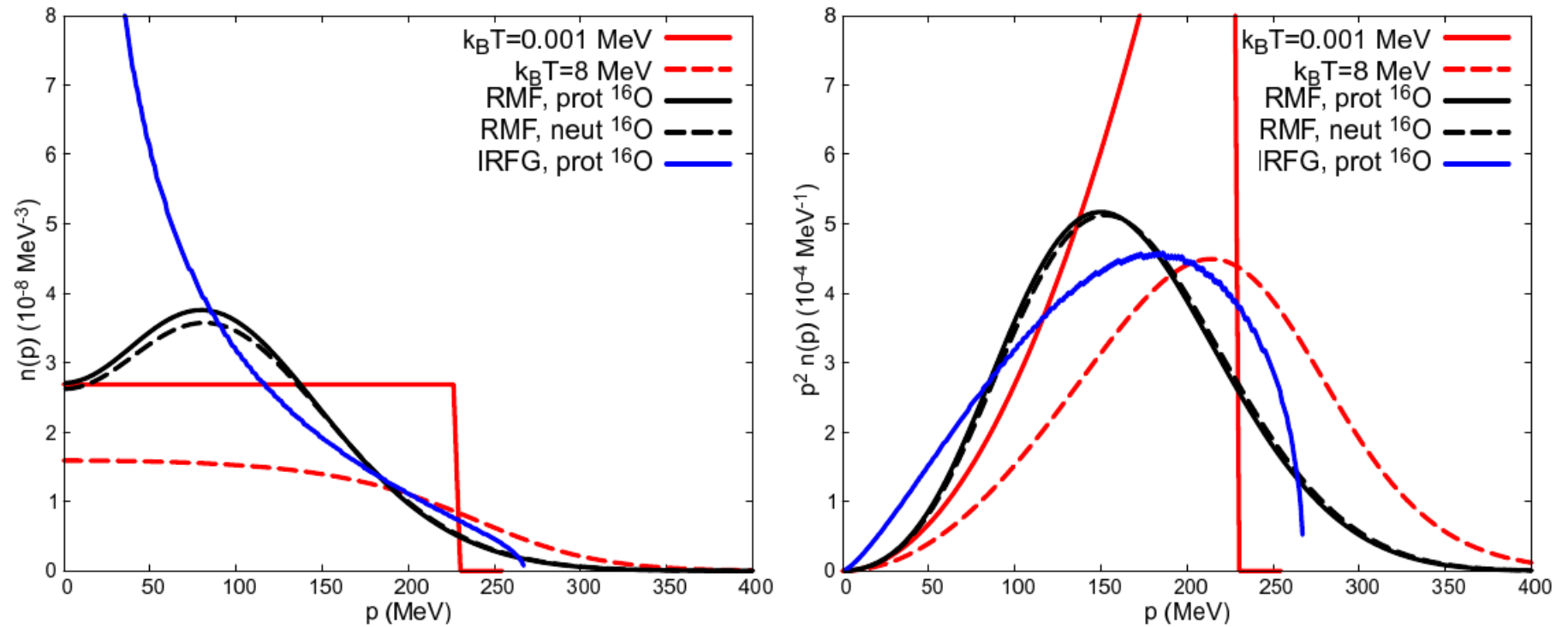
Thanks!!!

additional slides

Summary of separation energies (SE) for p, n, p+n, p+p, n+n,... for ^{12}C and ^{40}Ar

	One nucleon knockout		Two-nucleon knockout			Three-nucleon knockout				Deuterium knockout
	p SE (MeV)	n SE (MeV)	p+n SE (MeV)	p+p SE (MeV)	n+n SE (MeV)	3p SE (MeV)	2p+n SE (MeV)	p+2n SE (MeV)	3n SE (MeV)	^2H SE (MeV)
^{12}C	-15.96	-18.72	-27.41	-27.19	-31.84	-46.82	-34.00	-35.85	-53.12	-25.19
^{40}Ar	-12.53	-9.87	-20.60	-22.76	-16.47	-37.91	-26.74	-26.71	-32.36	-18.38

Momentum distributions



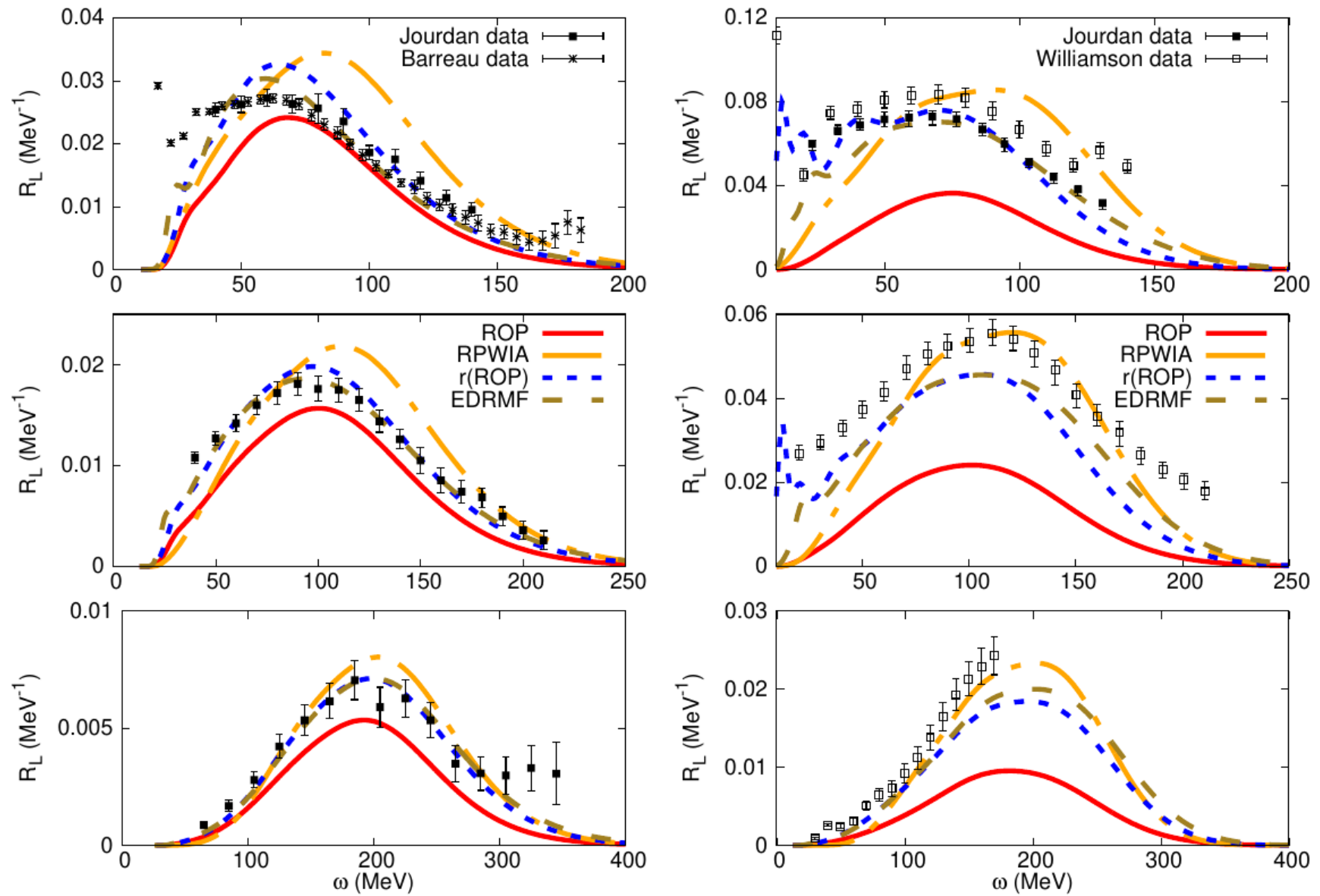


FIG. 4. EDRMF, $r(\text{ROP})$ and ROP predictions compared to R_L ^{12}C data from [60] [61] (left panels) and ^{40}Ca data from [61]–[64] (right panels). Each panel corresponds to a fixed q : 300, 380 and 570 MeV (left panels); and 300, 380 and 550 MeV (right panels).